

TCP/IP API

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15.1.1.1 API Introduction

[Robokit](#) provides users with the APIs related to robot operation. The entire APIs adopt the TCP request/response mode, and the robot as a server accepts the client's request and responds to the client. The API request consists of a header and a data area. The header is fixed-length, used to identify the basic information of the data packet and the length and type of the data area. The data area is serialized JSON data, and the corresponding JSON object can be obtained by deserializing according to the type information in the header. The response of the API also consists of a header and a data area. The header is a fixed-length byte obtained according to the request (For relationship between the response header and the corresponding request header, see below.), and the data area is also the serialized JSON data.

API Test and Examples

We open source API test tool on Github. The tool, [Source](#), is written in Qt, and its source code can be used as a example. The compiled windows executable file can also be downloaded in [Release](#).

For C# example, [Please click the link](#).

API Category

Types of API are divided into five, namely robot status API, robot control API, robot navigation API, robot configuration API, other API, and robot push API.

1. [Robot Status API](#) : Queries various state of the robot, such as position, speed, and alarm information.
2. [Robot Control API](#) : Sends open loop control commands to the robot, such as forward speed, rotate speed, relocating, etc.
3. [Robot Navigation API](#) : Sends navigation commands to the robot, such as path navigation, translation and rotation.
4. [Robot Config API](#) : Sets the parameters of the robot, as well as download and upload maps, etc.
5. [Other API](#) : Other parameters, such as audio control, etc.
6. [Robot Push API](#) : Uses by robot to actively push data to the currently connected client.

Each category uses different ports, independent of each other.

Category	Port	Number of Connections Allowed
Robot Status API	19204	10

Category	Port	Number of Connections Allowed
Robot Control API	19205	5
Robot Navigation API	19206	5
Robot Configuration API	19207	5
Other API	19210	5
Robot Push API	19301	10

Port

The fastest time interval of API requests for each port is recommended to be 100-200ms, please do not be too fast.

The robot will use the TCP KeepAlive mechanism for each established TCP connection to keep alive, and the dead connection will be cleared in time.

If the robot receives any erroneous data packets that do not conform to the header of this protocol, it will actively close the TCP connection (no reply will be received).

In other cases, the robot will not actively disconnect, and API users should control when the connection is interrupted.

For each connection, the robot is processed in the form of request and response, that is, the next request will not be processed before a response is made, therefore, it is recommended that when using it, wait for the response to the previous request before sending the next request.

15.1.1.1.2 API Message Structure

The structure is as follows:

Name	Content	Length (byte)	Description
Message synchronization header	0x5A	1 (uint8)	Message synchronization header, determines the beginning of the message header.
Protocol version	1	1 (uint8)	The major version number of the agreement, currently 0x01
Serial number		2 (uint16)	The serial number of the request and response (0 ~ 65535), this field of the request packet and the response packet is the same, and API users fill in the serial number by themselves. The robot's response to each request uses this sequence number.
Data area length		4(uint32)	Data area length, which is the length of JSON serialized data.
Message type (API number)		2(uint16)	The type of message, that is, the API number, see the API chapters in below .
Reserved area		6(uint8[6])	Reserved area (Do not focus on the content of the reserved area, and its content may not be 0 and may change.)
Data area		Depends on the length of the data area in the header.	JSON serialized data content.

```
// The C++ header structure is as follows:
#include <stdint.h>
struct ProtocolHeader {
    uint8_t m_sync;
```

```
uint8_t  m_version;  
uint16_t m_number;  
uint32_t m_length;  
uint16_t m_type;  
uint8_t  m_reserved[6];  
};
```

As shown in the table above, the length of the message header is $1 + 1 + 2 + 4 + 2 + 6 = 16$ bytes, and the length of the request and response message headers is the same. If a message type has no data area, the content of the data area length is 0x00000000. The message type (API number) must be one of the API numbers below. If the robot receives an undefined message type, it will respond with an error. If the format of the **message header** cannot be parsed, the robot will **disconnect** without any response. If the data area is parsed incorrectly, an error will also be responded.

Note 1: Each request with the correct header in the message will have a response. The API number is used to correspond the request to the response. In general, the response API number is the request API number plus 10000 (0x2710).

Note 2: The robot will not actively send data under any circumstances.

Note 3: If the robot receives a message type that does not belong to a port, it will respond with an error.

Note 4: The 6 bytes of the reserved area must be filled and cannot be omitted, generally filled with 0x000000000000.

Example of Request

Take the following request for 2002 (0x07D2) as an example:

The structure of request data area is shown as follows :

Field Name	Type	Description	Whether Default
x	Number	The x coordinate in the global coordinate system, unit m	Yes
y	Number	The y coordinate in the global coordinate system, unit m	Yes
angle	Number	The angle in the global coordinate system, unit rad	Yes

2002(0x07D2) is the relocation request, the parameters are x coordinate and y coordinate. The standard JSON object is as follows:

```
{
  "x": 10.0,
  "y": 3.0,
  "angle": 0
}
```

In the message, it is necessary to serialize the above JSON object and stuff it into the data area of the message, and fill the length of the serialization into the length of the data area of the message header.

The serialized data is {"x":10.0,"y":3.0,"angle":0}, which converts to hexadecimal as 7B 22 78 22 3A 31 30 2E 30 2C 22 79 22 3A 33 2E 30 2C 22 61 6E 67 6C 65 22 3A 30 7D. The length is 28(0x1c) bytes, so the header is 5A 01 00 01 00 00 00 1C 07 D2 00 00 00 00 00 00, and splice the header and data area together to get a request as follows (total 44 bytes):

```
5A 01 00 01 00 00 00 1C 07 D2 00 00 00 00 00 00
7B 22 78 22 3A 31 30 2E 30 2C 22 79 22 3A 33 2E 30 2C 22 61 6E 67 6C 65 22 3A 30 7D
```

Example of response

Take the following response of 11004 (0x2AFC) as an example:

The structure of response data is shown as follows:

Field Name	Type	Description	Whether Default
x	Number	The x-coordinate of the robot, unit m	Yes
y	Number	The y-coordinate of the robot, unit m	Yes
angle	Number	The angle of the robot, unit rad	Yes
confidence	Number	Location confidence of robot, range [0, 1]	Yes
ret_code	Number	API error code	Yes

The order of the keys in the JSON Object returned in each data area is not guaranteed, because JSON does not specify this. The current position of the robot is `x = 6.0`, `y = 2.0`, `angle = 1.57`, `confidence = 0.9`, represented as a JSON object as follows:

```
{
  "ret_code": 0,
  "x": 6.0,
  "y": 2.0,
  "angle": 1.57,
  "confidence": 0.9
}
```

The serialized data is {"ret_code":0,"x":6.0,"y":2.0,"angle":1.57,"confidence":0.9}, which converts to hexadecimal as following:

```
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 2C 22 78 22 3A 36 2E 30 2C 22 79 22 3A 32 2E 30 2C
22 61 6E 67 6C 65 22 3A 31 2E 35 37 2C 22 63 6F 6E 66 69 64 65 6E 63 65 22 3A 30 2E 39 7D
```

The length is 60(0x3C) bytes, so the response header is 5A 01 00 01 00 00 00 3C 2A FC 00 00 00 00 00 00. The data after splicing is as follows (a total of 76 bytes):

```
5A 01 00 01 00 00 00 3C 2A FC 00 00 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 2C 22 78 22 3A 36 2E 30 2C 22 79 22 3A 32 2E 30 2C
22 61 6E 67 6C 65 22 3A 31 2E 35 37 2C 22 63 6F 6E 66 69 64 65 6E 63 65 22 3A 30 2E 39 7D
```

Response Structure

API response data area usually contains following fields:

Field	Type	Description	Whether Default
ret_code	Number	API error code	Yes
create_on	String	API upload timestamp (added in version 3.3.3.59 or later)	Yes
err_msg	String	Error message	Yes

The API error code is the error code contained in the JSON object in the response data area (ret_code), which is used to indicate errors such as unsuccessful request execution or request error. If ret_code is 0 or default, no error occurs; if not 0, an error has occurred. For details, see error code in [Appendix A: API error code](#). When there is no error, err_msg will also be defaulted, that is, the data area will be empty. create_on is the timestamp when the API returns, added in version 3.3.3.59.

15.1.1.1.3 Coordinate System

In [Robokit](#), there are two sets of coordinate systems, namely the global coordinate system and the robot coordinate system. The global coordinate system is the right-handed coordinate system, that is, the coordinate system of the map and the operations related to the map and the location all use the global coordinate system. For example, the position of the robot (x, y, r) describes the x coordinate, y coordinate, and r angle of the robot in the global coordinate system. [Schematic diagram](#) see below.

For the movement of the robot itself, you need to refer to the robot coordinate system. The robot coordinate system is the right-hand coordinate system. For [details](#), see below.

Global Coordinate System

The global coordinate system is the right-handed coordinate system.

Origin of Global Coordinate System

Open a map in Roboshop to see the origin of the global coordinate system of the map. You can also change the position of the origin in the map arbitrarily through the coordinate transformation tool.

Robot Coordinate System

The robot coordinate system is the right-handed coordinate system.

Origin of Robot Coordinate System

For a robot moving in a differential manner, the origin of the robot coordinate system is the center point of the wheelbase of the two wheels, as shown in the following figure:

Note: When we talk about the speed of the robot, in general, v_x is the speed of the robot in the x-axis direction of the robot coordinate system, v_y is the speed of the robot in the y-axis direction of the robot coordinate system, and w is the angular velocity of the robot in robot coordinate system.

15.1.1.1.4 API Usage Precautions

1. International units are used, such as m, rad, m/s, etc, except those of some specified values.
2. The API number of response is the API number of request + 10000 (0x2710), and the request corresponds to the response respectively.
3. Avoid sending any non-ASCII characters, especially operations related to map names, because map names containing Chinese characters will be considered illegal.
4. The order of the keys in the JSON Object returned in each data area is not guaranteed, because JSON does not specify this.
5. If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly.

15.1.1.2 API User Guide

This guide introduces the use of this TCP API protocol to control the robot for continuous path navigation (This is the most common way.).

Path Navigation Process

1. Start the Robot

2. Wait for the robot to boot complete (After the robot is successfully connected using TCP, the boot is completed.)
3. Perform relocation [2002](#)
4. Query location status and wait for relocating completion [1021](#)
5. Confirm correct location [2003](#)
6. Perform path navigation [3051](#)
7. Query navigation status and wait for navigation to complete [1020](#)
8. Repeat steps 6 and 7 to achieve continuous path navigation

If the location was correct and did not move after the robot was turned off the last time, then step 3 can be omitted after the robot is turned on. After querying that the relocating is completed, directly confirm that the location is correct.

Robot Status API

The robot status API is used to query all status of the robot.

API List

Request

API Number	Name	Description
1000	robot_status_info_req	Robot information inquiry
1002	robot_status_run_req	Robot running information inquiry(Such as running time, mileage, etc.)
1004	robot_status_loc_req	Robot location inquiry
1005	robot_status_speed_req	Robot speed inquiry
1006	robot_status_block_req	Robot blocked status inquiry
1007	robot_status_battery_req	Robot battery status inquiry
1009	robot_status_laser_req	Robot laser status inquiry
1010	robot_status_path_req	Robot path status inquiry
1011	robot_status_area_req	Robot area status inquiry
1012	robot_status_emergency_req	Robot estop status inquiry

API Number	Name	Description
1013	robot_status_io_req	Robot I/O status inquiry
1014	robot_status_imu_req	Robot IMU data inquiry
1015	robot_status_rfid_req	Robot RFID data inquiry
1016	robot_status_ultrasonic_req	Robot ultrasonic status inquiry
1017	robot_status_pgv_req	Robot PGV data inquiry
1018	robot_status_encoder_req	Robot encoder status inquiry
1020	robot_status_task_req	Robot navigation status inquiry, such as navigation target, navigation related path, etc.
1021	robot_status_reloc_req	Robot localization status inquiry
1022	robot_status_loadmap_req	Robot map loading status inquiry
1025	robot_status_slam_req	Robot SLAM status inquiry

API Number	Name	Description
1027	robot_status_jack_req	Robot jack status inquiry
1028	robot_status_fork_req	Robot fork status inquiry
1029	robot_status_roller_req	Robot roller status inquiry
1030	robot_status_dispatch_req	Robot dispatching status inquiry
1040	robot_status_motor_req	Robot motor status inquiry
1050	robot_status_alarm_req	Robot alarm status inquiry
1060	robot_status_current_lock_req	Robot current lock inquiry
1100	robot_status_all1_req	Batch data inquiry 1
1101	robot_status_all2_req	Batch data inquiry 2
1102	robot_status_all3_req	Batch data inquiry 3

API Number	Name	Description
1110	robot_status_task_status_package_req	Robot task status package inquiry
1300	robot_status_map_req	Loaded map and stored map inquiry
1301	robot_status_station	Station information of currently loaded map
1302	robot_status_mapmd5_req	MD5 of specific map list inquiry
1400	robot_status_params_req	Robot parameters inquiry
...	...	...

Response

API Number	Name	Description
11000	robot_status_info_res	Response to 1000 request
11002	robot_status_run_res	Response to 1002 request
11004	robot_status_loc_res	Response to 1004 request
11005	robot_status_speed_res	Response to 1005 request
11006	robot_status_block_res	Response to 1006 request
11007	robot_status_battery_res	Response to 1007 request
11009	robot_status_laser_res	Response to 1009 request
11010	robot_status_path_res	Response to 1010 request
11011	robot_status_area_res	Response to 1011 request
11012	robot_status_emergency_res	Response to 1012 request
11013	robot_status_io_res	Response to 1013 request

API Number	Name	Description
11014	robot_status_imu_res	Response to 1014 request
11015	robot_status_rfid_res	Response to 1015 request
11016	robot_status_ultrasonic_res	Response to 1016 request
11017	robot_status_pgv_res	Response to 1017 request
11018	robot_status_encoder_res	Response to 1018 request
11020	robot_status_task_res	Response to 1020 request
11021	robot_status_reloc_res	Response to 1021 request
11022	robot_status_loadmap_res	Response to 1022 request
11025	robot_status_slam_res	Response to 1025 request
11027	robot_status_jack_res	Response to 1027 request

API Number	Name	Description
11028	robot_status_fork_res	Response to 1028 request
11029	robot_status_roller_res	Response to 1029 request
11030	robot_status_dispatch_res	Response to 1030 request
11040	robot_status_motor_res	Response to 1040 request
11050	robot_status_alarm_res	Response to 1050 request
11060	robot_status_current_lock_res	Response to 1060 request
11100	robot_status_all1_res	Response to 1100 request
11101	robot_status_all2_res	Response to 1101 request
11102	robot_status_all3_res	Response to 1102 request
11110	robot_status_task_status_package_res	Response to 1110 request

API Number	Name	Description
11300	robot_status_map_res	Response to 1300 request
11301	robot_status_station_res	Response to 1301 request
11302	robot_status_mapmd5_res	Response to 1302 request
11400	robot_status_params_res	Response to 1400 request
...	...	...

15.1.1.3.2 Robot Information Inquiry

Request

- API number: 1000 (0x03E8)
- Name: robot_status_info_req
- Description: Robot Information inquiry

- JSON data area: None

Request Example

The length of the data area is 0, and it only has the header:

```
5A 01 00 01 00 00 00 00 03 E8 00 00 00 00 00 00
```

Response

- API number: 11000 (0x2AF8)
- Name: robot_status_info_res
- Description: Response of robot Information inquiry
- JSON data area: See table below.

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

Field Name	Type	Description	Whether Default
id	string	Robot id	YES
vehicle_id	string	Robot name	YES
robot_note	string	Robot notes	YES
version	string	Robokit version	YES
model	string	Robot model name	YES
dsp_version	string	Firmware version	YES
gyro_version	string	Gyroscope version	YES
map_version	string	Map version	YES
model_version	string	Model version	YES

Field Name	Type	Description	Whether Default
netprotocol_version	string	Network protocol version	YES
modbus_version	string	ModbusTCP protocol version	YES
current_map	string	Current map name	YES
current_map_md5	string	MD5 value of the current map	YES
model_md5	string	MD5 value of the current model	YES
ssid	string	SSID of the currently connected Wifi (the robot needs to support connection to Wifi and is already connected, otherwise it is empty)	YES
rssi	number	The signal strength of the currently connected Wifi, 0-100% (the robot needs to support Wifi connection and is already connected, otherwise it is 0)	YES
ap_addr	string	MAC address of current connected AP	YES

Field Name	Type	Description	Whether Default
current_ip	string	Current IP (another IP address except 192.168.192.5)	YES
MAC	string	The MAC address of the dedicated Ethernet interface for the robot wireless client (only valid for SRC -2000), and the format is XXXXXXXXXXXX, with no separator in the middle.	YES
echoid_type	string	Robot machine code type (used for activation, usually 0x1800)	YES
echoid	string	Robot machine code (used for activation, usually *XXX XXXX XXXX XXXX)	YES
features	array	Robot function module name = Function module name active = Whether it is activated expiry_date = Expiry date (can be default)	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES

Field Name	Type	Description	Whether Default
err_msg	string	Error message	YES

Response Example

Suppose the response content is:

```
{"id":"S001","version":"v1.1.0","model":"S1","dsp_version":"v1.2.2","map_version":"v1.0.0","model_version":"v1.1.0","netp rotocol_version":"v1.2.0"}
```

Adding the header, it has a total of 163 (0xA3) bytes (Byte), so the message of the response is:

```
5A 01 00 01 00 00 00 93 2A F8 00 00 00 00 00
7B 22 69 64 22 3A 22 53 30 30 31 22 2C 22 76 65 72 73 69 6F
6E 22 3A 22 76 31 2E 31 2E 30 22 2C 22 6D 6F 64 65 6C 22 3A
22 53 31 22 2C 22 64 73 70 5F 76 65 72 73 69 6F 6E 22 3A 22
76 31 2E 32 2E 32 22 2C 22 6D 61 70 5F 76 65 72 73 69 6F 6E
22 3A 22 76 31 2E 30 2E 30 22 2C 22 6D 6F 64 65 6C 5F 76 65
72 73 69 6F 6E 22 3A 22 76 31 2E 31 2E 30 22 2C 22 6E 65 74
70 72 6F 74 6F 63 6F 6C 5F 76 65 72 73 69 6F 6E 22 3A 22 76
31 2E 32 2E 30 22 7D
```

15.1.1.3.3 Robot Running Information Inquiry

Request

- API number: 1002 (0x03EA)
- Name: robot_status_run_req
- Description: Robot running information inquiry(Such as running time, mileage, etc.)
- JSON data area: None

Request Example

The length of the data area is 0, and it only has the header:

```
5A 01 00 01 00 00 00 00 03 EA 00 00 00 00 00 00
```

Response

- API number: 11002 (0x2AFA)
- Name: robot_status_run_res
- Description: Response of robot running information inquiry
- JSON data area: See table below.
-

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

Field Name	Type		Description	Whether Default
odo	number		Accumulated mileage, unit: m	YES
today_odo	number		Today's cumulative mileage, unit: m	YES
time	number		The current running time (the time from boot to now), unit: ms	YES
total_time	number		Cumulative running time, unit: ms	YES
controller_temp	number		Controller temperature, unit: °C	YES
controller_humi	number		Controller humidity, unit: %	YES
controller_voltage	number		Controller voltage, unit: V	YES
ret_code	number		API error code	YES
create_on	string		API upload timestamp	YES

Field Name	Type		Description	Whether Default
err_msg	string		Error message	YES

Response Example

```
{
  "controller_humi": 50,
  "controller_temp": 30,
  "controller_voltage": 24,
  "odo": 146.34,
  "time": 794297,
  "total_time": 116606929
}
```

15.1.1.3.4 Robot Location Inquiry

The coordinate of the robot in the world coordinate system

Request

- API number: 1004 (0x03EC)
- Name: robot_status_loc_req
- Description: Robot location inquiry
- JSON data area: None

Request Example

5A 01 00 01 00 00 00 00 03 EC 00 00 00 00 00 00

Response

- API number: 11004 (0x2AFC)
- Name: robot_status_loc_res
- Description: Response of robot location inquiry
- JSON data area: See table below

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

Field Name	Type	Description	Whether Default
x	number	The x coordinate of the robot, unit: m	YES
y	number	The x coordinate of the robot, unit: m	YES
angle	number	The angle coordinate of the robo, unit: rad	YES
confidence	number	Confidence of robot's localization, range [0, 1]	YES

Field Name	Type	Description	Whether Default
current_station	string	The id of the station where the robot is currently located(this judgment is strict, the robot must be very close to a certain station(<50cm), otherwise it is a null character, that is, it is not at any station)	YES
last_station	string	id of the previous station where the robot is located	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "angle": -0.0064,
  "confidence": 0.637,
  "x": 3.5069,
  "y": 0.0687,
  "current_station": "LM1",
  "last_station": "LM2"
}
```

15.1.1.3.5 Robot Speed Inquiry

Request

- API number: 1005 (0x03ED)
- Name: robot_status_speed_req
- Description: Robot speed inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 ED 00 00 00 00 00 00
```

Response

- API number: 11005 (0x2AFD)
- Name: robot_status_speed_res
- Description: Response of robot speed inquiry
- JSON data area: See table below.

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

Field Name	Type	Description	Whether Default
vx	number	The actual speed of the robot in the x direction of the robot coordinate system, unit: m/s	YES
vy	number	The actual speed of the robot in the y direction of the robot coordinate system, unit: m/s	YES
w	number	The actual angular velocity of the robot in the robot coordinate system (i.e., turn clockwise to negative and counterclockwise to positive), unit: rad/s	YES
steer	number	Single steering wheel robot's current steering wheel angle, unit: rad	YES
spin	number	Robot tray(if exists) angle, unit: rad	YES
r_vx	number	The speed the robot receives in the x direction of the robot coordinate system, unit: m/s	YES
r_vy	number	The speed the robot receives in the y direction of the robot coordinate system, unit: m/s	YES
r_w	number	The received angular velocity of the robot in the robot coordinate system (i.e., turn clockwise to negative and counterclockwise to positive), unit: rad/s	YES

Field Name	Type	Description	Whether Default
r_steer	number	The steering wheel angle received by the single steering wheel robot, unit: rad	YES
r_spin	number	The rotation speed of the tray(if exists) received by robot, unit: rad/s	YES
steer_angles	array[number]	The current steer wheel angle of a multi-steering wheel robot, double-steering wheel robot = [front steering wheel angle, rear steering wheel angle], unit: rad	YES
r_steer_angles	array[number]	The steer wheel angle received by the multi-steer wheel robot, double-teering wheel robot = [front steering wheel angle, rear steering wheel angle], unit: rad	YES
is_stop	boolean	Whether the robot chassis is stationary (walking motor)	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "is_stop": true,
  "motor_steel_angles": [],
  "r_spin": 0,
  "r_steel": 0,
  "r_steel_angles": [
    0
  ],
  "r_vx": 0,
  "r_vy": 0,
  "r_w": 0,
  "ret_code": 0,
  "spin": 0,
  "steel": 0,
  "steel_angles": [
    0
  ],
  "vx": 0,
  "vy": 0,
  "w": 0
}
```

15.1.1.3.6 Robot Blocked Status Inquiry

Request

- API number: 1006 (0x03EE)
- Name: robot_status_block_req
- Description: Robot blocked status inquiry

- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 EF 00 00 00 00 00 00
```

Response

- API number: 11006 (0x2AFE)
- Name: robot_status_block_res
- Description: Response of robot blocked status inquiry
- JSON data area: See table below.

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

Field Name	Type	Description	Whether Default
blocked	boolean	Whether the robot is blocked	YES
block_reason	number	Reasons for being blocked, 0 = Ultrasonic, 1 = Laser, 2 = Fallingdown, 3 = Collision, 4 = Infrared, 5 = Lock, 6 = dynamic obstacle, 7 = virtual laser, 8 = 3D camera	YES
block_x	number	The x coordinate of the nearest obstacle(world coordinate system), unit: m	YES
block_y	number	The y coordinate of the nearest obstacle(world coordinate system), unit: m	YES
block_ultrasonic_id	number	The blocked ultrasonic id number, only makes sense when block_reason = 0, It is abandoned by 3.2.8.0, and changes to block_id.	YES
block_di	number	The ID number of the blocked DI, only makes sense when block_reason = 2, 3, 4 It is abandoned by 3.2.8.0, and changes to block_id.	YES
block_id	number	The blocked ID, only makes sense when the block_reason = 0、2、3、4、9、10 It is added from 3.2.8.0.	是

Field Name	Type	Description	Whether Default
slowed	boolean	Whether the robot slows down.	YES
slow_reason	number	Reasons for slowing down, 0 = Ultrasonic, 1 = Laser, 2 = Fallingdown, 3 = Collision, 4 = Infrared, 5 = Lock, 6 = dynamic obstacle, 7 = virtual laser, 8 = 3D camera	YES
slow_x	number	The x coordinate of nearest obstacle that caused the slow down, unit: m	YES
slow_y	number	The y coordinate of nearest obstacle that caused the slow down, unit: m	YES
slow_ultrasonic_id	number	Ultrasonic id number which caused slow down, only makes sense when slow_reason = 0 It is abandoned by 3.2.8.0, and changes to block_id.	YES
slow_di	number	The id number of the DI which caused slow down, only makes sense when slow_reason = 2, 3, 4 It is abandoned by 3.2.8.0, and changes to block_id.	YES
slow_id	number	The slowed ID, only makes sense when slow_reason = 0、2、3、4、9、10. It is added from 3.2.8.0.	是

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{  
  "block_x": 15.2845,  
  "block_y": -8.1834,  
  "blocked": false,  
  "ret_code": 0,  
  "slowed": false  
}
```

15.1.1.3.7 Robot Battery Status Inquiry

Request

- API number: 1007 (0x03EF)
- Name: robot_status_battery_req
- Description: Robot battery status inquiry
- JSON data area: See table below.

Field name	Type	Description	Whether Default
simple	boolean	Whether to response only simple data, true = yes, false = no, and default is no.	YES

Request Example

```
5A 01 00 01 00 00 00 00 03 EF 00 00 00 00 00 00
// Response only simple data.
// {"simple":true}
5A 01 00 01 00 00 00 0F 03 EF 00 00 00 00 00 00
7B 22 73 69 6D 70 6C 65 22 3A 74 72 75 65 7D
```

Response

- API number: 11007 (0x2AFF)
- Name: robot_status_battery_res
- Description: Response of robot battery status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
battery_level	number	Robot battery level, range [0, 1]	YES
battery_temp	number	Robot battery temperature, unit: °C	YES
charging	boolean	Whether the battery is charging.	YES
voltage	number	Voltage, unit: V	YES
current	number	Current, Unit: A	YES
max_charge_voltage	number	The maximum voltage allowed to charge (-1 = the battery does not support this function), unit: V	YES
max_charge_current	number	Maximum current allowed to charge (-1 = the battery does not support this function), unit: A	YES
manual_charge	boolean	It only indicates whether the robot is connected to a manual charger, cannot ensure whether the charger power is turned on, and robot cannot move when the manual charger is connected (only supported by SRC -2000)	YES
auto_charge	boolean	It only indicates whether the robot is connected to the automatic charging pile, and cannot ensure whether the charger power is turned on (only supported by SRC -2000)	YES

Field Name	Type	Description	Whether Default
battery_cycle	number	Number of battery cycles(obtained from battery BMS, not guaranteed to be correct)	YES
battery_user_data	string	Battery user-defined data	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "auto_charge": false,
  "battery_cycle": 9,
  "battery_level": 0.87,
  "battery_temp": 35,
  "battery_user_data": "",
  "charging": false,
  "current": 2,
  "manual_charge": false,
  "max_charge_current": 5,
  "max_charge_voltage": 48,
```

```
"ret_code": 0,  
"voltage": 24.5  
}
```

If only simple data is responded (the data area only has battery level), the result is as follows:

```
{"battery_level":0.8}
```

15.1.1.3.8 Robot Motor Status Inquiry

Request

- API number: 1040 (0x0410)
- Name: robot_status_motor_req
- Description: Robot motor status inquiry
- JSON data area: See table below.

Field name	Type	Description	Whether Default
motor_names	array[string]	Get the specified motor status information according to the motor name If this field is defaulted (empty), it means to query all motor status information	YES

Request Example

Field Name	Type	Description	Whether Default
motor_name	string	Motor name	NO
type	number	Motor type 0=WALK 1=SPEED 2=SPIN 3=LINEAR 4=ROTATION 5=DO	YES
can_router	number	canPort corresponding to the model file	YES
can_id	number	canId corresponding to the model file	YES
position	number	position, unit: (linear is m; steer, spin, rotation are rad)	YES
speed	number	speed, unit: (linear is m/s; steer, spin, rotation are rad/s)	YES
current	number	Current, A	YES
voltage	number	Voltage, V	YES
stop	boolean	Stop or not	YES

Field Name	Type	Description	Whether Default
error_code	number	error code	YES
err	number	error or not	YES
emc	boolean	Emergency stop or not	YES
temperature	number	temperature	YES
encoder	number	encoder	YES
passive	boolean	Whether it is a passive motor (refers to a motor that is not executed, only feedback)	YES
calib	boolean	Whether the calibration is completed	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

1. Get the status information of the motor named "motor1"

```
5A 01 00 01 00 00 00 1B 04 10 00 00 00 00 00
// {"motor_names": ["motor1"]}
5A 01 00 01 00 00 00 1B 04 10 00 00 00 00 00 7B 22 6D 6F 74 6F 72 5F 6E 61 6D 65 73 22 3A 20 5B 22 6D 6F 74 6F 72 31
22 5D 7D
```

1. Get all motor status information

```
5A 01 00 01 00 00 00 00 04 10 00 00 00 00 00 00
```

Response

- API number: 11040 (0x2B20)
- Name: robot_status_motor_res
- Description: Response of robot motor status inquiry.
- JSON data area: See table below.

Response Example

1. Response to "motor1" inquiry

```
{
  "motor_info": [
    {
      "can_id": 2,
      "can_router": 2,
      "emc": false,
      "err": false,
      "error_code": 0,
      "motor_name": "motor1",

```

```
    "position": 0,  
    "speed": 0,  
    "stop": false  
  }  
],  
"ret_code": 0  
}
```

1. All motor status data

```
2. {  
3. "motor_info": [  
4. {  
5.   "can_id": 1,  
6.   "can_router": 1,  
7.   "emc": false,  
8.   "err": false,  
9.   "error_code": 0,  
10.  "motor_name": "motor",  
11.  "position": 0,  
12.  "speed": 0,  
13.  "stop": false  
14. },  
15. {  
16.   "can_id": 2,  
17.   "can_router": 2,  
18.   "emc": false,  
19.   "err": false,  
20.   "error_code": 0,  
21.   "motor_name": "motor1",  
22.   "position": 0,  
23.   "speed": 0,  
24.   "stop": false  
25. }  
26. ],  
27. "ret_code": 0  
}
```

15.1.1.3.9 Robot Laser Status Inquiry

Request

- API number: 1009 (0x03F1)
- Name: robot_status_laser_req
- Description: Robot laser status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
return_beams3D	boolean	Whether to return multi-line laser data, true = return, false = no return, and default is false.	YES

Request Example

```
5A 01 00 01 00 00 00 00 03 F1 00 00 00 00 00 00
```

Response

- API number: 11009 (0x2B01)
- Name: robot_status_laser_res
- Description: Response of robot laser status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
lasers	array	Laser data, see below for data examples	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Data Example

The following is the data of three laser points:

```
{
  "lasers": [
    {
      "beams": [
        {
          "angle": -90,
          "dist": 11.722,
          "is_obstacle": false,
          "is_virtual": false,
          "rssi": 220,
          "valid": true
        },
        {
          "angle": -89.5,
          "dist": 12.3252,
          "is_obstacle": false,
```

```
    "is_virtual": false,
    "rssi": 220,
    "valid": true
  },
  {
    "angle": -89,
    "dist": 11.7744,
    "is_obstacle": false,
    "is_virtual": false,
    "rssi": 220,
    "valid": true
  }
],
"device_info": {
  "device_name": "laser_sim",
  "max_angle": 90,
  "max_range": 20,
  "min_angle": -90,
  "min_range": 0,
  "pub_step": 0.5,
  "real_step": 0.5,
  "scan_freq": 50,
  "time_increment": 0
},
"header": {
  "data_nsec": "1523965240342673100",
  "frame_id": "/scan_0",
  "pub_nsec": "1523965240342673100",
  "seq": "0"
},
"install_info": {
  "upside": true,
  "x": 0,
  "y": 0,
  "yaw": 0,
  "z": 0.34
}
}
```

```
]
}
```

For the convenience of reading, the JSON data here is in an expanded form. In actual use, to reduce the data flow, the JSON data area is compact, and there will be no extra spaces and line breaks. The same applies to the following text.

Response example

```
{
  "lasers": [
    {
      "beams": [
        {
          "angle": -90,
          "dist": 2.762,
          "rssi": 44.5273,
          "valid": true
        },
        {
          "angle": -89.6143,
          "dist": 2.75,
          "rssi": 44.9334,
          "valid": true
        },
        {
          "angle": -89.1,
          "dist": 2.755,
          "rssi": 44.8016,
          "valid": true
        },
        {
          "angle": -88.5857,
          "dist": 2.747,
          "rssi": 44.7736,
          "valid": true
        },
        {
          "angle": -88.0714,
```



```

        "dist": 2.758,
        "rssi": 44.6627,
        "valid": true
    }
],
"device_info": {
    "device_name": "laser",
    "max_angle": 90,
    "max_range": 30,
    "min_angle": -90,
    "pub_step": 0.5,
    "real_step": 0.5,
    "scan_freq": 30
},
"header": {
    "data_nsec": "888484363086750",
    "frame_id": "Pepper1R2000-HD-TCP",
    "pub_nsec": "888484363087000"
},
"install_info": {
    "upside": true,
    "x": 0.3386,
    "z": 0.229
},
"use_forLoc": true
}
],
"ret_code": 0
}

```

15.1.1.3.10 Robot Path Status Inquiry

Request

- API number: 1010 (0x03F2)
- Name: robot_status_path_req
- Description: Robot path status inquiry
- JSON data area: None

Note: The path data of the robot is meaningful only during free navigation. The robot moves along the drawn Bezier curve during path navigation. The path data queried through this API is meaningless when the robot moves during path navigation. If you need to query the path data during path navigation, please refer to [1020](#) Navigation status.

Request Example

```
5A 01 00 01 00 00 00 00 03 F2 00 00 00 00 00 00
```

Response

- API number: 11010 (0x2B02)
- Name: robot_status_path_res
- Description: Response of robot path status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
path	array[array[number]], Array includes array, which consists of two numbers -- the coordinate of path represent by X and Y in world coordinate.	Sequential path point data, and the sequential connection is the planned robot path. For data examples, see below.	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

The following is the data of five path points:

```
{
  "path": [[1.0, 1.2], [2.0, 1.3], [3.2, 2.1], [3.4, 1.4], [3.9, 1.5]]
}
```

The above path consists of 5 points, and the 5 points are (1.0, 1.2), (2.0, 1.3), (3.2, 2.1), (3.4, 1.4), (3.9, 1.5), which represent the five coordinates in the world coordinate system. Connecting these 5 points in sequence is the current planned path of the robot.

15.1.1.3.11 Robot Area Status Inquiry

Request

- API number: 1011 (0x03F3)
- Name: robot_status_area_req
- Description: Robot area status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 F3 00 00 00 00 00 00
```

Response

- API number: 11011 (0x2B03)
- Name: robot_status_area_res
- Description: Response of robot area status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
area_ids	array[string]	The array of the IDs of area where the robot is located (Since the areas on the map can overlap, the robot may be in multiple areas at the same time.), and the array may be empty.	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

The following shows that the robot is currently in the area with id numbers "1" and "2"

```
{
  "area_ids": [
    "1",
    "2"
  ]
}
```

15.1.1.3.12 Robot Estop Status Inquiry

Request

- API number: 1012 (0x03F4)
- Name: robot_status_emergency_req
- Description: Robot Estop status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 F4 00 00 00 00 00 00
```

Response

- API number: 11012 (0x2B04)
- Name: robot_status_emergency_res
- Description: Response of robot Estop status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
emergency	boolean	True means the emergency stop button is activated (pressed), and false means the emergency stop button is inactive (unpressed).	YES
driver_emc	boolean	True means that the motor driver has an emergency stop, and false means that the motor driver has not emergency stop.	YES
electric	boolean	True indicates that the relay is on. False indicates that the relay is off.	YES
soft_emc	boolean	True indicates the motor driver has a soft emergency stop. False indicates the motor driver does not have a soft emergency stop.	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "driver_emc": false,
```

```
"electric": false,  
"emergency": false,  
"ret_code": 0,  
"soft_emc": false  
}
```

15.1.1.3.13 Robot I/O Status Inquiry

Request

- API number: 1013 (0x03F5)
- Name: robot_status_io_req
- Description: Robot I/O status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 F5 00 00 00 00 00 00
```

Response

- API number: 11013 (0x2B05)
- Name: robot_status_io_res
- Description: Response of robot I/O status inquiry
- JSON data area: See table below.

Field name	Type	Description	Whether Default
DI	array[Object]	id: corresponding id number source: normal = normal DI virtual = virtual DI modbus = modbus DI status: indicates high or low level true = high level false = low level valid: Whether the corresponding DI is enabled true = enabled false = disabled	YES
DO	array[Object]	id: corresponding id number source: normal = normal DO modbus = modbus DO status: indicates high or low level true = high level false = low level	YES

Field name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "DI": [
    {
      "id": 0,
      "source": "normal",
      "status": true,
      "valid": true
    },
    {
      "id": 1,
      "source": "normal",
      "status": true,
      "valid": true
    },
    {
      "id": 2,
      "source": "normal",
      "status": true,
      "valid": false
    },
    {
      "id": 3,
```

```
    "source": "normal",
    "status": true,
    "valid": true
  },
  {
    "id": 4,
    "source": "normal",
    "status": false,
    "valid": true
  },
  {
    "id": 5,
    "source": "normal",
    "status": true,
    "valid": true
  },
  {
    "id": 6,
    "source": "normal",
    "status": true,
    "valid": true
  },
  {
    "id": 7,
    "source": "normal",
    "status": true,
    "valid": true
  },
  {
    "id": 8,
    "source": "normal",
    "status": true,
    "valid": true
  },
  {
    "id": 9,
    "source": "virtual",
    "status": true,
```

```
    "valid": true
  }
],
"DO": [
  {
    "id": 0,
    "source": "normal",
    "status": true
  },
  {
    "id": 1,
    "source": "normal",
    "status": true
  },
  {
    "id": 2,
    "source": "normal",
    "status": true
  },
  {
    "id": 3,
    "source": "normal",
    "status": true
  },
  {
    "id": 4,
    "source": "normal",
    "status": true
  },
  {
    "id": 5,
    "source": "normal",
    "status": true
  },
  {
    "id": 6,
    "source": "normal",
    "status": true
  }
]
```

```
}  
],  
"ret_code": 0  
}
```

15.1.1.3.14 Robot IMU Data Inquiry

Request

- API number: 1014 (0x03F6)
- Name: robot_status_imu_req
- Description: Robot IMU data inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 F6 00 00 00 00 00 00
```

Response

- API number: 11014 (0x2B06)
- Name: robot_status_imu_res
- Description: Response of robot IMU data inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
yaw	number	yaw, unit: rad	YES
roll	number	roll, unit: rad	YES
pitch	number	pitch, unit: rad	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "pitch": 0,
  "ret_code": 0,
  "roll": 0,
  "yaw": 0
}
```

15.1.1.3.15 Robot RFID Data Inquiry

Request

- API number: 1015 (0x03F7)
- Name: robot_status_rfid_req
- Description: Robot RFID data inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 F7 00 00 00 00 00 00
```

Response

- API number: 11015 (0x2B07)
- Name: robot_status_rfid_res
- Description: Response of robot RFID data inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
rfids	array[int]	All RFID tag IDs scanned, if no RFID tag is scanned, it is an empty array.	YES
ret_code	number	API error code	YES

Field Name	Type	Description	Whether Default
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "ret_code": 0,
  "rfids": []
}
```

15.1.1.3.16 Robot Ultrasonic Status Inquiry

Request

- API number: 1016 (0x03F8)
- Name: robot_status_ultrasonic_req
- Description: Robot ultrasonic status inquiry
- JSON data area: None

Request Example

5A 01 00 01 00 00 00 00 03 F8 00 00 00 00 00 00

Response

- API number: 11016 (0x2B08)
- Name: robot_status_ultrasonic_res
- Description: Response of robot ultrasonic status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ultrasonic_nodes	array[object]	Ultrasonic data, see below for data examples	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

object form is as follows:

```
{
  "id": 0,      // ultrasonic node ID
  "dist": 0.2,  // the distance that the ultrasonic detected(Unit:m)
  "valid": true // valid or not
}
```

After obtaining `id` and `dist`, it needs to be converted into values in the world coordinate system according to the corresponding ultrasound configuration in the model file.

Response Example

The following is the data of five ultrasound nodes:

```
{
  "ultrasonic_nodes": [
    {
      "id": 0,
      "dist": 0.1,
      "valid": true
    },
    {
      "id": 1,
      "dist": 0.12,
      "valid": true
    },
    {
      "id": 2,
      "dist": 0.14,
      "valid": false
    },
    {
      "id": 3,
      "dist": 0.16,
      "valid": true
    },
    {
      "id": 4,
      "dist": 0.03,
      "valid": false
    }
  ]
}
```

For the ease of reading, the JSON data here is in an expanded form. In practice, the JSON data area is compact in order to reduce the data flow, and there will be no extra spaces and line breaks. The same applies to the following text.

15.1.1.3.17 Robot PGV Data Inquiry

Note: PGV means Position Guided Vision.

Request

- API number: 1017 (0x03F9)
- Name: robot_status_pgv_req
- Description: Robot PGV data inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 F9 00 00 00 00 00 00
```

Response

- API number: 11017 (0x2B09)
- Name: robot_status_pgv_res

- Description: Response of robot PGV data inquiry
- JSON data area: See table below

Field Name	Type	Description	Whether Default
pgvs	array[object]	QR code identification data, see below for data examples	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

The format of object is as follows: Mainly focusing on device_address (id of PGV device) and tag_value (data of QR code tag)

```
{
  // id of PGV device
  "device_address": 1,
  "error_code": 0,
  "header": {
    "data_nsec": "1750000088331",
    "frame_id": "",
    "pub_nsec": "0",
    "seq": "0"
  },
  "is_DMT_detected": true,
  "is_absolute_X_pos_valid": true,
  "pgv_info": {
    "func": 1,

```

```

        "r": 0,
        "x": 0,
        "y": 0,
        "z": 0
    },
    "tag_diff_angle": 1744,
    "tag_diff_x": 256,
    "tag_diff_y": 47,
    // data of QR code tag
    "tag_value": 1,
    "warning_code": 0
}

```

Response Example

The data for the two PGV devices is as follows:

```

{
  "pgvs": [
    {
      "device_address": 0,
      "error_code": 0,
      "header": {
        "data_nsec": "1749997924588",
        "frame_id": "",
        "pub_nsec": "0",
        "seq": "0"
      },
      "is_DMT_detected": false,
      "is_absolute_X_pos_valid": false,
      "pgv_info": {
        "func": 1,
        "r": 0,
        "x": 0,
        "y": 0,
        "z": 0
      },
      "tag_diff_angle": 0,
    }
  ]
}

```

```

    "tag_diff_x": 0,
    "tag_diff_y": 0,
    "tag_value": 0,
    "warning_code": 0
  },
  {
    "device_address": 1,
    "error_code": 0,
    "header": {
      "data_nsec": "1749997924588",
      "frame_id": "",
      "pub_nsec": "0",
      "seq": "0"
    },
    "is_DMT_detected": true,
    "is_absolute_X_pos_valid": true,
    "pgv_info": {
      "func": 2,
      "r": 0,
      "x": 0,
      "y": 0,
      "z": 0
    },
    "tag_diff_angle": 1790,
    "tag_diff_x": 54,
    "tag_diff_y": 3,
    "tag_value": 1,
    "warning_code": 0
  }
]
}

```

For ease of reading, the JSON data is expanded here. In practice, the JSON data area is compact in order to reduce the data flow, and there will be no extra spaces or line break

15.1.1.3.18 Robot Encoder Status Inquiry

Request

- API number: 1018 (0x03FA)
- Name: robot_status_encoder_req
- Description: Robot encoder status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 FA 00 00 00 00 00 00
```

Response

- API number: 11018 (0x2B0A)
- Name: robot_status_encoder_res
- Description: Response of robot encoder status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
encoder	array[int]	Encoder pulse value Abandoned; It has changed to <code>motor_encoder</code> .	YES
motor_encoder	array[object]	The queried motor encoder, including the motor name and encoder pulse value	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "encoder": [],
  "motor_encoder": [],
  "ret_code": 0
}
```

15.1.1.3.19 Robot Navigation Status Inquiry

Request

- API number: 1020 (0x03FC)
- Name: robot_status_task_req
- Description: Robot navigation status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
simple	boolean	Whether to return only simple data, true = yes, false = no, and the default is no.	YES

Request Example

```
5A 01 00 01 00 00 00 00 03 FC 00 00 00 00 00 00
// Only return simple data
// {"simple":true}
5A 01 00 01 00 00 00 0F 03 FC 00 00 00 00 00 00
7B 22 73 69 6D 70 6C 65 22 3A 74 72 75 65 7D
```

Response

- API number: 11020 (0x2B0C)
- Name: robot_status_task_req
- Description: Response of robot navigation status inquiry
- JSON data area: See table below.

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

If only simple data is returned, only task_status in the fields in the table below.

Field Name	Type	Description	Whether Default
task_status	number	0 = NONE, 1 = WAITING(This state is currently impossible), 2 = RUNNING, 3 = SUSPENDED, 4 = COMPLETED, 5 = FAILED, and 6 = CANCELED.	YES
task_type	number	navigation type, 0 = no navigation, 1 = free navigation to a coordinate, 2 = free navigation to a station, 3 = path navigation to a station, 7 = translational/rotation, and 100 = other.	YES
target_id	string	The current navigation target, this field is valid only when task_type is 2 or 3, when task_status is RUNNING, it means that robot are going to this station, when task_status is COMPLETED, it means that robot have reached this station, when task_status is FAILED, it means that robot have failed to go to this station, if task_status is SUSPENDED, then navigation to this station is paused.	YES

Field Name	Type	Description	Whether Default
target_point	array[number]	The current navigation coordinate. It is an array including 3 numbers, coordinate of X, Y,and R in world coordinate respectively. This field is valid only when task_type is 1, and when task_status is RUNNING, robot are going to this station; when task_status is COMPLETED, robot have reached this station; when task_status is FAILED, robot have failed to go to this station; when task_status is SUSPENDED, navigation to this station is paused.	YES
finished_path	array[string]	The stations that have passed on the current navigation path, it is an array of stations. This field is valid only when task_type is 3 or 4. It will list all the intermediate stations that have passed.	YES
unfinished_path	array[string]	The stations that have not passed on the current navigation path,it is an array of stations. This field is valid only when task_type is 3 or 4. It will list all the intermediate stations that have not been passed.	YES
move_status_info	string	Additional information about navigation tasks	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES

Field Name	Type	Description	Whether Default
err_msg	string	error message	YES

Response Example

The following data indicates that the robot is using path navigation to go to the station with the id "LM6", the stations that have passed are LM1, LM2, LM3, and the stations that have not passed are LM4, LM5, LM6

```
{
  "task_status": 2,
  "task_type": 3,
  "target_id": "LM1",
  "finished_path": [
    "LM1",
    "LM2",
    "LM3"
  ],
  "unfinished_path": [
    "LM4",
    "LM5",
    "LM6"
  ]
}
```

If only simple data is returned (there are only 17 characters in the data area):

```
{"task_status":4}
```

15.1.1.3.20 Robot Task Status Package Inquiry

Request

- API number: 1110 (0x456)
- Name: robot_status_task_status_package_req
- Description: Robot task status package inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
task_ids	array[string]	Specify the task_id to be queried in the array. If the array is empty, the response is also empty; If this field is default, it will return the status of a task that the robot has recently completed and all uncompleted tasks.	YES

Request Example

```
// Query the last completed task and all uncompleted tasks
5A 01 00 01 00 00 00 00 04 56 00 00 00 00 00 00
// Query task status with task_id SEER78909
// {"task_ids": ["SEER78909"]}
5A 01 00 01 00 00 00 1B 04 56 00 00 00 00 00 7B 22 74 61 73 6B 5F 69 64 73 22 3A 20 5B 22 53 45 45 52 37 38 39 30 39
22 5D 7D
```

Response

- API number: 11110 (0x2B66)
- Name: robot_status_task_status_package_res
- Description: Response of robot task status package inquiry
- JSON data area: See table below

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly without paying attention to its meaning. I won't repeat them later.

Field Name	Type	Description	Whether Default
closest_target	string	The closest station to the robot within a certain distance. It may null, which indicates there is no station around the robot.	YES
source_name	string	The last station the robot visited	YES
target_name	string	Target station the robot is currently going to	YES
percentage	number	The percentage of the part that the robot has walked to the whole path during corresponding	YES

Field Name	Type	Description	Whether Default
		path navigation task	
distance	number	The shortest distance from the current path	YES
task_status_list	array[object]	Task status list (field description show table below)	YES
info	string	Some prompt messages reported by the robot to users in the navigation process. It can be output to the front desk. This field does not invovles in the actual logical judgment.	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

The related parameter description in task_status_list

Field Name	Type	Description	Whether Default
task_id	string	Task id	YES
status	number	Task status StatusNone = 0 Waiting = 1 Running = 2 Suspended = 3 Completed = 4 Failed = 5 Canceled = 6 OverTime = 7 NotFound = 404	YES

Response Example

The following data indicates that the robot is using `path navigation` method to go to AP62 -> AP5 -> AP67 in turn, and return to the list of the last completed tasks and all unfinished tasks.

```
{
  "ret_code": 0,
  "task_status_package": {
    "closest_target": "AP67",
    "distance": 0.00287,
    "info": "",
    "percentage": 0.93491,
  }
}
```



```
"source_name": "AP5",
"target_name": "AP67",
"task_status_list": [
  {
    "status": 4,
    "task_id": "SEER78914",
    "type": 3
  },
  {
    "status": 2,
    "task_id": "SEER78915",
    "type": 3
  }
]
}
```

Only return the task status package with task_id "SEER78914".

```
{
  "ret_code": 0,
  "task_status_package": {
    "closest_target": "AP67",
    "distance": -1,
    "info": "",
    "percentage": 0,
    "source_name": "",
    "target_name": "",
    "task_status_list": [
      {
        "status": 4,
        "task_id": "SEER78914",
        "type": 3
      }
    ]
  }
}
```

15.1.1.3.21 Robot Localization Status Inquiry

Request

- API number: 1021 (0x03FD)
- Name: robot_status_reloc_req
- Description: Robot localization status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 FD 00 00 00 00 00 00
```

Response

- API number: 11021 (0x2B0D)
- Name: robot_status_reloc_res
- Description: Response of robot localization status inquiry
- JSON data area: See table below.

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

Field Name	Type	Description	Whether Default
reloc_status	number	0 = FAILED(Localization failed), 1 = SUCCESS(Localization is correct), 2 = RELOCING(Relocating), 3=COMPLETED(Localization completed)	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Note: The localization status (1021) is used to indicate whether the current robot localization status is correct. If the localization status is COMPLETED, the relocation has been completed, but the user has not confirmed. In this case, you need to confirm (2003) or relocate. After the user confirms that the localization is correct, the localization status will change to SUCCESS.

Response Example

```
{
  "reloc_status": 1
}
```

15.1.1.3.22 Robot Map Loading Status Inquiry

Request

- API number: 1022 (0x03FE)
- Name: robot_status_loadmap_req
- Description: Robot map loading status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 03 FE 00 00 00 00 00 00
```

Response

- API number: 11022 (0x2B0E)
- Name: robot_status_loadmap_res
- Description: Response of robot map loading status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
loadmap_status	number	0 = FAILED(Failed to load map), 1 = SUCCESS(Map loading succeeded), 2 = LOADING>Loading map)	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Note: The map loading status (1022) is used to indicate whether the current map is loaded successfully. When the robot has just booted or the map switching operation occurs, the map loading status will be indicated as LOADING, and the relocation operation cannot be performed at this time. You must wait for the map loading status to change to SUCCESS before relocating.

Response example

```
{
  "loadmap_status": 1
}
```

15.1.1.3.23 Robot SLAM Status Inquiry

Request

- API number: 1025 (0x0401)
- Name: robot_status_slam_req
- Description: Robot SLAM status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 04 01 00 00 00 00 00 00
```

Response

- API number: 11025 (0x2B11)
- Name: robot_status_slam_res
- Description: Response of robot SLAM status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
slam_status	number	0 = None, 1 = Scanning(Offline), 2 = Real-time scanning(SLAM)	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "slam_status": 0
}
```

15.1.1.3.24 Robot Jacking Status Inquiry

Only makes sense for robots with jacking.

Request

- API number: 1027 (0x0403)
- Name: robot_status_jack_req
- Description: Robot jacking status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 04 03 00 00 00 00 00 00
```

Response

- API number: 11027 (0x2B13)
- Name: robot_status_jack_res
- Description: Response of robot jacking status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
jack_mode	boolean	Operating mode, false = manual, true = automatic	YES
jack_state	number	Running status, 0x00 = loading, 0x01 = loading in place, 0x02 = unloading, 0x03 = unloading in place, 0x04 = stop, 0xFF = execution failed	YES
jack_enable	boolean	Whether the jack is enabled, false = disabled, true = enabled	YES
jack_error_code	number	Jacking error code	YES
jack_isFull	boolean	Whether the jacking has goods, false = no goods, true = has goods	YES
jack_speed	number	Motor speed of jacking, unit: mm/s	YES
jack_emc	boolean	Whether the jacking in Estop, false = not Estop, true = Estop	YES
jack_height	number	Jacking real-time height, unit: m	YES
peripheral_data	array[number]	User-defined data in a peripheral.	YES

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "jack_emc": false,
  "jack_enable": false,
  "jack_error_code": 0,
  "jack_height": 0,
  "jack_isFull": false,
  "jack_mode": false,
  "jack_speed": 0,
  "jack_state": 0,
  "peripheral_data": [],
  "ret_code": 0
}
```

15.1.1.3.25 Robot fork status inquiry

Only makes sense for robots with forks, such as forklifts, etc.

Request

- API number: 1028 (0x0404)
- Name: robot_status_fork_req
- Description: Robot fork status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 04 04 00 00 00 00 00 00
```

Response

- API number: 11028 (0x2B14)
- Name: robot_status_fork_res
- Description: Response of robot fork status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
fork_height	number	Fork height, unit: m	YES
fork_height_in_place	boolean	Whether the height of the fork is in place, true = in place, false = not in place	YES
fork_auto_flag	boolean	Forklift's control mode (switched via physical buttons on the forklift), ture = automatic control (controller driving), false = manual control (manual driving)	YES
forward_val	number	Forward and backward distance of fork, unit: m	YES
forward_in_place	boolean	Whether the fork moves forward or backward in place, true = in place, false = not in place	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
{
  "fork_auto_flag": true,
```

```
"fork_height": 0,  
"fork_height_in_place": false,  
"forward_in_place": false,  
"forward_val": 0,  
"ret_code": 0  
}
```

15.1.1.3.26 Robot Roller Status Inquiry

Only makes sense for robots equipped with rollers or belts.

Request

- API number: 1029 (0x0405)
- Name: robot_status_roller_req
- Description: Robot roller status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 04 05 00 00 00 00 00 00
```

Response

- API number: 11029 (0x2B15)
- Name: robot_status_roller_res
- Description: Response of robot roller status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
roller_mode	boolean	Operating mode, false = manual, true = automatic	YES
roller_state	number	Running status, 0x01 = standby, 0x02 = executing, 0x03 = executing completed, 0xFF = executing failed	YES
roller_enable	boolean	Whether the roller is enabled, false = disabled, true = enabled	YES
roller_error_code	number	Roller error code	YES
roller_isFull	boolean	Whether the roller has goods, false = no goods, true = has goods	YES
roller_speed	number	Motor speed of roller	YES
roller_emc	boolean	Whether the roller in Estop, false = not Estop, true = Estop	YES
peripheral_data	array[number]	User-defined data in a peripheral	YES
ret_code	number	API error code	YES

Field Name	Type	Description	Whether Default
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

roller_error_code Definition:

```
0x00 (00) ,      // No error
0x11 (17) ,      // Motor error
0x01 (01) ,      // Roller has goods (Load)
0x02 (02) ,      // Roller is empty (Unload)
0x03 (03) ,      // Load timeout
0x04 (04) ,      // Unload timeout
0x05 (05) ,      // The goods are too long
```

Response Example

```
{
  "peripheral_data": [],
  "ret_code": 0,
  "roller_emc": false,
  "roller_enable": false,
  "roller_error_code": 0,
  "roller_isFull": false,
  "roller_mode": false,
  "roller_speed": 0,
  "roller_state": 0
}
```

15.1.1.3.32 Robot Dispatching Status Inquiry

Request

- API number: 1030 (0x0406)
- Name: robot_status_dispatch_req
- Description: Robot dispatching status inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 04 06 00 00 00 00 00
```

Response

- API number: 11030 (0x2B16)
- Name: robot_status_dispatch_res
- Description: Response of robot dispatching status inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
dispatch_mode	number	0 = stand-alone mode, 1 = dispatching mode, 2 = dispatching mode but lost connection with dispatch system	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response example

```
{
  "dispatch_mode": 0
}
```

15.1.1.3.28 Robot Alarm Status Inquiry

Request

- API number: 1050 (0x041A)
- Name: robot_status_alarm_req

- Description: Query the current alarm status of the robot.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 04 1A 00 00 00 00 00 00
```

Response

- API number: 11050 (0x2B2A)
- Name: robot_status_alarm_res
- Description: Query the response of the current alarm state of the robot.
- JSON data area: See the table below.

Field Name	Type	Description	Whether Default
fatals	Array[Object]	Fatal alarm array, all Fatal alarms will appear in the data, see the object format below.	Yes
errors	Array[Object]	Error alarm array, all Error alarms will appear in the data, see the object format below.	Yes
warnings	Array[Object]	Warning alarm array, all Warning alarm will appear in the data, the object format is described below.	Yes
notices	Array[Object]	Notice alarm array, all Warning alarm will appear in the data, the object format is described below.	Yes
ret_code	Number	API error code	Yes
create_on	String	API upload timestamp	Yes
err_msg	String	Error message	Yes

The object format is as follows:

```
{"50000":1497698400,"desc":"xxxxx","times":1}
```

// Key is the alarm code. The corresponding value is the number of seconds from Thu Jan 01 08:00:00 1970 to the time when the alarm code is generated, desc is the English description corresponding to the error code, and times is the number of times.

Response Example

```
{
  "fatals": [
    {
      "50000": 1497698400,
      "desc": "xxxxx",
      "times": 1
    }
  ],
  "errors": [
    {
      "52201": 1497698402,
      "desc": "xxxxx",
      "times": 1
    },
    {
      "52118": 1497698404,
      "desc": "xxxxx",
      "times": 1
    }
  ],
  "warnings": [
    {
      "54003": 1497698405,
      "desc": "xxxxx",
      "times": 1
    }
  ],
  "notices": [
    {
      "56000": 1497698505,
```

```
    "desc": "xxxxx",  
    "times": 20  
  }  
]  
}
```

15.1.1.3.29 Batch Data Inquiry 1

Request

- API number: 1100 (0x044C)
- Name: robot_status_all1_req
- Description: Batch data inquiry 1
- JSON data area: See table below.

All1 is to get most of the previously mentioned status data in one request,

including 1000, 1001, 1002, 1003, 1004, 1005, 1006, 1007, 1008, 1009, 1010, 1011, 1012, 1013, 1014, 1015, 1016, 1017, 1019, 1020, 1021, 1022, 1025, 1027, 1028, 1029, 1030, 1050, etc

Field Name	Type	Description	Whether Default
keys	array[string]	Only return the specified key-value in the array, and the default is to return all data.	YES
return_laser	boolean	Whether to return the laser data (the data of 1009), true = return, false = not return, and the default is return.	YES
return_beams3D	boolean	Whether to return the multi-beam 3D laser data (the data of 1009), true = return, false = not return, and the default is false.	NO

Request Example

```
// Return all data.
5A 01 00 01 00 00 00 00 04 4C 00 00 00 00 00 00
// Return only the inquired data.
{"keys":["last_station","current_map"],"return_laser":false,"return_beams3D":false}
5A 01 00 01 00 00 00 53 04 4C 00 00 00 00 00 00
7B 22 6B 65 79 73 22 3A 5B 22 6C 61 73 74 5F 73
74 61 74 69 6F 6E 22 2C 22 63 75 72 72 65 6E 74
5F 6D 61 70 22 5D 2C 22 72 65 74 75 72 6E 5F 6C
61 73 65 72 22 3A 66 61 6C 73 65 2C 22 72 65 74
75 72 6E 5F 62 65 61 6D 73 33 44 22 3A 66 61 6C
73 65 7D
```

Response

- API number: 11100 (0x2B5C)

- Name: robot_status_all1_res
- Description: Response of batch data inquiry 1
- JSON data area: See table below.

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

The JSON data area will contain all key-values

of 11000, 11001, 11002, 11003, 11004, 11005, 11006, 11007, 11008, 11009, 11010, 11011, 11012, 11013, 11014, 11015, 11016, 11017, 11019, 11020, 11021, 11022, 11025, 11027, 11028, 11029, 11030, 11050 and so on.

Field Name	Type	Description	Whether Default
odo	number	Accumulated mileage, unit: m	YES
today_odo	number	Today's cumulative mileage, unit: m	YES
time	number	The current running time (the time from boot to now), unit: ms	YES
total_time	number	Cumulative running time, unit: ms	YES
controller_temp	number	Controller temperature, unit: °C	YES
controller_humi	number	Controller humidity, unit: %	YES
controller_voltage	number	Controller voltage, unit: V	YES
x	number	The x coordinate of the robot, unit: m	YES
y	number	The x coordinate of the robot, unit: m	YES

Field Name	Type	Description	Whether Default
angle	number	The angle coordinate of the robo, unit: rad	YES
confidence	number	Confidence of robot's localization, range [0, 1]	YES
current_station	string	The id of the station where the robot is currently located (this judgment is strict, and the robot must be very close to a certain station(<50cm), otherwise it is a null character, that is, it is not at any station.)	YES
last_station	string	id of the previous station where the robot is located	YES
vx	number	The actual speed of the robot in the x direction of the robot coordinate system, unit: m/s	YES
vy	number	The actual speed of the robot in the y direction of the robot coordinate system, unit: m/s	YES
w	number	The actual angular velocity of the robot in the robot coordinate system (Clockwise is negative while counterclockwise is positive.),	YES

Field Name	Type	Description	Whether Default
		unit: rad/s	
steer	number	Single steering wheel robot's current steering wheel angle, unit: rad	YES
spin	number	Robot tray(if exists) angle, unit: rad	YES
r_vx	number	The speed the robot receives in the x direction of the robot coordinate system, unit: m/s	YES
r_vy	number	The speed the robot receives in the y direction of the robot coordinate system, unit: m/s	YES
r_w	number	The received angular velocity of the robot in the robot coordinate system (Clockwise is negative while counterclockwise is positive.), unit: rad/s	YES
r_steer	number	The steering wheel angle received by the single steering wheel robot, unit: rad	YES

Field Name	Type	Description	Whether Default
r_spin	number	The rotation speed of the tray(if exists) received by robot, unit: rad/s	YES
steer_angles	array[number]	The current steer wheel angle of a multi-steering wheel robot, double-steering wheel robot = [front steering wheel angle, rear steering wheel angle], unit: rad	YES
r_steer_angles	array[number]	The steer wheel angle received by the multi-steer wheel robot, double-teering wheel robot = [front steering wheel angle, rear steering wheel angle], unit: rad	YES
is_stop	boolean	Whether the robot chassis is stationary (walking motor)	YES
blocked	boolean	Whether the robot is blocked	YES
block_reason	number	Reasons for being blocked, 0 = Ultrasonic, 1 = Laser, 2 = Fallingdown, 3 = Collision, 4 = Infrared, 5 = Lock, 6 = dynamic obstacle, 7 = virtual laser, 8 = 3D camera	YES

Field Name	Type	Description	Whether Default
block_x	number	The x coordinate of the nearest obstacle(world coordinate system), unit: m	YES
block_y	number	The y coordinate of the nearest obstacle(world coordinate system), unit: m	YES
block_ultrasonic_id	number	The blocked ultrasonic id number, only meaningful when block_reason = 0	YES
block_di	number	The ID number of the blocked DI, only meaningful when block_reason = 2, 3, 4	YES
slowed	boolean	Whether the robot slow down	YES
slow_reason	number	Reasons for slowing down, 0 = Ultrasonic, 1 = Laser, 2 = Fallingdown, 3 = Collision, 4 = Infrared, 5 = Lock, 6 = dynamic obstacle, 7 = virtual laser, 8 = 3D camera	YES

Field Name	Type	Description	Whether Default
slow_x	number	The x coordinate of nearest obstacle that caused the slow down, unit: m	YES
slow_y	number	The y coordinate of nearest obstacle that caused the slow down, unit: m	YES
slow_ultrasonic_id	number	Ultrasonic id number which caused slow down, only meaningful when slow_reason = 0	YES
slow_di	number	The id number of the DI which caused slow down, only available when slow_reason = 2, 3, 4.	YES
battery_level	number	Robot battery level, range [0, 1]	YES
battery_temp	number	Robot battery temperature, unit: °C	YES
charging	boolean	Whether the battery is charging.	YES

Field Name	Type	Description	Whether Default
voltage	number	Voltage, unit: V	YES
current	number	Current, Unit: A	YES
max_charge_voltage	number	The maximum voltage allowed to charge (-1 = the battery does not support this function), unit: V	YES
max_charge_current	number	Maximum current allowed to charge (-1 = the battery does not support this function), unit: A	YES
manual_charge	boolean	It only indicates whether the robot is connected to a manual charger, and cannot ensure whether the charger power is turned on, and robot cannot move when the manual charger is connected (only supported by SRC-2000).	YES
auto_charge	boolean	It only indicates whether the robot is connected to the automatic charging pile, and cannot ensure whether the charger power is turned on (only supported by SRC-2000)	YES

Field Name	Type	Description	Whether Default
battery_cycle	number	Number of battery cycles(obtained from battery BMS, not guaranteed to be correct)	YES
lasers	array	Laser point cloud data	YES
path	array[array[number]], Array includes array, which consists of two numbers -- the coodinante of path represent by X and Y in world coordinate	Sequential waypoint data, sequential connection is the robot planned path.	YES
area_ids	array[string]	An array of ids for the regions where the robot is located (Since regions on the map can overlap, the robot may be in more than one region at a time.). The array may be empty.	YES
emergency	boolean	True means the emergency stop button is activated (pressed), and false means the emergency stop button is inactive (unpressed).	YES

Field Name	Type	Description	Whether Default
driver_emc	boolean	True means that the motor driver has an emergency stop, and false means that the motor driver has not emergency stop.	YES
DI	array[Object]	id: corresponding id number source: normal = normal DI virtual = virtual DI modbus = modbus DI status: indicates high or low level true = high level false = low level valid: Whether the corresponding DI is enabled true = enabled false = disabled	YES
DO	array[Object]	id: corresponding id number source: normal = normal DO modbus = modbus DO	YES

Field Name	Type	Description	Whether Default
		status: indicates high or low level true = high level false = low level	
yaw	number	yaw, unit: rad	YES
roll	number	roll, unit: rad	YES
pitch	number	pitch, unit: rad	YES
ultrasonic_nodes	array[object]	Ultrasonic node data	YES
pgvs	array[object]	QR code identification data	YES
task_status	number	0 = NONE, 1 = WAITING(This state is currently impossible), 2 = RUNNING, 3 = SUSPENDED, 4 = COMPLETED, 5 = FAILED, 6 = CANCELED	YES

Field Name	Type	Description	Whether Default
task_type	number	navigation type, 0 = no navigation, 1 = free navigation to a coordinate, 2 = free navigation to a station, 3 = path navigation to a station, 7 = translational/rotation, 100 = other	YES
target_id	string	The current navigation target. This field is valid only when task_type is 2 or 3, and when task_status is RUNNING, robot are going to this station; when task_status is COMPLETED, robot have reached this station; when task_status is FAILED, robot have failed to go to this station; when task_status is SUSPENDED, navigation to this station is paused.	YES
target_point	array[number]	The current navigation coordinate. It is an array including 3 numbers, coordinate of X, Y, and R in world coordinate respectively. This field is valid only when task_type is 1, and when task_status is RUNNING, robot are going to this station; when task_status is COMPLETED, robot have reached this station; when task_status is FAILED, robot have failed to go to this station; when task_status is SUSPENDED, navigation to this station is paused.	YES

Field Name	Type	Description	Whether Default
finished_path	array[string]	The stations that have passed on the current navigation path, it is an array of stations. This field is valid only when task_type is 3 or 4. It will list all the intermediate stations that have passed	YES
unfinished_path	array[string]	The stations that have not passed on the current navigation path,it is an array of stations. This field is valid only when task_type is 3 or 4. It will list all the intermediate stations that have not been passed	YES
reloc_status	number	0 = FAILED(Localization failed), 1 = SUCCESS(Localization is correct), 2 = RELOCING(Relocating), 3=COMPLETED(Localization completed)	YES
loadmap_status	number	0 = FAILED(Failed to load map), 1 = SUCCESS(Map loading succeeded), 2 = LOADING>Loading map)	YES
slam_status	number	0 = None, 1 = Scanning(Offline), 2 = Real-time scanning(SLAM)	YES
tasklist_status	object	The JSON format of Message_TaskStatus in message_task.proto	YES

Field Name	Type	Description	Whether Default
fork_height	number	Fork height, unit: m	YES
fork_height_in_place	boolean	Whether the height of the fork is in place, true = in place, false = not in place	YES
fork_auto_flag	boolean	Forklift's control mode (switched via physical buttons on the forklift), true = automatic control (controller driving), false = manual control (manual driving)	YES
forward_val	number	Forward and backward distance of fork, unit: m	YES
forward_in_place	boolean	Whether the fork moves forward or backward in place, true = in place, false = not in place	YES
roller_mode	boolean	Operating mode, false = manual, true = automatic	YES
roller_state	number	Running status, 0x01 = standby, 0x02 = executing, 0x03 = executing completed, 0xFF = executing failed	YES

Field Name	Type	Description	Whether Default
roller_enable	boolean	Whether the roller is enabled, false = disabled, true = enabled	YES
roller_error_code	number	Roller error code	YES
roller_isFull	boolean	Whether the roller has goods, false = no goods, true = has goods	YES
roller_speed	number	Motor speed of roller	YES
roller_emc	boolean	Whether the roller in Estop, false = not Estop, true = Estop	YES
jack_mode	boolean	Operating mode, false = manual, true = automatic	YES
jack_state	number	Running status, 0x00 = loading, 0x01 = loading in place, 0x02 = unloading, 0x03 = unloading in place, 0x04 = stop, 0xFF = execution failed	YES
jack_enable	boolean	Whether the jacking is enabled, false = disabled, true = enabled	YES

Field Name	Type	Description	Whether Default
jack_error_code	number	Jacking error code	YES
jack_isFull	boolean	Whether the jack has goods, false = no goods, true = has goods	YES
jack_speed	number	Motor speed of jack, unit: mm/s	YES
jack_emc	boolean	Whether the jack in Estop, false = not Estop, true = Estop	YES
jack_height	number	Jacking real-time height, unit: m	YES
jack_load_times	number	Jacking load times	YES
peripheral_data	array[number]	User-Defined data in a peripheral	YES
dispatch_mode	number	0 = stand-alone mode, 1 = dispatching mode, 2 = dispatching mode but lost connection with dispatch system	YES
fatals	array[object]	Array of Fatal alarm , all Fatal alarms will appear in the array	YES

Field Name	Type	Description	Whether Default
errors	array[object]	Array of Error alarm , all Error alarms will appear in the array	YES
warnings	array[object]	Array of Warning alarm , all Warning alarms will appear in the array	YES
notices	array[object]	Array of Notice , all Notice will appear in the array	YES
current_map	string	Current map name	YES
current_map_md5	string	MD5 value of current map	YES
ssid	string	SSID currently connected to Wifi (The robot needs to support Wifi connection and has been connected, otherwise it is empty.)	YES
rsi	number	Signal strength currently connected to Wifi, 0-100 percent (The robot needs to support Wifi connection and has been connected, otherwise it is empty.)	YES
current_ip	string	Current IP address (other than 192.168.192.5)	YES

Field Name	Type	Description	Whether Default
MAC	string	The MAC address of the robot wireless client (valid for SRC -2000 controllers only), and its format is XXXXXXXXXXXX with no separator in the middle.	YES
rfids	array[int]	Ids of all scanned RFID tags. If no RFID tag is detected, the array is empty.	YES
battery_user_data	string	Battery user-defined data	YES
vehicle_id	string	Robot name	YES
robot_note	string	Robot remark	YES
move_status_info	string	Additional information about navigation tasks	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES

Field Name	Type	Description	Whether Default
err_msg	string	Error message	YES

Response Example

```
{
  "19204": 1,
  "19204_ip": [
    "192.168.192.55",
    "192.168.4.236",
    "192.168.4.241",
    "192.168.4.59",
    "192.168.4.84"
  ],
  "DI": [
    {
      "id": 0,
      "source": "normal",
      "status": true,
      "valid": true
    },
    {
      "id": 1,
      "source": "normal",
      "status": true,
      "valid": true
    },
    {
      "id": 2,
      "source": "normal",
      "status": true,
      "valid": true
    }
  ]
}
```

```
},
{
  "id": 3,
  "source": "normal",
  "status": true,
  "valid": true
},
{
  "id": 4,
  "source": "normal",
  "status": true,
  "valid": true
},
{
  "id": 5,
  "source": "normal",
  "status": true,
  "valid": true
},
{
  "id": 6,
  "source": "normal",
  "status": true,
  "valid": true
},
{
  "id": 7,
  "source": "normal",
  "status": true,
  "valid": true
},
{
  "id": 8,
  "source": "normal",
  "status": true,
  "valid": true
},
{

```

```
    "id": 9,  
    "source": "normal",  
    "status": true,  
    "valid": true  
  },  
  {  
    "id": 10,  
    "source": "normal",  
    "status": true,  
    "valid": true  
  }  
],  
"DO": [  
  {  
    "id": 0,  
    "source": "normal",  
    "status": true  
  },  
  {  
    "id": 1,  
    "source": "normal",  
    "status": false  
  },  
  {  
    "id": 2,  
    "source": "normal",  
    "status": false  
  },  
  {  
    "id": 3,  
    "source": "normal",  
    "status": false  
  }  
],  
"MAC": "8A54D1848DCC",  
"angle": 3.0294,  
"block_x": -1.6795,  
"block_y": 0.5164,
```

```
"blocked": false,
"brake": false,
"calib_status": {
  "desc": "No calibration task",
  "status": 0
},
"charging": false,
"confidence": 0.9806,
"correction_errs": [
  -0.00008646898603853969,
  0.0002113406009003569,
  1.0012185057479304
],
"current_ip": "192.168.192.5",
"current_lock": {
  "locked": false
},
"current_map": "20210115160931443",
"current_map_md5": "e77a4eab89ba906668ab4e55f53a6e2c",
"current_station": "CP2",
"di_max_node": 24,
"do_max_node": 16,
"finished_path": [
  "LM1"
],
"fork_auto_flag": true,
"fork_height_in_place": false,
"forward_in_place": false,
"forward_val": 0,
"head": 0.42,
"hook_angle": 0,
"hook_clamping_state": false,
"hook_emc": false,
"hook_enable": false,
"hook_isFull": false,
"hook_mode": false,
"hook_state": 0,
"in_forbidden_area": false,
```

```
"is_stop": false,
"jack_emc": false,
"jack_enable": false,
"jack_error_code": 0,
"jack_height": 0,
"jack_isFull": false,
"jack_load_times": 2,
"jack_mode": false,
"jack_speed": 0,
"jack_state": 0,
"lasers": [
  {
    "beams": [
      {
        "angle": -90,
        "dist": 2.753,
        "rssi": 44.496,
        "valid": true
      },
      {
        "angle": -89.6143,
        "dist": 2.742,
        "rssi": 44.6069,
        "valid": true
      },
      {
        "angle": -89.1,
        "dist": 2.754,
        "rssi": 44.4994,
        "valid": true
      },
      {
        "angle": -88.5857,
        "dist": 2.742,
        "rssi": 44.756,
        "valid": true
      },
      {
```

```
    "angle": -88.0714,  
    "dist": 2.761,  
    "rssi": 44.3744,  
    "valid": true  
  },  
  {  
    "angle": -87.5571,  
    "dist": 2.757,  
    "rssi": 44.6592,  
    "valid": true  
  },  
  {  
    "angle": -87.0429,  
    "dist": 2.748,  
    "rssi": 44.7771,  
    "valid": true  
  },  
  {  
    "angle": -86.5286,  
    "dist": 2.753,  
    "rssi": 44.6453,  
    "valid": true  
  },  
  {  
    "angle": -86.0143,  
    "dist": 2.761,  
    "rssi": 44.8226,  
    "valid": true  
  },  
  {  
    "angle": -85.5,  
    "dist": 2.757,  
    "rssi": 45.1073,  
    "valid": true  
  },  
  {  
    "angle": -85.1143,  
    "dist": 2.761,
```

```
    "rssi": 44.5238,  
    "valid": true  
  },  
  {  
    "angle": 60.5571,  
    "dist": 1.137,  
    "rssi": 46.0842,  
    "valid": true  
  },  
  {  
    "angle": 61.0714,  
    "dist": 1.14,  
    "rssi": 45.9568,  
    "valid": true  
  },  
  {  
    "angle": 61.5857,  
    "dist": 1.165,  
    "rssi": 46.1372,  
    "valid": true  
  },  
  {  
    "angle": 79.0714,  
    "dist": 1.384,  
    "rssi": 45.4692,  
    "valid": true  
  },  
  {  
    "angle": 89.1,  
    "dist": 1.368,  
    "rssi": 103.0789,  
    "valid": true  
  },  
  {  
    "angle": 89.6143,  
    "dist": 1.362,  
    "rssi": 105.997,  
    "valid": true  
  }
```

```

    }
  ],
  "device_info": {
    "device_name": "laser",
    "max_angle": 90,
    "max_range": 30,
    "min_angle": -90,
    "pub_step": 0.5,
    "real_step": 0.5,
    "scan_freq": 30
  },
  "header": {
    "data_nsec": "966597904606000",
    "frame_id": "Pepper1R2000-HD-TCP",
    "pub_nsec": "966597904606500"
  },
  "install_info": {
    "upside": true,
    "x": 0.3386,
    "z": 0.229
  },
  "use_forLoc": true
}
],
"last_station": "LM1",
"loadmap_status": 1,
"loc_state": 0,
"mode": 1,
"model_md5": "3ec73a093211d2b1987d1a085944d0c8",
"odo": 134766.944,
"pitch": 0.11170107379211278,
"point_cloud": null,
"r_spin": 0,
"r_steer": 0,
"r_steer_angles": [
  0
],
"r_vx": 0.02,

```



```
"r_vy": 0,
"r_w": 0.0008,
"radius": 0,
"ref_pos": [
  {
    "type": "cylinder",
    "width": 0.08,
    "x": -1.5212,
    "y": 0.1804
  }
],
"reliabilities": [
  0.9896586782861317,
  0,
  0,
  0.9716049382716068
],
"reloc_status": 1,
"removed_regions": [],
"ret_code": 0,
"rfids": [],
"robot_note": "",
"robot_region": {
  "name": "",
  "point": [
    {
      "x": -1.3296384614432126,
      "y": -0.21090255905554606
    },
    {
      "x": -0.5346855105804782,
      "y": -0.30062326967818387
    },
    {
      "x": -0.4494508354889723,
      "y": 0.4545820336414137
    },
    {
```

```
    "x": -1.2444037863517066,  
    "y": 0.5443027442640516  
  }  
]  
}
```

15.1.1.3.30 Batch Data Inquiry 2

Request

- API number: 1101 (0x044D)
- Name: robot_status_all2_req
- Description: Batch data inquiry 2
- JSON data area: See table below.

All2 is to get much frequently updated data (such as location, speed, etc.) in one request, including 1004, 1005, 1006, 1009, 1010, 1011, 1012, 1013, 1014, 1016, 1017, 1020, 1050

Field Name	Type	Description	Whether Default
return_laser	boolean	Whether to return the laser data (the data of 1009), true = return, false = not return, and the default is return.	YES

Request Example

```
// Return all data.  
5A 01 00 01 00 00 00 00 04 4D 00 00 00 00 00 00  
// Don't return laser data.  
{ "return_laser": false }  
5A 01 00 01 00 00 00 16 04 4D 00 00 00 00 00 00  
7B 22 72 65 74 75 72 6E 5F 6C 61 73 65 72 22 3A  
66 61 6C 73 65 7D
```

Response

- API number: 11101 (0x2B5D)
- Name: robot_status_all2_res
- Description: Response of batch data inquiry 2
- JSON data area: See table below

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

The JSON data area will contain all key-values

of 11004, 11005, 11006, 11008, 11009, 11010, 11011, 11012, 11013, 11014, 11016, 11017, 11020, 11050

Field Name	Type	Description	Whether Default
mode	number	Robot running status, manual=0, and automatic =1	YES
x	number	The x coordinate of the robot, unit: m	YES
y	number	The x coordinate of the robot, unit: m	YES
angle	number	The angle coordinate of the robo, unit: rad	YES
confidence	number	Confidence of robot's localization, range [0, 1]	YES
current_station	string	The id of the station where the robot is currently located(this judgment is strict, the robot must be very close to a certain station(<50cm), otherwise it is a null character, that is, it is not at any station)	YES
last_station	string	id of the previous station where the robot is located	YES
vx	number	The actual speed of the robot in the x direction of the robot	YES

Field Name	Type	Description	Whether Default
		coordinate system, unit: m/s	
vy	number	The actual speed of the robot in the y direction of the robot coordinate system, unit: m/s	YES
w	number	The actual angular velocity of the robot in the robot coordinate system (i.e., turn clockwise to negative and counterclockwise to positive), unit: rad/s	YES
steer	number	Single steering wheel robot's current steering wheel angle, unit: rad	YES
spin	number	Robot tray(if exists) angle, unit: rad	YES
r_vx	number	The speed the robot receives in the x direction of the robot coordinate system, unit: m/s	YES
r_vy	number	The speed the robot receives in the y direction of the robot coordinate system, unit: m/s	YES

Field Name	Type	Description	Whether Default
r_w	number	The received angular velocity of the robot in the robot coordinate system (i.e., turn clockwise to negative and counterclockwise to positive), unit: rad/s	YES
r_steer	number	The steering wheel angle received by the single steering wheel robot, unit: rad	YES
r_spin	number	The rotation speed of the tray(if exists) received by robot, unit: rad/s	YES
steer_angles	array[number]	The current steer wheel angle of a multi-steering wheel robot, double-steering wheel robot = [front steering wheel angle, rear steering wheel angle], unit: rad	YES
r_steer_angles	array[number]	The steer wheel angle received by the multi-steer wheel robot, double-teering wheel robot = [front steering wheel angle, rear steering wheel angle], unit: rad	YES
is_stop	boolean	Whether the robot chassis is stationary (walking motor)	YES

Field Name	Type	Description	Whether Default
blocked	boolean	Whether the robot is blocked	YES
block_reason	number	Reasons for being blocked, 0 = Ultrasonic, 1 = Laser, 2 = Fallingdown, 3 = Collision, 4 = Infrared, 5 = Lock, 6 = dynamic obstacle, 7 = virtual laser, 8 = 3D camera	YES
block_x	number	The x coordinate of the nearest obstacle(world coordinate system), unit: m	YES
block_y	number	The y coordinate of the nearest obstacle(world coordinate system), unit: m	YES
block_ultrasonic_id	number	The blocked ultrasonic id number, only meaningful when block_reason = 0	YES
block_di	number	The ID number of the blocked DI, only meaningful when block_reason = 2, 3, 4	YES

Field Name	Type	Description	Whether Default
slowed	boolean	Whether the robot slow down	YES
slow_reason	number	Reasons for slow down, 0 = Ultrasonic, 1 = Laser, 2 = Fallingdown, 3 = Collision, 4 = Infrared, 5 = Lock, 6 = dynamic obstacle, 7 = virtual laser, 8 = 3D camera	YES
slow_x	number	The x coordinate of nearest obstacle that caused the slow down, unit: m	YES
slow_y	number	The y coordinate of nearest obstacle that caused the slow down, unit: m	YES
slow_ultrasonic_id	number	Ultrasonic id number which caused slow down, only meaningful when slow_reason = 0	YES
slow_di	number	The id number of the DI which caused slow down, only meaningful when slow_reason = 2, 3, 4	YES

Field Name	Type	Description	Whether Default
lasers	array	Laser point cloud data	YES
path	array[array[number]], Array includes array, which consists of two numbers -- the coodinante of path represent by X and Y in world coordinate	Sequential waypoint data, sequential connection is the robot planned path.	YES
area_ids	array[string]	An array of ids for the regions where the robot is located (Since regions on the map can overlap, the robot may be in more than one region at a time.). The array may be empty.	YES
emergency	boolean	true means the emergency stop button is activated (pressed), false means the emergency stop button is inactive (unpressed)	YES
driver_emc	boolean	true means that the motor driver has an emergency stop, false means that the motor driver has not emergency stop	YES

Field Name	Type	Description	Whether Default
DI	array[Object]	id: corresponding id number source: - normal = normal DI - virtual = virtual DI - modbus = modbus DI status: indicates high or low level - true = high level - false = low level valid: Whether the corresponding DI is enabled - true = enabled - false = disabled	YES
DO	array[Object]	id: corresponding id number source: - normal = normal DO - modbus = modbus DO status: indicates high or low level - true = high level	YES

Field Name	Type	Description	Whether Default
		false = low level	
yaw	number	yaw, unit: rad	YES
roll	number	roll, unit: rad	YES
pitch	number	pitch, unit: rad	YES
ultrasonic_nodes	array[object]	Ultrasonic node data	YES
pgvs	array[object]	QR code identification data	YES
task_status	number	0 = NONE, 1 = WAITING(This state is currently impossible), 2 = RUNNING, 3 = SUSPENDED, 4 = COMPLETED, 5 = FAILED, 6 = CANCELED	YES
task_type	number	navigation type, 0 = no navigation, 1 = free navigation to a coordinate, 2 = free navigation to a station, 3 = path navigation to	YES

Field Name	Type	Description	Whether Default
		a station, 7 = translational/rotation, 100 = other	
target_id	string	The current navigation target, this field is valid only when task_type is 2 or 3, when task_status is RUNNING, it means that robot are going to this station, when task_status is COMPLETED, it means that robot have reached this station, when task_status is FAILED, it means that robot have failed to go to this station, if task_status is SUSPENDED, then navigation to this station is paused	YES
target_point	array[number]	The current navigation coordinate. It is an array including 3 numbers, coordinate of X, Y, and R in world coordinate respectively. This field is valid only when task_type is 1, and when task_status is RUNNING, robot are going to this station; when task_status is COMPLETED, robot have reached this station; when task_status is FAILED, robot have failed to go to this station; when task_status is SUSPENDED, navigation to this station is paused.	YES
finished_path	array[string]	The stations that have passed on the current navigation path, it is an array of stations. This field is valid only when task_type is 3 or 4.	YES

Field Name	Type	Description	Whether Default
		It will list all the intermediate stations that have passed	
unfinished_path	array[string]	The stations that have not passed on the current navigation path,it is an array of stations. This field is valid only when task_type is 3 or 4. It will list all the intermediate stations that have not been passed	YES
fatals	array[object]	Array of Fatal alarm , all Fatal alarms will appear in the array	YES
errors	array[object]	Array of Error alarm , all Error alarms will appear in the array	YES
warnings	array[object]	Array of Warning alarm , all Warning alarms will appear in the array	YES
notices	array[object]	Array of Notice , all Notice will appear in the array	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES

Field Name	Type	Description	Whether Default
err_msg	string	Error message	YES

Response Example

```
{
  "DI": [
    {
      "id": 0,
      "source": "normal",
      "status": true,
      "valid": true
    },
    {
      "id": 1,
      "source": "normal",
      "status": true,
      "valid": true
    },
    {
      "id": 2,
      "source": "normal",
      "status": true,
      "valid": true
    },
    {
      "id": 3,
      "source": "normal",
      "status": true,
      "valid": true
    },
    {

```

```
    "id": 4,  
    "source": "normal",  
    "status": true,  
    "valid": true  
  },  
  {  
    "id": 5,  
    "source": "normal",  
    "status": true,  
    "valid": true  
  },  
  {  
    "id": 6,  
    "source": "normal",  
    "status": true,  
    "valid": true  
  },  
  {  
    "id": 7,  
    "source": "normal",  
    "status": true,  
    "valid": true  
  },  
  {  
    "id": 8,  
    "source": "normal",  
    "status": true,  
    "valid": true  
  }  
],  
"DO": [  
  {  
    "id": 0,  
    "source": "normal",  
    "status": true  
  },  
  {  
    "id": 1,
```

```
    "source": "normal",
    "status": false
  },
  {
    "id": 2,
    "source": "normal",
    "status": false
  },
  {
    "id": 3,
    "source": "normal",
    "status": false
  },
  {
    "id": 4,
    "source": "normal",
    "status": false
  },
  {
    "id": 5,
    "source": "normal",
    "status": true
  },
  {
    "id": 6,
    "source": "normal",
    "status": true
  },
  {
    "id": 7,
    "source": "normal",
    "status": true
  },
  {
    "id": 8,
    "source": "normal",
    "status": true
  }
}
```



```
],  
"angle": 3.0178,  
"area_ids": [],  
"block_x": 1.1998,  
"block_y": -1.4924,  
"blocked": false,  
"brake": false,  
"confidence": 0.9039,  
"current_station": "",  
"di_max_node": 24,  
"do_max_node": 16,  
"driver_emc": false,  
"electric": false,  
"emergency": false,  
"errors": [],  
"fatals": [],  
"finished_path": [],  
"last_station": "LM3",  
"mode": 1,  
"notices": [],  
"path": [],  
"pgvs": [],  
"pitch": 0.12391837522712172,  
"r_spin": 0,  
"r_steer": 0,  
"r_vx": 1,  
"r_vy": 0,  
"r_w": 0.0041,  
"ret_code": 0,  
"roll": 0.12391837522712172,  
"running_status": 1,  
"slowed": false,  
"soft_emc": false,  
"spin": 0,  
"steer": 0,  
"target_dist": 0,  
"target_id": "1",  
"task_status": 2,
```

```

"task_type": 3,
"ultrasonic_nodes": [],
"unfinished_path": [
  "LM1"
],
"vx": 1.0303,
"vy": 0,
"w": 0.1801,
"warnings": [],
"x": 1.7051,
"y": -0.174,
"yaw": -1.6249016401331637
}

```

Batch Data Inquiry 3

Request

- API number: 1102 (0x044E)
- Name: robot_status_all3_req
- Description: Batch data inquiry 3
- JSON data area: None

All 3 is to get infrequently updated status data in one request, including 1002, 1007, 1021, 1022, 1025

Request Example

```

// Return all data.
5A 01 00 01 00 00 00 00 04 4E 00 00 00 00 00 00

```

Response

- API number: 11102 (0x2B5E)
- Name: robot_status_all3_res
- Description: Response of batch data inquiry 3
- JSON data area: See table below.

If there is a key-value in the JSON Object which is not mentioned in the API response data area, you can ignore it directly, which also applies for the following content.

The JSON data area will contain all key-values of 11002, 11007, 11021, 11022, 11025

Field Name	Type	Description	Whether Default
odo	number	Accumulated mileage, unit: m	YES
today_odo	number	Today's cumulative mileage, unit: m	YES
time	number	The current running time (the time from boot to now), unit: ms	YES
total_time	number	Cumulative running time, unit: ms	YES
controller_temp	number	Controller temperature, unit: °C	YES
controller_humi	number	Controller humidity, unit: %	YES
controller_voltage	number	Controller voltage, unit: V	YES
battery_level	number	Robot battery level, range [0, 1]	YES
battery_temp	number	Robot battery temperature, unit: °C	YES

Field Name	Type	Description	Whether Default
charging	boolean	Whether the battery is charging	YES
voltage	number	Voltage, unit: V	YES
current	number	Current, Unit: A	YES
max_charge_voltage	number	The maximum voltage allowed to charge (-1 = the battery does not support this function), unit: V	YES
max_charge_current	number	Maximum current allowed to charge (-1 = the battery does not support this function), unit: A	YES
manual_charge	boolean	It only indicates whether the robot is connected to a manual charger, cannot ensure whether the charger power is turned on, and robot cannot move when the manual charger is connected (only supported by SRC -2000)	YES
auto_charge	boolean	It only indicates whether the robot is connected to the automatic charging pile, and cannot ensure whether the charger power is turned on (only supported by SRC -2000)	YES

Field Name	Type	Description	Whether Default
battery_cycle	number	Number of battery cycles(obtained from battery BMS, not guaranteed to be correct)	YES
reloc_status	number	0 = FAILED(Localization failed), 1 = SUCCESS(Localization is correct), 2 = RELOCING(Relocating), 3=COMPLETED(Localization completed)	YES
loadmap_status	number	0 = FAILED(Failed to load map), 1 = SUCCESS(Map loading succeeded), 2 = LOADING>Loading map)	YES
slam_status	number	0 = None, 1 = Scanning(Offline), 2 = Real-time scanning(SLAM)	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "auto_charge": false,
  "battery_cycle": 0,
```

```
"battery_level": 0,  
"battery_temp": 0,  
"calib_status": {  
  "desc": "No calibration task",  
  "status": 0  
},  
"charging": false,  
"controller_humi": 0,  
"controller_temp": 0,  
"controller_voltage": 0,  
"current": 0,  
"loadmap_status": 1,  
"manual_charge": false,  
"max_charge_current": 0,  
"max_charge_voltage": 0,  
"odo": 135191.236,  
"reloc_status": 1,  
"ret_code": 0,  
"slam_status": 0,  
"time": 948965592,  
"today_odo": 6365.95,  
"total_time": 2004939186,  
"voltage": 0  
}
```

15.1.1.3.32 Robot Current Lock Inquiry

Request

- API number: 1060 (0x0424)

- Name: robot_status_current_lock_req
- Description: Robot current lock inquiry
- JSON data area: None

Field Name	Type	Description	Whether Default
locked	boolean	Whether current control is locked	NO
ip	string	Control owner IP address	YES
port	Interger	Control owner port number	YES
type	uint8_t	Types of control right owner 0x00=default 0x02=roboshop 0xDD=srd	YES
nick_name	string	Control owner nickname	YES
time_t	number	Timestamp of locked control (s)	YES

Field Name	Type	Description	Whether Default
desc	string	Description of control right owner	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Request Example

```
5A 01 00 01 00 00 00 00 03 EF 00 00 00 00 00 00
```

Response

- API number: 11060 (0x2B34)
- Name: robot_status_current_lock_res
- Description: Response of robot current lock inquiry
- JSON data area: See table below.

Response Example

```
{
  "desc": "",
  "ip": "183.192.159.190",
```

```
"locked": true,  
"nick_name": "srd-seer-mizhan",  
"port": 52681,  
"ret_code": 0,  
"time_t": 1611562267,  
"type": 0  
}
```

15.1.1.3.33 Loaded Map and Stored Map Inquiry

Request

- API number: 1300 (0x0514)
- Name: robot_status_map_req
- Description: Loaded map and stored map inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 05 14 00 00 00 00 00 00
```

Response

- API number: 11300 (0x2C24)
- Name: robot_status_map_res

- Description: Response of loaded map and stored map inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
current_map	string	The currently loaded map	YES
maps	array[string]	An array of all map names stored on the robot	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "current_map": "301",
  "maps": [
    "301",
    "302",
    "303"
  ]
}
```

15.1.1.3.34 Status Station Information of Current Map Inquiry

This API is used to get the coordinates, angles and type information of all the station in the map.

Request

- API number: 1301 (0x0515)
- Name: robot_status_station_req
- Description: Station information of currently loaded map inquiry
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 05 15 00 00 00 00 00 00
```

Response

- API number: 11301 (0x2C25)
- Name: robot_status_station_res
- Description: Response of station information of currently loaded map inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
stations	array[object]	station array, if there are no stations in the map, it will be an empty array, see object form below	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

The form of object in the following:

```
{
  "id": "LM1",          // station id
  "type": "LocationMark", // Station type: LocationMark, ChargePoint ...
  "x": 1.23,            // The x coordinate (m) of the station in the world coordinate system
  "y": 4.56,            // The y coordinate (m) of the station in the world coordinate system
  "r": 1.57,            // The orientation angle (rad) of the station in the world coordinate system
  "desc": "xxxxx"       // Remarks of the station, if it is Chinese, it is Unicode encoding, such as: \u5907\u6ce8
}
```

Response Example

```
{
  "stations": [
    {
      "id": "LM1",
      "type": "LocationMark",
```

```

    "x": 1.23,
    "y": 4.56,
    "r": 1.57,
    "desc": "xxxxx"
  },
  {
    "id": "LM2",
    "type": "LocationMark",
    "x": -1.23,
    "y": 4.56,
    "r": 3.14,
    "desc": ""
  },
  {
    "id": "CP1",
    "type": "ChargePoint",
    "x": -1.23,
    "y": -4.56,
    "r": 0,
    "desc": ""
  }
],
"ret_code": 0
}

```

15.1.1.3.36 Robot Parameters Inquiry

Request

- API number: 1400 (0x0578)

- Name: robot_status_params_req
- Description: Robot parameters inquiry
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
plugin	string	The name of the plug-in. If it defaults (empty), it indicates to inquire all the parameters of all plug-ins.	YES
param	string	Parameter name. If plugin exists, but param defaults (empty), it means to inquire all the parameters of the plugin. Otherwise it is to inquire the specified parameters of the plugin.	YES

Data Example:

Inquiring the MaxAcc parameter of the MoveFactory plug-in is shown in the following.

```
{
  "plugin": "MoveFactory",
  "param": "MaxAcc"
}
```

Request Example

Omitted

Response

- API number: 11400 (0x2C88)
- Name: robot_status_params_res

- Description: Response of robot parameters inquiry
- JSON data area: see below.

A: If both `plugin` and `param` in the request are not default, it means to query a parameter of a specific plug-in. Let's take the MoveFactory plugin as an example to illustrate the format of the data area.

`plugin = "MoveFactory"`, `param = "MaxAcc"`, the response data area of inquiring the MaxAcc parameter of the MoveFactory plug-in is as follows:

```
{
  "MoveFactory": {
    // Only when the query is successful, the plug-in does not exist, the parameter does not exist or other errors occur,
    please see ret_code
    "MaxAcc": {
      "value": 0.5,    // The value field indicates its value, and the value type varies with the parameter. For the
value type of each parameter, refer to Appendix 3
      "defaultValue": 0.5,    // default value
      "minValue": 0.0,    // minimum value
      "maxValue": 10.0,    // maximum value
      "desc": "maximum acceleration",    // description
      "group": "Navigation",    // group
      "advanced": false,    // advanced parameters
      "type": "double"    // parameter's type
    }
  },
  "ret_core": 0,    // default
  "err_msg": ""    // default
}
```

B: If `plugin` is not default but `param` is default in the request, it means to query all the parameters of a specific `plugin`. Let's take NetProtocol plugin as an example to illustrate the format of the data area.

`plugin = "NetProtocol"`, `param` default or `param=""`, querying all parameters of the NetProtocol plugin is shown as follows:

```
{
  "NetProtocol": {
```



```

// Only when the query is successful, the plug-in does not exist or other errors occur, please see ret_code
// suppose NetProtocol has two parameters: defaultAutoMode, ControlMotionDuration
"defaultAutoMode": {
    "value": false,
    "defaultValue": false,
    "desc": "The default mode is automatic after start",
    "group": "Network",
    "advanced": false,
    "type": "bool"
},
"ControlMotionDuration": {
    "value": 500,
    "defaultValue": 500,
    "minValue": 0,
    "maxValue": 5000,
    "desc": "The duration of open-loop motion command delivery speed(ms)",
    "group": "Network",
    "advanced": true,
    "type": "uint32"
}
},
"ret_core": 0, // default
"err_msg": "" // default
}

```

C: If `plugin` is default in the request, it means query all parameters.

plugin default or plugin="", query all parameters:

```

{
    // If there is no error, all plug-ins and all the parameters included in their runtime will be listed in the
    // following form. This is just an example. There will be many plug-ins and parameters in actual runtime. Please refer to
    // Appendix 3 for specific plug-in and parameter descriptions.
    "NetProtocol": {
        "defaultAutoMode": {
            "value": false,
            "defaultValue": false,
            "desc": "The default mode is automatic after start",

```

```
        "group": "Network",
        "advanced": false,
        "type": "bool"
    },
    "ControlMotionDuration": {
        "value": 500,
        "defaultValue": 500,
        "minValue": 0,
        "maxValue": 5000,
        "desc": "Duration of open-loop motion command delivery speed(ms)",
        "group": "Network",
        "advanced": true,
        "type": "uint32"
    }
},
"MoveFactory": {
    "MaxAcc": {
        "value": 0.5,
        "defaultValue": 0.5,
        "minValue": 0.0,
        "maxValue": 10.0,
        "desc": "maximum acceleration",
        "group": "Navigation",
        "advanced": false,
        "type": "double"
    },
    "MaxSpeed": {
        "value": 0.3,
        "defaultValue": 0.5,
        "minValue": 0.0,
        "maxValue": 10.0,
        "desc": "maximum speed",
        "group": "Navigation",
        "advanced": false,
        "type": "double"
    },
    "MaxRotAcc": {
        "value": 60,
```

```

        "defaultValue": 60,
        "minValue": 0.0,
        "maxValue": 360.0,
        "desc": "maximum angular acceleration",
        "group": "Navigation",
        "advanced": false,
        "type": "double"
    },
    "MaxRotSpeed": {
        "value": 90,
        "defaultValue": 60,
        "minValue": 0.0,
        "maxValue": 360.0,
        "desc": "maximum angular velocity",
        "group": "Navigation",
        "advanced": false,
        "type": "double"
    }
    // ...
},
// ....
"ret_core": 0, // default
"err_msg": "" // default
}

```

Response example

```

{
  "NetProtocol": {
    "ArmIP": {
      "advanced": true,
      "defaultValue": "192.168.192.200",
      "desc": "IP address of the robot arm",
      "group": "Network",
      "type": "string",
      "value": "192.168.192.200"
    },
    "ConfidenceLowConfirm": {
      "advanced": false,

```

```

    "defaultValue": 0.3,
    "desc": "If the confidence level is lower than the value, do not confirm the location.",
    "group": "Localization",
    "maxValue": 0.95,
    "minValue": 0,
    "type": "double",
    "value": 0.3
  },
  "ControlMotionDuration": {
    "advanced": true,
    "defaultValue": 500,
    "desc": "Duration of open loop motion command delivery speed (ms).",
    "group": "Network",
    "maxValue": 5000,
    "minValue": 0,
    "type": "uint32",
    "value": 500
  }
},
"ret_code": 0
}

```

15.1.1.4 Robot Control API

The robot control API is mainly used to send open-loop control commands.

Note: Confirm location correct([2003](#)) can only be used when Robot map loading status([1022](#)) is SUCCESS, and Robot map loading status([1021](#)) is COMPLETED. After confirming correct location, Robot location status([1021](#)) will become SUCCESS.

15.1.1.4.1 API list

Request

Serial Number	Name	Description
2000	robot_control_stop_req	Stop open loop
2002	robot_control_reloc_req	Relocation
2003	robot_control_confirmloc_req	Confirm correct location
2004	robot_control_cancelreloc_req	Cancel relocation
2010	robot_control_motion_req	Open loop
2022	robot_control_loadmap_req	Toggle loaded map
...	...	...

Response

The response from the robot control API is returned to the client immediately and does not wait for the robot to complete. If `ret_code` does not exist, the command will be executed, but it does not mean that execution of the command has been completed. For some control commands, such as (2022) switch map, the state of loading map can be queried by (1022). Most of the other commands are instantaneous, and the customer doesn't need to care when the robot will finish, just know that the robot will run the command.

Serial Number	Name	Description
12000	robot_control_stop_res	Response to request 2000
12002	robot_control_reloc_res	Response to request 2002
12003	robot_control_comfirmloc_res	Response to request 2003
12004	robot_control_cancelreloc_res	Response to request 2004
12010	robot_control_motion_res	Response to request 2010
12022	robot_control_loadmap_res	Response to request 2022
...	...	...

- 15.1.1.4.1 API list

- Request
- Response

15.1.1.4.2 Stop Open Loop Motion

Request

- API number: 2000 (0x07D0)
- Name: robot_control_stop_req
- Description: Stop open loop motion.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 07 D0 00 00 00 00 00 00
```

Response

- API number: 12000 (0x2EE0)
- Name: robot_control_stop_res
- Description: Response of stopping open loop motion
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
5A 01 00 01 00 00 00 0E 2E E0 07 D0 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.4.3 Relocation

Request

- API number: 2002 (0x07D2)
- Name: robot_control_reloc_req
- Description: Relocation
- JSON data area: See table below.

Note: All default parameters are not allowed

Field Name	Type	Description	Whether Default
x	Number	X coordinate in global coordinate system, unit: m	YES
y	Number	Y coordinate in global coordinate system, unit: m	YES
angle	Number	The angle in the global coordinate system, unit: rad	YES
length	number	Radius of relocation area, unit: m	YES
home	Bool	Relocate in RobotHome (if true, the first three parameters are invalid, and relocate from RobotHome1-5 in Roboshop parameter configuration, if RobotHome1-5 is not configured, then do nothing. The default is false.)	YES

Request Example

```
5A 01 00 01 00 00 00 1C 07 D2 00 00 00 00 00 00
{"x":10.0,"y":3.0,"angle":0}
// After serialization, the hex is:
7B 22 78 22 3A 31 30 2E 30 2C 22 79 22 3A 33 2E 30 2C 22 61 6E 67 6C 65 22 3A 30 7D
```

Response

- API number: 12002 (0x2EE2)

- Name: robot_control_reloc_res
- Description: Response of Relocation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
5A 01 00 01 00 00 00 00 2E E2 00 00 00 00 00
```

Confirm Correct Location

Request

- API number: 2003 (0x07D3)
- Name: robot_control_comfirmloc_req
- Description: Confirm correct location.

- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 07 D3 00 00 00 00 00 00
```

Response

- API number: 12003 (0x2EE3)
- Name: robot_control_comfirmloc_res
- Description: Response of confirming correct location
- JSON data area: See table below.

Field Name	Type	Description	Whether Ddefault
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
5A 01 00 01 00 00 00 00 2E E3 00 00 00 00 00 00
```

15.1.1.4.5 Cancel Relocation

It is used to cancel the relocation process. If the relocation is currently being performed (that is, when `reloc_status = 2`), the relocation will be canceled, the position of the robot will return to the position before the relocation instruction is sent, and the `reloc_status` will equal to 3 (relocation success). If there is no relocation currently in progress, this command has no effect.

Request

- API number: 2004 (0x07D4)
- Name: `robot_control_cancelreloc_req`
- Description: Cancel relocation.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 07 D4 00 00 00 00 00 00
```

Response

- API number: 12004 (0x2EE4)
- Name: `robot_control_cancelreloc_res`
- Description: Response of cancelling relocation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

15.1.1.4.6 Open Loop Motion

Controls onlt through vx, vy,and w for multi-steering wheel.

Request

- API number: 2010 (0x07DA)
- Name: robot_control_motion_req
- Description: Open loop motion.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
vx	Number	The speed of the robot in the x-axis direction in the robot coordinate system, default is 0, unit: m/s	YES
vy	Number	The speed of the robot in the y-axis direction in the robot coordinate system, default is 0, unit: m/s	YES
w	Number	The angular velocity of the robot in the robot coordinate system (i.e., turn clockwise to negative and counterclockwise to positive), default is 0, unit: rad/s	YES
steer	number	-2= back to zero, 1=15° increment, -1=15° decrement (robot coordinate system), and this value is valid only for the robots with a single steering wheel. When this field exists, it does not respond to vy and w, and vx indicates the line speed of the steering wheel.	是
real_steer	number	The target rudder angle value (robot coordinate system), unit: rad; This value is only valid for the robots with a single steering wheel. When this field exists, it does not respond to vy and w, and vx indicates the line speed of the steering wheel, but its priority is higher than <code>steer</code> .	是
duration	Number	The duration of the robot's open-loop speed movement, unit ms, 0 = keep the current open-loop speed movement, this parameter defaults to the value of ControlMotionDuration in the parameter configuration.	YES

Request Example

```
5A 01 00 01 00 00 00 13 07 DA 00 00 00 00 00 00
{"vx":0.5,"vy":0.5}
// After serialization, the hex is:
7B 22 76 78 22 3A 30 2E 35 2C 22 76 79 22 3A 30 2E 35 7D
```

Response

- API number: 12010 (0x2EEA)
- Name: robot_control_motion_res
- Description: Response of open loop motion
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

15.1.1.4.7 Switch Map

Request

- API number: 2022 (0x07E6)
- Name: robot_control_loadmap_req
- Description: Switch map.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
map_name	String	Map name to switch (Contains only 0-9, a-z, A-Z, -, _ instead of illegal characters such as Chinese characters.)	NO

Request Example

```
5A 01 00 01 00 00 00 15 07 E6 00 00 00 00 00 00
{"map_name":"middle"}
// After serialization, the hex is:
7B 22 6D 61 70 5F 6E 61 6D 65 22 3A 22 6D 69 64 64 6C 65 22 7D
```


Response

- API number: 12022 (0x2EF6)
- Name: robot_control_loadmap_res
- Description: Response of switch map
- JSON data area: See table below

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "ret_code": 0  
}
```

API Number	Name	Description
3001	robot_task_pause_req	Pause navigation
3002	robot_task_resume_req	Resume navigation
3003	robot_task_cancel_req	Cancel navigation
3051	robot_task_gotarget_req	Path navigation(Navigate according to the station and path on the map)
3052	robot_task_gotarget2_req	Path navigation 2
3053	robot_task_target_path_req	Get the path of the path navigation
3055	robot_task_translate_req	Translation, move a fixed distance in a straight line at a fixed speed
3056	robot_task_turn_req	Rotation, rotate a fixed angle at a fixed angular speed
3057	robot_task_spin_req	Tray rotation
3058	robot_task_circular_req	Circular motion
3066	robot_task_gotargetlist_req	Designated path navigation(navigate according to the path navigation sequence)
3067	robot_task_cleartargetlist_req	Clear Designated path navigation

API Number	Name	Description
3101	robot_tasklist_status_req	Query a robot task list
3106	robot_tasklist_name_req	Execute pre-stored tasklist
...	...	...

15.1.1.5 Robot Navigation API

The robot navigation API is mainly used to send navigation related commands.

15.1.1.5.1 API List

Request

Note: All commands cannot be performed simultaneously.

Response

The robot navigation API will immediately respond to the client without waiting for the robot to perform navigation. If ret_code does not exist or is 0, it means that the navigation will be executed, but it does not mean that the navigation has been completed.

The user needs to query the status of the navigation through the robot status API (1020). A robot can only have one navigation at a time, and new navigation instructions will interrupt the previous navigation.

API Number	Name	Description
13001	robot_task_pause_res	Response to 3001 request
13002	robot_task_resume_res	Response to 3002 request
13003	robot_task_cancel_res	Response to 3003 request
13051	robot_task_gotarget_res	Response to 3051 request
13052	robot_task_gotarget2_res	Response to 3052 request
13053	robot_task_target_path_res	Response to 3053 request
13055	robot_task_translate_res	Response to 3055 request
13056	robot_task_turn_res	Response to 3056 request
13057	robot_task_spin_res	Response to 3057 request
13058	robot_task_circular_res	Response to 3058 request

API Number	Name	Description
13066	robot_task_gotargetlist_res	Response to 3066 request
13067	robot_task_cleartargetlist_res	Response to 3067 request
13101	robot_tasklist_status_res	Response to 3101 request
13106	robot_tasklist_name_res	Response to 3106 request
...	...	...

- [15.1.1.5.1 API List](#)
- [Request](#)
- [Response](#)

15.1.1.5.2 Pause Navigation

Request

- API number: 3001 (0x0BB9)
- Name: robot_task_pause_req
- Description: Pause navigation
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0B B9 00 00 00 00 00 00
```

Response

- API number: 13001 (0x32C9)
- Name: robot_task_pause_res
- Description: Response of pause navigation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES

Field Name	Type	Description	Whether Default
err_msg	string	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 32 C9 0B B9 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.3 Resume Navigation

Request

- API number: 3002 (0x0BBA)
- Name: robot_task_resume_req
- Description: Resume navigation
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0B BA 00 00 00 00 00 00
```


Response

- API number: 13002 (0x32CA)
- Name: robot_task_resume_res
- Description: Response of resume navigation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 32 CA 0B BA 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.4 Cancel Navigation

Request

- API number: 3003 (0x0BBB)
- Name: robot_task_cancel_req
- Description: Cancel navigation
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0B BB 00 00 00 00 00 00
```

Response

- API number: 13003 (0x32CB)
- Name: robot_task_cancel_res
- Description: Response of cancel navigation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES

Field Name	Type	Description	Whether Default
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 32 CB 0B BB 00 00 00 00
// {"ret_code":0}
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.5 Path Navigation

Path navigation is to send a target station to the robot who automatically plans the path to run, and passes through the intermediate station without stopping.

Request

- API number: 3051 (0x0BEB)

- Name: robot_task_gotarget_req
- Description: Path navigation
- JSON data area: See table below.

1. Fixed Path Navigation

```
// Example 1.1 The robot goes to LM1 along the line LM2->LM1
{
  "source_id": "LM2",
  "id": "LM1",
  "task_id": "87654321"
}
```

Fixed path navigation supports user-defined optional attributes. For example, it supports the configuration of parameters such as maximum speed, maximum acceleration, and angle of arrival. A complete fixed path navigation command is shown in example 1.2.

```
// Example 1.2 fixed path navigation command
{
  "source_id": "LM2",
  "id": "LM1",
  "task_id": "87654321",
  "angle": 1.57,
  "method": "backward",
  "max_speed": 0.2,
  "max_wspeed": 0.2,
  "max_acc": 0.1,
  "max_wacc": 0.1,
  "duration": 100,
  "orientation": 90,
  "spin": true
}
```

:-: Table 1.1. Fixed Path Navigation Command

Field Name	Type	Description	Whether Default
id	string	Target station name When the robot performs operation in place, its value is "SELF_POSITION"	NO
source_id	string	Source station name When the starting position is not on a station but the current position of the robot, its value is "SELF_POSITION"	NO
task_id	string	Task id	NO
angle	number	The angle of target station in world coordinate system, unit rad	YES
method	string	Movement method, can only be "forward" or "backward", if this value is default, the movement method will be set to the method configured on the path	YES
max_speed	number	Maximum speed, unit m/s	YES
max_wspeed	number	Maximum angular speed, unit rad/s	YES

Field Name	Type	Description	Whether Default
max_acc	number	Maximum acceleration, unit m/s^2	YES
max_wacc	number	Maximum angular acceleration, unit rad/s^2	YES
duration	number	Waiting time after path navigation, unit ms	YES
orientation	number	The angle maintained by the omnidirectional car (not used temporarily)	YES
spin	boolean	Spin or not	YES
delay	number	Delay time to end navigation status, unit ms, default=0	YES
rot_dir	number	-1: clockwise 0: rotate in the near direction 1: counterclockwise	YES
skill_name	string	"Action" expresses an action at the current point, or "GotoSpecifiedPose" expresses path navigation	YES

2. Fixed Path Navigation with PGV Secondary Location Adjustment

After the robot reaches the target station, it can perform a secondary adjustment of its position based on the QR code on the ground or on the shelf.

There are four ways to adjust PGV. For specific adjustment methods, see

[\[https://shimo.im/docs/r16q8MwZOWos6k7D\]](https://shimo.im/docs/r16q8MwZOWos6k7D)[\]\(https://shimo.im/docs/r16q8MwZOWos6k7D\)](https://shimo.im/docs/r16q8MwZOWos6k7D). Among them, you can only choose either `pgv_x_adjust` or `pgv_adjust_dist` or neither.

```
// Example 2.1. From LM2 to AP1, the robot uses pgv to perform a secondary adjustment of its position.
{
  "id": "AP1",
  "source_id": "LM2",
  "task_id": "123456",
  "use_pgv": true,           //adjust position by using upview PGV
  "pgv_adjust_dist": 0.3,   //maximum adjustment radius
  "pgv_adjust_cx": -0.3,    //center offset of adjustment range
  "pgv_adjust_cy": 0
}
```

:-: Table 2.1. Second Adjustment Command of Path Navigation Based on QR Code

Field name	Type	Description	Whether default
use_pgv	bool	Whether to use the upview PGV to identify the QR code for the secondary adjustment of the position	YES, false by default
pgv_adjust_dist	number	Maximum adjustment radius (used for the third and fourth adjustment methods)	YES
pgv_adjust_cx	number	The offset between x coordinate of the center point in secondary adjustment range and x coordinate of the center point in the QR code coordinate system (this coordinate system is parallel to the current robot coordinate system), (used for the fourth adjustment method)	YES
pgv_ajdust_cy	number	The offset between y coordinate of the center point in secondary adjustment range and y coordinate of the center point in the QR code coordinate system (this coordinate system is parallel to the current robot coordinate system), (used for the fourth adjustment method)	YES
pgv_x_adjust	number	In the PGV adjustment mode, only the error in the x direction of the car is adjusted (used for the second adjustment method)	YES
use_dow_pgv	bool	Whether to use the down PGV to identify the QR code for the secondary adjustment of the position	YES

3. Fixed Path Navigation with Operation

The robot supports the control of jack equipment, fork equipment, spin equipment and hook equipment.

3.1 Jacking

The robot can jack up the goods or put the goods down through recognition.

```
// Example 3.1.1. The robot reaches AP1 by identifying the shelf, and then uses the QR code on the shelf to perform
secondary positioning, and then lift the goods up
{
  "id": "AP1",
  "source_id": "LM2",
  "task_id": "12344321",
  "operation": "JackLoad",
  "recognize": true,
  "use_pgv": true
}
// Example 3.1.2 The robot arrives at AP1, and then adjusts its position through the down PGV, and then puts down the
goods
{
  "id": "AP1",
  "source_id": "LM2",
  "task_id": "12344321",
  "operation": "JackUnload",
  "use_down_pgv": true
}
// Example 3.1.3. The robot reaches AP1 and raises the jack equipment to a height of 0.2m
{
  "id": "AP1",
  "source_id": "LM2",
  "task_id": "12344321",
```

```

"operation": "JackHeight",
"jack_height": 0.2
}

```

:- Table 3.1. Path navigation with jacking equipment

Field Name	Type	Description	Whether Default
operation	string	Control the actions of jack equipment, supported actions JackLoad (Lift up and load cargo) JackUnload (Jack down) JackHeight (Jack up) Wait (No action)	YES, default is Wait.
jack_height	number	When operation equals to JackHeight, the height of jack	YES

3.2 Fork

The robot (fork or hand hydraulic carrier) can lift up or put down the pallet through recognition.

```

// Example 3.2.1. The robot moves the fork to the start_height height, and then determines the position of the target
point by identifying the pallet, and then the robot moves to the target point, and then the robot lifts the shelf and
loads the goods
{
  "id": "AP1",
  "source_id": "LM2",
  "task_id": "12344321",

```

```

    "operation": "ForkLoad",
    "start_height": 0.1,
    "end_height": 1.0,
    "recognize": true
}
// Example 3.2.2. The robot moves the fork to start_height, then the robot moves to the target point, and then the robot
lowers the fork and unloads the goods
{
    "id": "AP1",
    "source_id": "LM2",
    "task_id": "12344321",
    "operation": "ForkUnload",
    "start_height": 0.5,
    "end_height": 0.1
}
// Example 3.2.3. The robot raises the fork to a height of 0.5m on the way to AP1
{
    "id": "AP1",
    "source_id": "LM2",
    "task_id": "12344321",
    "operation": "ForkHeight",
    "fork_mid_height": 0.5
}
// Example 3.2.4. The robot raises the fork to a height of 0.1m after going to AP1
{
    "id": "AP1",
    "source_id": "LM2",
    "task_id": "12344321",
    "operation": "ForkHeight",
    "end_height": 0.1
}
// Example 3.2.5. The robot moves forward the fork to 0.1m after going to AP1
{
    "id": "AP1",
    "source_id": "LM2",
    "task_id": "12344321",
    "operation": "ForkForward",
    "fork_dist": 0.1
}

```

```

}
// Example 3.2.6. The robot moves the fork to start_height and determines the target position by recognizing the pallet
(shelf/ S0002. Shelf)
{
  "id": "AP1",
  "source_id": "LM2",
  "task_id": "12344321",
  "operation": "ForkLoad",
  "start_height": 0.1,
  "end_height": 1.0,
  "recognize": true,
  "recfile": "shelf/s0002.shelf"
}

```

:-: Table 3.2. Path Navigation with Fork

Field Name	Type	Description	Whether Default
operation	string	Control the movement of the fork equipment, supported movement ForkLoad (Lift the fork to load the cargo) ForkUnload (Fork down) ForkHeight (Fork up) ForkForward (Fork move forward) Wait (No action)	YES, default is Wait.
start_height	number	Lifting height before robot start path navigation	YES
fork_mid_height	number	Lifting height during robot navigation	YES

Field Name	Type	Description	Whether Default
end_height	number	Lifting height after robot reach the target station	YES
fork_dist	number	Forward moving distance for forward moving fork	YES

3.3 Roller

The robot supports the control of single-roller.

```
// Example 3.3.1, the robot loads goods on the left through the roller
{
  "id": "AP1",
  "source_id": "LM2",
  "task_id": "12345",
  "operation": "RollerLoad",
  "direction ": "left"
}
// Example 3.3.2, the robot unloads goods on the right side through the roller
{
  "id": "AP1",
  "source_id": "LM2",
  "task_id": "12345",
  "operation": "RollerUnload",
  "direction": "right"
}
```

:- Table 3.3. Path Navigation with Roller

Field Name	Type	Description	Whether Default
operation	string	Control the actions of the roller equipment, supported actions RollerLoad (Roller loaded cargo) RollerRoll (Roller roll) RollerPreLoad (Roller pre-load cargo) RollerStop (Roller stop) RollerPass RollerInverse RollerInverseReset	YES, default is Wait.
direction	string	left, right, front, back	YES

3.4 Hook

Robot supports hook.

```
// Example 3.4.1, the robot hooks the goods through recognition
{
```

```

    "id": "AP1",
    "operation": "HookLoad",
    "source_id": "LM2",
    "task_id": "12345",
    "recognize": true
  }
  // Example 3.4.2, the robot puts down the hooked goods
  {
    "id": "AP1",
    "operation": "HookUnload",
    "source_id": "LM2",
    "task_id": "12345"
  }

```

:- Table 3.4. Path Navigation with Hook

Field Name	Type	Description	Whether Default
operation	string	Control the movement of the hook equipment, supported movement HookLoad (Hook cargo) HookUnload (Hook unload cargo) Wait (No action)	YES, default is Wait.

3.5 Audio

```

// // Example 3.5.1, when the robot reaches AP1 station, the navigation music will be played in a loop.
{
  "id": "AP1",
  "operation": "sound",
  "source_id": "LM2",

```

```

"task_id": "12345",
"sounds_args": {
  "name": "navigation",
  "loop": 1
}
}
// // Example 3.5.2, the robot stops the music after reaching AP1.
{
  "id": "AP1",
  "operation": "sound",
  "source_id": "LM2",
  "task_id": "12345",
  "sounds_args": {
    "stop": 1
  }
}
{
  "id": "SELF_POSITION",
  "operation": "sound",
  "sounds_args": {
    "name": "navigation",
    "loop": 1
  }
}
}

```

:-: Table 3.5 Description of sound_args parameters

Field Name	Type	Description	Whether Default
name	string	Audio name	YES, default is empty.
loop	int	0: play only once 1: loop play	YES, default is 0

Field Name	Type	Description	Whether Default
stop	int	0: no action 1: shut down the audio	YES, default is 0

3.6 Waiting for DI Trigger

Example 3.5.1: After the robot reaches AP1 station, the robot waits for DI1 to become true and DI2 to false, or waits for 10s to complete the task.

```
{
  "id": "AP1",
  "operation": "WaitDI",
  "source_id": "LM2",
  "task_id": "12345",
  "args": {
    "DI": [
      {
        "id": 1,
        "status": true
      },
      {
        "id": 2,
        "status": false
      }
    ]
  },
  "timeout": 10
}
```

You also can use the following way, but this is not recommended.

```
{
  "id": "AP1",
  "operation": "WaitDI",
  "source_id": "LM2",
  "task_id": "12345",
  "1": true,
  "2": false
}
```

:- Table 3.5 WaitDI Parameter Description

Field Name	Type	Description	Whether Default
DI	object	JSON object of DI includes <code>id</code> and <code>status</code>	NO
timeout	double	Timeout s for task completion	YES

3.7 Controlling DO

Example 3.7.1 shows that after the robot reaches AP1, it starts to control DO state. The passed parameter means that the robot opens DO1 and closes DO2.

```
{
  "id": "AP1",
  "operation": "SetDO",
  "source_id": "LM2",
  "task_id": "12345",

```

```

"args": {
  "DO": [
    {
      "id": 1,
      "status": true
    },
    {
      "id": 2,
      "status": false
    }
  ]
}

```

You also can use the following way, but this is not recommended.

```

{
  "id": "AP1",
  "operation": "SetDO",
  "source_id": "LM2",
  "task_id": "12345",
  "1": true,
  "2": false
}

```

:-: Table 3.7.1 Description of SetDO parameters

Field Name	Type	Description	Whether Default
DO	object	JSON object of DO includes <code>id</code> and <code>status</code>	NO

3.8 Waiting for Triggler to Trigger Script Action

// Example 3.9.1 shows that after the robot reaches AP1, it waits for Triggler to trigger and executes the corresponding script action. If the execution is completed, the task is completed

```
{
  "id": "AP1",
  "script_name": "syspy/trigglerScript.py",
  "script_args": {},
  "operation": "Script",
  "source_id": "LM2",
  "task_id": "12345"
}
```

3.9 Single Motor Control

- [] TODO

3.10 Adjusting on the QR Code

Control the robot to adjust the position on the QR code.

```
// Example 3.10.1 shows the upview PGV adjustment at the current point
{
  "id": "SELF_POSITION",
  "operation": "GoPGV",
  "source_id": "SELF_POSITION",
  "task_id": "123",
  "use_pgv": true,
  "use_down_pgv": false
}
```

```

}
// Example 3.10.2 shows the downview PGV adjustment at the current point
{
  "id": "SELF_POSITION",
  "operation": "GoPGV",
  "source_id": "SELF_POSITION",
  "task_id": "1234",
  "use_pgv": false,
  "use_down_pgv": true
}

```

:-: Table 3.10.1 Description of adjusting PGV parameters

Field Name	Type	Description	Whether Default
use_pgv	bool	Use upview PGV adjustment	NO
use_down_pgv	bool	Use downview PGV adjustment	NO

Note that if both use_pgv and use_down_pgv are true, adjust according to the up PGV adjustment.

4. Performing "operation" in Place

The **operation** of the robot not only can be executed before, during and after the path navigation, but also can be executed in place. In-place execution needs to set both `id` and `source_id` to "SELF_POSITION"

```

// Example 4.1 shows the operation of JackLoad at the current place
{
  "id": "SELF_POSITION",
  "source_id": "SELF_POSITION",

```

```
"task_id": "12345678abcdefgh",  
"operation": "JackLoad"  
}
```

5. Fixed Path Execution Script

:-: Table 5.0.0 Fixed path execution script, description of script related fields

Field Name	Type	Description	Whether Default
operation	string	Fixed value <code>Script</code>	NO
script_name	string	Script name. It's the relative path of script. E.g: "syspy/goPath.py"	NO
script_args	object	The parameters actually defined in the script. E.g: { "x": 1.0, "y": 0.0, "coordinate": "robot" }	According to the actual situation of the script

5.1 Universal Script

The robot can execute the script before, during, or after path navigation.

```
// Example 5.1.1 shows that the robot **after** goes to the LM1, executes the script firstScript.py, the input parameter of the script is {"hello":"world"}.
{
  "script_name": "firstScript.py",
  "script_args": {
    "hello": "world"
  },
  "operation": "Script",
  "id": "LM1",
  "source_id": "LM2",
  "task_id": "12344321",
  "script_stage": 2
}
// Example 5.1.2 shows that the robot **during** going to LM1, executes the script firstScript.py, the input parameter of the script is {"hello":"world"}.
{
  "script_name": "firstScript.py",
  "script_args": {
    "hello": "world"
  },
  "operation": "Script",
  "id": "LM1",
  "source_id": "LM2",
  "task_id": "12344321",
  "script_stage": 1
}
// Example 5.1.3 shows that the robot **before** goes to LM1 and executes the script firstScript.py. The input parameter of the script is {"hello":"world"}.
{
  "script_name": "firstScript.py",
  "script_args": {
    "hello": "world"
  },
```

```

},
"operation": "Script",
"id": "LM1",
"source_id": "LM2",
"task_id": "12344321",
"script_stage": 0
}

```

:-: Table 5.1.1 Description of universal script

Field Name	Type	Description	Whether Default
operation	string	"script"	NO
script_name	string	Script name. It's the relative path of script.	NO
script_args	object	The parameters actually defined in the script. It's JSON string.	YES
script_stage	int	0: execute before navigation 1: execute during navigation 2: execute after navigation	YES, default is 2

5.2 Built-in Script goPath.py, Navigating by Custom Path

The **goPath.py** is used to send a path in the world coordinate system or robot coordinate system to the robot.

// Example 5.2.1 shows that the robot is at the current position, execute the script goPath.py, the input parameter of the script means that the robot walks 1m forward in its own coordinate system.

```
{
  "script_name": "syspy/goPath.py",
  "script_args": {
    "x": 1.0,
    "y": 0.0,
    "coordinate": "robot"
  },
  "operation": "Script",
  "id": "SELF_POSITION",
  "source_id": "SELF_POSITION",
  "task_id": "12344321"
}
```

// Example 5.2.2 shows that the robot is at the current position, and the script goPath.py is executed. The incoming parameters of the script indicate that the robot walks to the position of x:1.0, y:0.0 in the world coordinate system.

```
{
  "script_name": "syspy/goPath.py",
  "script_args": {
    "x": 1.0,
    "y": 0.0,
    "coordinate": "world"
  },
  "operation": "Script",
  "id": "SELF_POSITION",
  "source_id": "SELF_POSITION",
  "task_id": "12344321"
}
```

:-: Table 5.2.1 Description of goPath.py input parameters

Field Name	Type	Description	Whether default
x	double	x coordinate, unit m	NO
y	double	y coordinate, unit m	NO
coordinate	string	"robot"=robot coordinate system "world"=world coordinate system	NO
theta	double	Angle in the world coordinate system, unit rad	YES
reachAngle	double	Angle accuracy in reach point, unit rad	YES
reachDist	double	Position accuracy in reach point, unit m	YES
backMode	int	1: backward 0: forward	YES, default is 0
useOdo	int	1: run according to mileage 0: run according to positioning	YES, default is 0

5.3 Built-in Script setDO.py, Controlling Multiple DO Together

The **setDO.py** script is used to control multiple Do switches at the same time.

```
// Example 5.3.1 shows that the robot is at the current position, execute the script setDO.py, the input parameter of the script means that the robot opens D01 and closes D02.
{
  "script_name": "syspy/setDO.py",
  "script_args": {
    "DO": [
      {
        "id": 1,
        "status": true
      },
      {
        "id": 2,
        "status": false
      }
    ]
  },
  "operation": "Script",
  "id": "SELF_POSITION",
  "source_id": "SELF_POSITION",
  "task_id": "12344321"
}
```

:- Table 5.3.1 Description of setDO.py Input Parameters

Field Name	Type	Description	Whether Default
DO	object	JSON object of DO includes <code>id</code> and <code>status</code>	NO

5.4 Built-in Script waitDI.py

The waitDI.py script is used to make the car wait for certain DIs to reach a certain state, or reach a timeout condition, and then the task is completed

// Example 5.4.1 shows that the robot is at the current position and executes the script waitDI.py. The input parameter of the script means that the robot plays wind audio, and then waits in place for DI1 to become true and DI2 to false, or waits for 10s to complete the task. The audio ends.

```
{
  "script_name": "syspy/waitDI.py",
  "script_args": {
    "DI": [
      {
        "id": 1,
        "status": true
      },
      {
        "id": 2,
        "status": false
      }
    ],
    "timeout": 10,
    "soundName": "wind"
  },
  "operation": "Script",
  "id": "SELF_POSITION",
  "source_id": "SELF_POSITION",
  "task_id": "12344321"
}
```

:-: Table 5.4.1 Description of waitDI.py Input Parameters

Field Name	Type	Description	Whether Default
DI	object	JSON object of DI includes <code>id</code> and <code>status</code>	NO
timeout	double	Timeout s for task completion	YES
soundName	string	The name of the audio when waiting for DI	YES

5.5 Built-in Script `sound.py` to Control Audio

The **`sound.py`** script is used to control audio playback and stop.

```
// Example 5.5.1 shows that the robot is at the current position, and the script sound.py is executed. The incoming
parameters of the script indicate that the robot starts to loop play the navigation audio.
{
  "script_name": "syspy/sound.py",
  "script_args": {
    "name": "navigation",
    "loop": 1
  },
  "operation": "Script",
  "id": "SELF_POSITION",
  "source_id": "SELF_POSITION",
  "task_id": "12344321"
}
// Example 5.5.2 shows that the robot is at the current position and executes the script sound.py. The incoming parameter
of the script indicates that the robot stops playing audio.
{
```

```

"script_name": "syspy/sound.py",
"script_args": {
  "stop": 1
},
"operation": "Script",
"id": "SELF_POSITION",
"source_id": "SELF_POSITION",
"task_id": "12344321"
}

```

:- Table 5.5.1 Description of input parameters of sound.py

Field name	Type	Description	Whether default
name	string	Audio name	YES, default is empty
loop	int	0: only play once 1: loop	YES, default is 0
stop	int	0: no action 1: stop	YES, default is 0

5.6 Built-in Script switchMap.py

switchMap.py script is used for map switching.

```
// Example 5.6.1 shows that the robot is at the current position, and the script switchMap.py is executed. The input
parameter of the script indicates that the robot switches to the map of mizhan.
{
  "script_name": "syspy/switchMap.py",
  "script_args": {
    "map": "mizhan"
  },
  "operation": "Script",
  "id": "SELF_POSITION",
  "source_id": "SELF_POSITION",
  "task_id": "12344321"
}
```

:- Table 5.6.1 Description of the input parameters of switchMap.py

Field Name	Type	Description	Whether Default
map	string	Map name	NO

Response

- API number: 13051 (0x32FB)
- Name: robot_task_gotarget_res
- Description: Response of path navigation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 32 FB 0B EB 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.6 Path Navigation 2

The path navigation is a given target station, which is automatically planned by the robot to run **along the Bezier curve** path, during the path navigation robot will pass through other intermediate stations but will not stop.

Request

- API number: 3052 (0x0BEC)
- Name: robot_task_gotarget2_req
- Description: Path navigation 2
- JSON data area: See below.

Note 1: Angle is the arrival direction of the robot, and the value set on the station will be used by default.

Note 2: The max_speed, max_wspeed, max_acc, max_wacc fields are not recommended. It is better to control these values through the path and area attributes or robot parameters on the map, and these fields may be deleted in the future.

Field Name	Type	Description	Whether Default
id	string	<code>id</code> of target station	NO
angle	number	The angle of target station in world coordinate system, unit rad	YES
method	string	Movement method, can only be "forward" or "backward", if this value is default, the movement method will be set to the method configured on the path	YES
operation	string	Action, see below for details	YES
direction	string	Direction, see below for details	YES
recognize	boolean	Whether to recognize, see below for details	YES
layer	number	layer, see below for details	YES
jack_height	number	Jack height, see below for details	YES
start_height	number	Fork start height, see below for details	YES

Field Name	Type	Description	Whether Default
end_height	number	Fork end height, see below for details	YES
fork_dist	number	Forward moving distance for forward moving fork, unit m	YES
use_pgv	boolean	Whether to use the up PGV to identify the QR code for the secondary adjustment of the position, see below for details	YES
max_speed	number	Maximum speed, unit m/s	YES
max_wspeed	number	Maximum angular speed, unit rad/s	YES
max_acc	number	Maximum acceleration, unit m/s ²	YES
max_wacc	number	Maximum angular acceleration, unit rad/s ²	YES
duration	number	Waiting time after path navigation, unit ms	YES
orientation	number	The angle maintained by the omnidirectional car (not used temporarily)	YES

Field Name	Type	Description	Whether Default
spin	boolean	Spin or not	YES
pgv_x_adjust	boolean	In the PGV adjustment mode, only the error in the x direction of the car is adjusted	YES
pgv_adjust_dist	number	Maximum adjustment radius in the PGV adjustment mode, unit m	YES
delay	number	Delay time to end navigation status, unit ms, default=0	YES
audio_name	string	The name of the audio file to be played after the navigation is completed (without file extension)	YES
audio_loop	boolean	Whether to loop play the audio after navigation, the default is not loop	YES
peripheral_data	array[number]	Write user-defined data in the peripheral, the length of the array is 0~16, directly transfer to the peripheral PLC after receiving the instruction	YES

Request Example

```

{"id": "LM1"}
// go to LM1
// The hexadecimal after serialization is:

```

```

5A 01 00 01 00 00 00 0C 0B EC 00 00 00 00 00
7B 22 69 64 22 3A 22 4C 4D 31 22 7D

{"id":"LM2"}
// go to LM2
// The hexadecimal after serialization is:
5A 01 00 01 00 00 00 0C 0B EC 00 00 00 00 00
7B 22 69 64 22 3A 22 4C 4D 32 22 7D

{"id":"LM10"}
// go to LM10
// The hexadecimal after serialization is:
5A 01 00 01 00 00 00 0D 0B EC 00 00 00 00 00
7B 22 69 64 22 3A 22 4C 4D 31 30 22 7D

{"id":"LM102"}
// go to LM102
// The hexadecimal after serialization is:
5A 01 00 01 00 00 00 0E 0B EC 00 00 00 00 00
7B 22 69 64 22 3A 22 4C 4D 31 30 32 22 7D

```

Operation Description When navigating to an AP (or TL, WL) station using path navigation in Roboshop, you can choose to execute

an action (meaning navigating to the AP and executing the action), as shown in the figure below:

Various actions can be

performed according to the execution object and recognition model file configured on the AP point

At this time, more

parameters need to be added to the path navigation commands to make the robot navigate to the target station and complete the corresponding actions. The commands are as follows:

1. Fork The execution object of the AP station needs to be configured as fork, and the recognition model file is configured as a pallet model file or not configured as required. **operation:** action can be "ForkLoad" or "ForkUnload" **recognize:** Indicates whether recognition is required when inserting the pallet (if recognition is required, please ensure that the recognition model file is configured correctly) **layer:** The number of pallet layers that need to be inserted into the fork (only meaningful when recognize = true) **start_height:** When the action starts, the starting height of the fork (only meaningful when recognize = false) **end_height:** After the action is over, the end height of the fork (only meaningful when recognize = false)

2. An example of sending data area is as follows:

```
3. // recognize = false
4. {
5.   "id": "AP1",
6.   "operation": "ForkLoad",
7.   "layer": 0,
8.   "recognize": true
9. }
10. // recognize = true
11. {
12.   "id": "AP1",
13.   "operation": "ForkUnload",
14.   "start_height": 1.0,
15.   "end_height": 0,
16.   "recognize": false
17. }
```

17. Roller The execution object of the AP station needs to be configured as roller, and the recognition model file does not need to be configured (the roller does not currently have recognition). **operation:** action can be "RollerLoad" or "RollerUnload" **direction:** directions of action for roller, can be "left" or "right" **recognize:** Whether need to recognition or not (For roller, recognition is currently invalid)

18. An example of sending data area is as follows:

```
19. {
20.   "id": "AP1",
21.   "operation": "RollerLoad",
22.   "direction": "left",
23.   "recognize": false // You can also omit the recognize field
24. }
```

24. Jack The execution object of the AP station needs to be configured as jack, and the recognition model file is configured as a shelf model file or not configured as required **operation:** action can be "JackLoad" or "JackUnload" or "JackHeight" **jack_height:** set height for jack (only meaningful when operation = "JackHeight") **recognize:** Represents whether recognition is required when drilling into the shelf (if recognition is required, please ensure that the recognition model file is configured correctly, for the use of recognition the shelf while drilling, please refer to this [tutorial](#)) **use_pgv:** represents whether the PGV sensor is used to correct the position after drilling into the shelf

25. An example of sending data area is as follows:

```
26.{
27."id": "AP1",
28."operation": "JackLoad",    // jack up
29."recognize": false,
30."use_pgv": false
31.}
32.or
33.{
34."id": "AP1",
35."operation": "JackUnload",  // jack down
36."recognize": false,
37."use_pgv": false
38.}
39.or
40.{
41."id": "AP1",
42."operation": "JackHeight",  // set fixed height to jack
43."jack_height": 0.2,
44."recognize": false,
45."use_pgv": false
}
```

46. Hook The execution object of the AP station needs to be configured as hook, and the recognition model file is configured as the material truck model file or not configured as required. **operation:** actoin can be "HookLoad" or "HookUnload" **recognize:** represents whether it needs to be recognized before hooking the material truck

47. An example of sending data area is as follows:

```
48.{
49."id": "AP1",
50."operation": "HookLoad",    // hook load
51."recognize": false
52.}
53.or
54.{
55."id": "AP1",
56."operation": "HookUnload",  // hook down
57."recognize": false
}
```

Response

- API number: 13052 (0x32FC)
- Name: robot_task_gotarget2_res
- Description: Response of path navigation 2
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 32 FC 0B EC 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```


15.1.1.5.7 Get the Path of Path Navigation

The request only returns the path planned using path navigation to the target station, and does not actually perform navigation.

Request

- API number: 3053 (0x0BED)
- Name: robot_task_target_path_req
- Description: Get the path of the path navigation.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
id	string	id of the target station	NO

Request Example

```
5A 01 00 01 00 00 00 0D 0B ED 00 00 00 00 00 00
// path inquiry to AP25 {"id":"AP25"}, and the data area after serialization is:
7B 22 69 64 22 3A 22 41 50 32 35 22 7D
```

Response

- API number: 13053 (0x32FD)

- Name: robot_task_target_path_res
- Description: Response of get the path of the path navigation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
path	array[string]	station on the path in sequence	YES
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
// Get the path to AP25, but don't execute the path navigation.
{
  "path": [
    "AP10",
    "AP11",
    "AP69",
    "AP13",
    "AP65",
    "AP14",
    "AP15",
    "AP16",
    "AP17",
    "LM41",
```

```

    "AP68",
    "AP63",
    "LM38",
    "LM58",
    "AP22",
    "AP23",
    "AP24",
    "AP25"
  ],
  "ret_code": 0
}
5A 01 00 01 00 00 00 95 32 FD 0B ED 00 00 00 00
// data area
7B 22 70 61 74 68 22 3A 5B 22 41 50 31 30 22 2C
22 41 50 31 31 22 2C 22 41 50 36 39 22 2C 22 41
50 31 33 22 2C 22 41 50 36 35 22 2C 22 41 50 31
34 22 2C 22 41 50 31 35 22 2C 22 41 50 31 36 22
2C 22 41 50 31 37 22 2C 22 4C 4D 34 31 22 2C 22
41 50 36 38 22 2C 22 41 50 36 33 22 2C 22 4C 4D
33 38 22 2C 22 4C 4D 35 38 22 2C 22 41 50 32 32
22 2C 22 41 50 32 33 22 2C 22 41 50 32 34 22 2C
22 41 50 32 35 22 5D 2C 22 72 65 74 5F 63 6F 64
65 22 3A 30 7D

```

15.1.1.5.8 Translation

Request

- API number: 3055 (0x0BEF)
- Name: robot_task_translate_req
- Description: Translation; Move a fixed distance in a straight line at a fixed speed.

- JSON data area: See table below.

Note: 3055(Translation), 3056(Rotation), 3057 cannot be performed simultaneously

Field Name	Type	Description	Whether Default
dist	number	Linear movement distance, absolute value, unit: m	NO
vx	number	The movement speed for X direction in the robot coordinate system, positive is forward, negative is backward, unit: m/s	YES
vy	number	The movement speed for Y direction in the robot coordinate system, positive is to the left, negative is to the right, unit: m/s	YES
mode	number	0 = mileage mode (Motion based on mileage), 1 = localization mode, default is mileage mode	YES

If vx and vy both have values, then will use composition of velocities. Mileage mode does not require precise localization, but it may produce large errors for long-distance movements. The error increases with distance. The localization mode requires stable localization in the current environment, and the error is related to the current environment and localization accuracy.

Request Example

```
{"dist":5.0,"vx":0.5,"vy":0.5}
```

```
// move straight for 5m
5A 01 00 01 00 00 00 1E 0B EF 00 00 00 00 00
7B 22 64 69 73 74 22 3A 35 2E 30 2C 22 76 78 22
3A 30 2E 35 2C 22 76 79 22 3A 30 2E 35 7D
```

Response

- API number: 13055 (0x32FF)
- Name: robot_task_translate_res
- Description: Response of translation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 32 FF 0B EF 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.9 Rotation

Request

- API number: 3056 (0x0BF0)
- Name: robot_task_turn_req
- Description: Rotation, rotate a fixed angle at a fixed angular speed
- JSON data area: See table below.

Note: 3055(Translation), 3056(Rotation), 3057 cannot be performed simultaneously

Field Name	Type	Description	Whether Default
angle	number	Rotation angle (robot coordinate system), absolute value, unit: rad, can be greater than 2π	NO
vw	number	Angular speed of rotation (robot coordinate system), positive for counterclockwise rotation, negative for clockwise rotation, unit: rad/s	NO
mode	number	0 = mileage mode (Motion based on mileage), 1 = localization mode, default is mileage mode	YES

The mileage mode does not require precise localization, but for a large rotation angle, a large error may occur, and the error increases as the rotation angle increases. The localization mode requires stable localization in the current environment, and the error is related to the current environment and localization accuracy.

Request Example

```
{"angle":3.14,"vw":1.6}  
// rotate 3.14 rad with rotation speed 1.6rad/s  
5A 01 00 01 00 00 00 17 0B F0 00 00 00 00 00  
7B 22 61 6E 67 6C 65 22 3A 33 2E 31 34 2C 22 76  
77 22 3A 31 2E 36 7D
```

Response

- API number: 13056 (0x3300)
- Name: robot_task_turn_res
- Description: Response of rotation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 33 00 0B F0 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.10 Pallet Rotation

Request

- API number: 3057 (0x0BF1)
- Name: robot_task_spin_req
- Description: Tray rotation
- JSON data area: See table below.

Note 1: 3055(Translation), 3056(Rotation), 3057 cannot be performed simultaneously

Note 2: All angles of tray rotation are in "radians"

Note 3: When the tray is not in following mode, if the jack is in the down state, the robot will automatically reset the tray angle to zero before navigating. If the jack is in the up state, the robot will automatically reset the tray angle to zero or turn to 180 degrees before navigating. This feature can be turned off by setting the parameters AutoSpinToZeroInUnload (down state) and AutoSpinToZero (up state) to false.

Field Name	Type	Description	Whether Default
increase_spin_angle	number	Add an angle to the current tray angle. If the angle is positive, rotate counterclockwise, and if it is negative, rotate clockwise, unit: rad	YES
robot_spin_angle	number	Rotate the angle of the tray to an angle under the robot coordinate system. spin_direction is 0, then turn to the nearest; spin_direction is 1, then turn counterclockwise; spin_direction is -1, then turn clockwise, unit: rad	YES
global_spin_angle	number	Rotate the angle of the tray to an angle in the world coordinate system. spin_direction is 0, then turn to the nearest; spin_direction is 1, then turn counterclockwise; spin_direction is -1, then turn clockwise	YES
spin_direction	number	0 = nearest, 1 = counterclockwise -1 = clockwise	YES

Request Example

```
{"increase_spin_angle":1.57}
// Rotate 1.57 radians counterclockwise
```

```

5A 01 00 01 00 00 00 00 0B F1 00 00 00 00 00 00
7B 22 69 6E 63 72 65 61 73 65 5F 73 70 69 6E 5F 61 6E 67 6C 65 22 3A 31 2E 35 37 7D
{"robot_spin_angle":-1.57,"spin_direction":0.0}
// Turn nearest to -1.57 radians (robot coordinate system)
5A 01 00 01 00 00 00 00 0B F1 00 00 00 00 00 00
7B 22 72 6F 62 6F 74 5F 73 70 69 6E 5F 61 6E 67 6C 65 22 3A 2D 31 2E 35 37 2C 22 73 70 69 6E 5F 64 69 72 65 63 74 69 6F
6E 22 3A 30 2E 30 7D
{"global_spin_angle":-1.57,"spin_direction":0.0}
// Turn nearest to -1.57 radians (world coordinate system)
5A 01 00 01 00 00 00 00 0B F1 00 00 00 00 00 00
7B 22 67 6C 6F 62 61 6C 5F 73 70 69 6E 5F 61 6E 67 6C 65 22 3A 2D 31 2E 35 37 2C 22 73 70 69 6E 5F 64 69 72 65 63 74 69
6F 6E 22 3A 30 2E 30 7D

```

Response

- API number: 13057 (0x3301)
- Name: robot_task_spin_res
- Description: Response of tray rotation
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 33 01 0B F1 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.11 Circular Motion

Request

- API number: 3058 (0x0BF2)
- Name: robot_task_circular_req
- Description: Circular motion
- JSON data area: See table below.

Note: 3055 (translation), 3056 (rotation), 3057 (spin), 3058 cannot be performed simultaneously.

Field Name	Type	Description	Whether Default
rot_radius	number	Rotation radius, unit m; the center of the rotation circle located on the left side of the robot coordinate system is positive, and the right side is negative.	YES
rot_degree	number	Rotation angle, unit °	YES

Field Name	Type	Description	Whether Default
rot_speed	number	Rotation speed, the unit is rad/s, negative number means clockwise.	YES
mode	number	0 = mileage mode (move according to mileage), 1 = positioning mode, the default is the mileage mode.	YES

Request Example

```
{
  "rot_radius": 1,
  "rot_degree": 360,
  "rot_speed": 0.5,
  "mode": 0
}
```

// Rotate 360 degrees with a rotation radius of 1m, 0.5 rad/s

```
5A 01 00 01 00 00 00 00 0B F1 00 00 00 00 00 00
7B 22 72 6F 74 5F 72 61 64 69 75 73 22 3A 31 2C 22 72 6F 74 5F 64 65 67 72 65 65 22 3A 33 36 30 2C 22 72 6F 74 5F 73 70
65 65 64 22 3A 30 2E 35 2C 20 22 6D 6F 64 65 22 3A 30 7D
```

Response

- API number: 13058 (0x3302)
- Name: robot_task_circular_res
- Description: Response of circular motion
- JSON data area: See table below

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES

Field Name	Type	Description	Whether Default
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 33 02 0B F2 00 00 00 00
// {"ret_code":0}
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.12 Designated Path Navigation

The designated path navigation is to send a set of station sequences to the robot. The robot will navigate according to this sequence (no longer autonomously plan the route), and will not stop when pass through the intermediate station.

[Robokit](#) 3.3 and above version must contain three parameters: "source_id" represents the starting point of the path, "id" represents the end of the path, and "task_id" represents the unique number of the task.

The sequence of stations in the specified path must be continuous stations in the map. There must be no gaps between the stations, nor can it contain disconnected stations, otherwise the robot will report an error.

Request

- API number: 3066 (0x0BFA)
- Name: robot_task_gotargetlist_req
- Description: Designated path navigation
- JSON data area: See below

Refer to Path Navigation [3051](#), where task_id, id, and source_id are required fields and cannot be omitted.

Request Example

```
// path: LM1 -> LM2 -> AP1 (then execute JackHeight)
{
  "move_task_list": [
    {
      "id": "LM2",
      "source_id": "LM1",
      "task_id": "12344321"
    },
    {
      "id": "AP1",
      "source_id": "LM2",
      "task_id": "12344322",
      "operation": "JackHeight",
      "jack_height": 0.2
    }
  ]
}
```

```
}  
// path: LM1 -> LM2 -> AP1 (then execute JackHeight)  
{  
  "move_task_list": [  
    {  
      "id": "LM2"  
    },  
    {  
      "id": "AP1",  
      "operation": "JackHeight",  
      "jack_height": 0.2  
    }  
  ]  
}
```

Response

- API number: 13066 (0x330A)
- Name: robot_task_gotargetlist_res
- Description: Response of designated path navigation
- JSON data area: See table below

Response Example

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 33 0A 0B FA 00 00 00 00  
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.13 Clear Designated Path Navigation

Clear the station sequence sent by the designated path navigation [3066], and the robot will stop moving.

Request

- API number: 3067 (0x0BFB)
- Name: robot_task_cleartargetlist_req

- Description: Clear designated path navigation.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0B FB 00 00 00 00 00 00
```

Response

- API number: 13067 (0x330B)
- Name: robot_task_cleartargetlist_res
- Description: Response of clear designated path navigation
- JSON data area: See table below

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
{  
  "ret_code": 0
```

```
}
5A 01 00 01 00 00 00 00 0B FB 00 00 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

15.1.1.5.15 Execute Pre-Stored Tasks

Request

- API number: 3106 (0x0C22)
- Name: robot_tasklist_name_req
- Description: Execute pre-stored tasks
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
name	string	The name of the pre-stored tasks (uploaded with Roboshop)	NO

Request Example

```
{"name": "task_666"}
// Execute the task_666 tasklist which pre-stored in robot
5A 01 00 01 00 00 00 13 0C 22 00 00 00 00 00 00
7B 22 6E 61 6D 65 22 3A 22 74 61 73 6B 5F 36 36
36 22 7D
```

Response

- API number: 13106 (0x3332)
- Name: robot_tasklist_name_res
- Description: Response of execute pre-stored tasks
- JSON data area: See table below

Field Name	Type	Description	Description
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 33 32 0C 22 00 00 00 00  
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Robot Configuration API

The robot configuration API is used to configure robots, switch maps, and so on.

API List

Request

Serial Number	Name	Description
4001	robot_config_require_req	Require control
4002	robot_config_release_req	Release control
4003	robot_config_src_require_req	SRC control mode
4004	robot_config_src_release_req	SRC monitor mode
4005	robot_config_lock_req	Release control

Serial Number	Name	Description
4006	robot_config_unlock_req	Release control
4009	robot_config_clearallerrors_req	Clear all current errors of the robot
4010	robot_config_uploadmap_req	Upload map to robot
4011	robot_config_downloadmap_req	Download map from robot
4012	robot_config_removemap_req	Delete map on robot
4091	robot_config_push_req	Configure robot push port
4130	robot_config_ultrasonic_req	Ultrasonic configuration
4140	robot_config_DI_req	Configure DI
4150	robot_config_motor_calib_req	Motor calibration
4151	robot_config_motor_clear_fault_req	Motor clearing error

Serial Number	Name	Description
4350	robot_config_addobstacle_req	Insert dynamic obstacles (robot coordinate system)
4351	robot_config_addgobstacle_req	Insert dynamic obstacle (global coordinate system)
4352	robot_config_removeobstacle_req	Remove dynamic obstacles
4356	robot_config_clear_goodsshape_req	Clear shelf description file
4357	robot_config_set_shelfshape_req	Set shelf description file
4800	robot_config_seterror_req	Set third-party error
4801	robot_config_clearerror_req	Clear third-party error
4802	robot_config_setwarning_req	Set up third-party warning
4803	robot_config_clearwarning_req	Clear third-party warning
...	...	...

Response

Serial Number	Name	Description
14001	robot_config_require_res	Response to 4001 request
14002	robot_config_release_res	Response to 4002 request
14003	robot_config_src_require_res	Response to 4003 request
14004	robot_config_src_release_res	Response to 4004 request
14005	robot_config_lock_res	Response to 4005 request
14006	robot_config_unlock_res	Response to 4006 request
14009	robot_config_clearallerrors_res	Response to 4009 request
14010	robot_config_uploadmap_res	Response to 4010 request
14011	robot_config_downloadmap_res	Response to 4011 request

Serial Number	Name	Description
14012	robot_config_removemap_res	Response to 4012 request
14091	robot_config_push_res	Response to 4091 request
14130	robot_config_ultrasonic_res	Response to 4130 request
14140	robot_config_DI_res	Response to 4140 request
14150	robot_config_motor_calib_res	Response to 4150 request
14151	robot_config_motor_clear_fault_res	Response to 4151 request
14350	robot_config_addobstacle_res	Response to 4350 request
14351	robot_config_addgobstacle_res	Response to 4351 request
14352	robot_config_removeobstacle_res	Response to 4352 request
14356	robot_config_clear_goodsshape_res	Response to 4356 request

Serial Number	Name	Description
14357	robot_config_set_shelfshape_res	Response to 4357 request
14800	robot_config_seterror_res	Response to 4800 request
14801	robot_config_clearerror_res	Response to 4801 request
14802	robot_config_setwarning_res	Response to 4802 request
14803	robot_config_clearwarning_res	Response to 4803 request
...	...	...

- [API List](#)
- [Request](#)
- [Response](#)

Require control

Require control from the dispatch system.

After the control is required, any navigation in progress currently will be canceled.

Request

- API number: 4001 (0x0FA1)
- Name: robot_config_require_req
- Description: Require control.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0F A1 00 00 00 00 00 00
```

Response

- API number: 14001 (0x36B1)
- Name: robot_config_require_res
- Description: Response of requiring control.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 36 B1 0F A1 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Release Control

Releases control to the dispatching system, and it can only be enabled when the location status is success. When the control is successfully released to the dispatch system, the robot will not respond to any commands except status query.

After the control is released, any navigation in progress currently will be canceled.

Request

- API number: 4002 (0x0FA2)
- Name: robot_config_release_req
- Description: Release control
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0F A2 00 00 00 00 00 00
```

Response

- API number: 14002 (0x36B2)
- Name: robot_config_release_res
- Description: Response of release control
- JSON data area: See table below.

Field name	Type	Description	Whether Default
success	boolean	false = fail, true = success	NO
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "ret_code": 0,
  "success": true
}
```

```
5A 01 00 01 00 00 00 1D 36 B2 0F A2 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 2C 22 73
75 63 63 65 73 73 22 3A 74 72 75 65 7D
```

Preempt Control

Preempt control

Request

- API number: 4005 (0x0FA5)
- Name: robot_config_lock_req
- Description: Preempt control
- JSON data area: See table below.

Field name	Type	Description	Whether Default
nick_name	String	Control preemptor name	NO

Request Example

```
{“nick_name”:“srd-seer-mizhan”}  
// preemptor name is srd-seer-mizhan  
5A 01 00 01 00 00 00 1F 0F A5 00 00 00 00 00 00  
7B 22 6E 69 63 6B 5F 6E 61 6D 65 22 3A 22 73 72  
64 2D 73 65 65 72 2D 6D 69 7A 68 61 6E 22 7D
```

Response

- API number: 14005 (0x36B5)
- Name: robot_config_lock_res
- Description: Response of preempt control
- JSON data area: See table below

Field name	Type	Description	Whether default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 36 B5 0F A5 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Release Control

You can only release the control that you have preempted, and you cannot release the control of others. If you want to gain control, you must preempt it.

Request

- API number: 4006 (0x0FA6)
- Name: robot_config_unlock_req

- Description: Release control.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0F A6 00 00 00 00 00 00
```

Response

- API number: 14006 (0x36B6)
- Name: robot_config_unlock_res
- Description: Response of releasing control
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "ret_code": 0
```



```
}
```

SRC Control Mode

Sets [SRC](#) to control mode, then no other controller can control the robot.

Request

- API number: 4003 (0x0FA3)
- Name: robot_config_src_require_req
- Description: [SRC](#) control mode
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0F A3 00 00 00 00 00 00
```

Response

- API number: 14003 (0x36B3)
- Name: robot_config_src_require_res
- Description: Response of [SRC](#) control mode
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 36 B3 0F A3 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

SRC Monitor Mode

Sets [SRC](#) to monitor mode, then the robot can be controlled by other controllers.

Request

- API number: 4004 (0x0FA4)
- Name: robot_config_src_release_req
- Description: [SRC](#) monitor mode

- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0F A4 00 00 00 00 00 00
```

Response

- API number: 14004 (0x36B4)
- Name: robot_config_src_release_res
- Description: Response of [SRC](#) monitor mode
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 36 B4 0F A4 00 00 00 00
```

7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Push Map to Robot

Request

- API number: 4010 (0x0FAA)
- Name: robot_config_uploadmap_req
- Description: Push map to robot.
- JSON data area: See it below.

The content in the data area is the entire json format map. For details, see map formats. [Appendix C: Map format](#)

Request Example

Omitted

Response

- API number: 14010 (0x36BA)
- Name: robot_config_uploadmap_res
- Description: Response of push map to robot
- JSON data area: See table below

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

Omitted.

Pull Map from Robot

Request

- API number: 4011 (0x0FAB)
- Name: robot_config_downloadmap_req
- Description: Pull map from robot.
- JSON data area: See table below.

Field name	Type	Description	Whether Default
map_name	String	Map name (Contains only 0-9, a-z, A-Z, -, _ instead of illegal characters such as Chinese characters.)	NO

Request Example

```

{"map_name":"default"}
// inquire the default map stored in the robot
5A 01 00 00 00 00 16 0F AB 00 00 00 00 00 00
7B 22 6D 61 70 5F 6E 61 6D 65 22 3A 22 64 65 66
61 75 6C 74 22 7D

```

Response

- API number: 14011 (0x36BB)
- Name: robot_config_downloadmap_res
- Description: Response of Pull map from robot
- JSON data area: See below

If no error occurs, the content in the data area is the entire map in json format. [Appendix C: Map format](#)

If an error occurs, the data area is as shown in the table below:

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "err_msg": "no this map file",
  "ret_code": 40051
}
```

```
5A 01 00 01 00 00 00 2F 36 BB 0F AB 00 00 00 00
7B 22 65 72 72 5F 6D 73 67 22 3A 22 6E 6F 20 74
68 69 73 20 6D 61 70 20 66 69 6C 65 22 2C 22 72
65 74 5F 63 6F 64 65 22 3A 34 30 30 35 31 7D
```

Delete Map in Robot

Request

- API number: 4012 (0x0FAC)
- Name: robot_config_removemap_req
- Description: Delete map in robot

- JSON data area: See table below.

Field name	Type	Description
map_name	String	Map name (cannot contain illegal characters such as Chinese, can only use 0-9, a-z, A-Z, -, _)

Request Example

```
{"map_name": "20210126173805395"}  
// Remove the map named 20210126173805395 in robot  
5A 01 00 01 00 00 00 20 0F AC 00 00 00 00 00 00  
7B 22 6D 61 70 5F 6E 61 6D 65 22 3A 22 32 30 32  
31 30 31 32 36 31 37 33 38 30 35 33 39 35 22 7D
```

Response

- API number: 14012 (0x36BC)
- Name: robot_config_removemap_res
- Description: Response of deleting map in robot
- JSON data area: See table below.

Field name	Type	Description	Whether Default
ret_code	Number	API error code	YES

Field name	Type	Description	Whether Default
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 36 BC 0F AC 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Configure Ultrasonic

Request

- API number: 4130 (0x1022)
- Name: robot_config_ultrasonic_req
- Description: Configure ultrasonic.
- JSON data area: See table below.

Configure whether the ultrasonic of a certain ID is enabled.

Field Name	Type	Description	Whether Default
id	Number	ID of ultrasonic	NO
valid	Boolean	Whether to enable, false = disable, true = enable	NO

Request example

```
5A 01 00 01 00 00 00 16 10 22 00 00 00 00 00
{"id":0,"valid":false}
// The hexadecimal after serialization is:
7B 22 69 64 22 3A 30 2C 22 76 61 6C 69 64 22 3A
66 61 6C 73 65 7D
```

Response

- API number: 14130 (0x3732)
- Name: robot_config_ultrasonic_res
- Description: Response of configuring ultrasonic
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 37 32 10 22 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Configure DI

Request

- API number: 4140 (0x102C)
- Name: robot_config_DI_req
- Description: Config DI
- JSON data area: See table below.

Configure whether an DI is enabled.

Field name	Type	Description	Whether default
id	Number	id of the DI	NO
valid	Boolean	Whether to enable, false = disable, true = enable	NO

Request Example

Omitted

Response

- API number: 14140 (0x373C)
- Name: robot_config_DI_res

- Description: Response of config DI
- JSON data area: See table below.

Field name	Type	Description	Whether default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response example

Omitted

Motor Calibration

Request

- API number: 4150 (0x1036)
- Name: robot_config_motor_calib_req
- Description: Motor Calibration
- JSON data area: See table below.

Currently this command support spin motors only.

Field name	Type	Description	Whether Default
motor_name	string	Motor name (in robot model)	NO

Request Example

```
{ "motor_name": "motor1" }  
// motor1  
5A 01 00 01 00 00 00 17 10 36 00 00 00 00 00  
7B 22 6D 6F 74 6F 72 5F 6E 61 6D 65 22 3A 22 6D  
6F 74 6F 72 31 22 7D
```

Response

- API number: 14150 (0x36E2)
- Name: robot_config_motor_calib_res
- Description: Response of motor calibration
- JSON data area: See table below.

Field name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES

Field name	Type	Description	Whether Default
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 37 46 10 36 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Motor Clear Fault

Request

- API number: 4151 (0x1037)
- Name: robot_config_motor_clear_fault_req
- Description: Motor clear fault
- JSON data area: See table below.

Field name	Type	Description	Whether Default
motor_name	String	Motor name (in the robot model)	NO

Request Example

```

{"motor_name": "motor1"}
// motor1
5A 01 00 01 00 00 00 17 10 37 00 00 00 00 00 00
7B 22 6D 6F 74 6F 72 5F 6E 61 6D 65 22 3A 22 6D
6F 74 6F 72 31 22 7D

```

Response

- API number: 14151 (0x36E3)
- Name: robot_config_motor_clear_fault_res
- Description: Response of motor clear fault
- JSON data area: See table below.

Field name	Type	Description	Whether Default
ret_code	Number	API error code	YES

Field name	Type	Description	Whether Default
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 37 47 10 37 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Add Dynamic Obstacles (Robot Coordinate System)

Request

- API number: 4350 (0x10FE)
- Name: robot_config_addobstacle_req
- Description: Add dynamic obstacles (robot coordinate system).
- JSON data area: See table below.

Add a rectangular dynamic obstacle in the robot coordinate system.

Field name	Type	Description
name	string	Obstacle name, unique
x1	double	The x coordinate of the first vertex of the rectangle (robot coordinate system), unit m
y1	double	The y coordinate of the first vertex of the rectangle (robot coordinate system), unit m
x2	double	The x coordinate of the second vertex of the rectangle (robot coordinate system), unit m
y2	double	The y coordinate of the second vertex of the rectangle (robot coordinate system), unit m
x3	double	The x coordinate of the third vertex of the rectangle (robot coordinate system), unit m
y3	double	The y coordinate of the third vertex of the rectangle (robot coordinate system), unit m
x4	double	The x coordinate of the fourth vertex of the rectangle (robot coordinate system), unit m
y4	double	The y coordinate of the fourth vertex of the rectangle (robot coordinate system), unit m

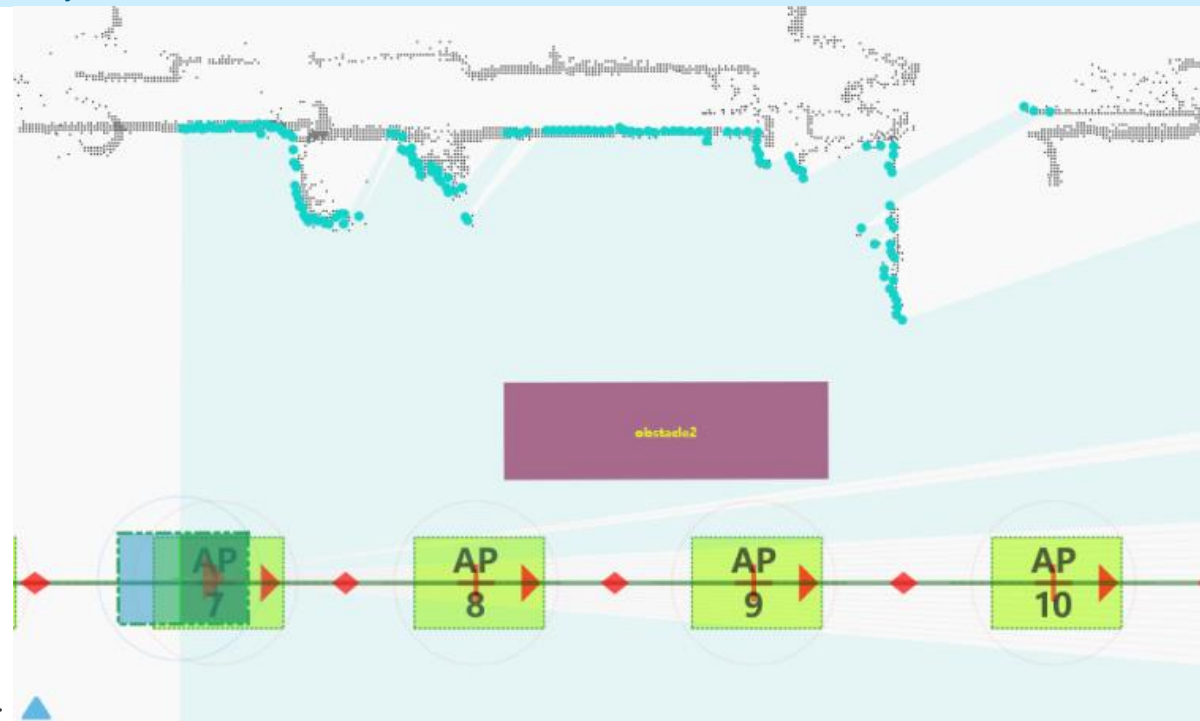
Request Example

```

{"name":"obstacle2","x1":2.0,"y1":0.6,"x2":4.0,"y2":0.6,"x3":4.0,"y3":1.2,"x4":2.0,"y4":1.2}
// Add dynamic obstacle2 in robot coordinate system.
5A 01 00 01 00 00 00 5C 10 FE 00 00 00 00 00
7B 22 6E 61 6D 65 22 3A 22 6F 62 73 74 61 63 6C
65 32 22 2C 22 78 31 22 3A 32 2E 30 2C 22 79 31
22 3A 30 2E 36 2C 22 78 32 22 3A 34 2E 30 2C 22
79 32 22 3A 30 2E 36 2C 22 78 33 22 3A 34 2E 30
2C 22 79 33 22 3A 31 2E 32 2C 22 78 34 22 3A 32
2E 30 2C 22 79 34 22 3A 31 2E 32 7D

```

When the server receives the detailed coordinates of the dynamic obstacle, it will add the obstacle to the corresponding coordinates in the robot coordinate system.



As shown in the following figure:

Response

- API number: 14350 (0x380E)
- Name: robot_config_addobstacle_res
- Description: Response of add dynamic obstacles (robot coordinate system)
- JSON data area: See table below.

Field name	Type	Description	Whether default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 38 0E 10 FE 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Add Dynamic Obstacles (World Coordinate System)

Request

- API number: 4351 (0x10FF)
- Name: robot_config_addgobstacle_req
- Description: Add dynamic obstacles (world coordinate system).
- JSON data area: See table below.

Add a rectangular dynamic obstacle in the global coordinate system.

Field name	Type	Description
name	string	Obstacle name, unique
x1	double	The x coordinate of the first vertex of the rectangle (world coordinate system), unit m
y1	double	The y coordinate of the first vertex of the rectangle (world coordinate system), unit m
x2	double	The x coordinate of the second vertex of the rectangle (world coordinate system), unit m
y2	double	The y coordinate of the second vertex of the rectangle (world coordinate system), unit m
x3	double	The x coordinate of the third vertex of the rectangle (world coordinate system), unit m

Field name	Type	Description
y3	double	The y coordinate of the third vertex of the rectangle (world coordinate system), unit m
x4	double	The x coordinate of the fourth vertex of the rectangle (world coordinate system), unit m
y4	double	The y coordinate of the fourth vertex of the rectangle (world coordinate system), unit m

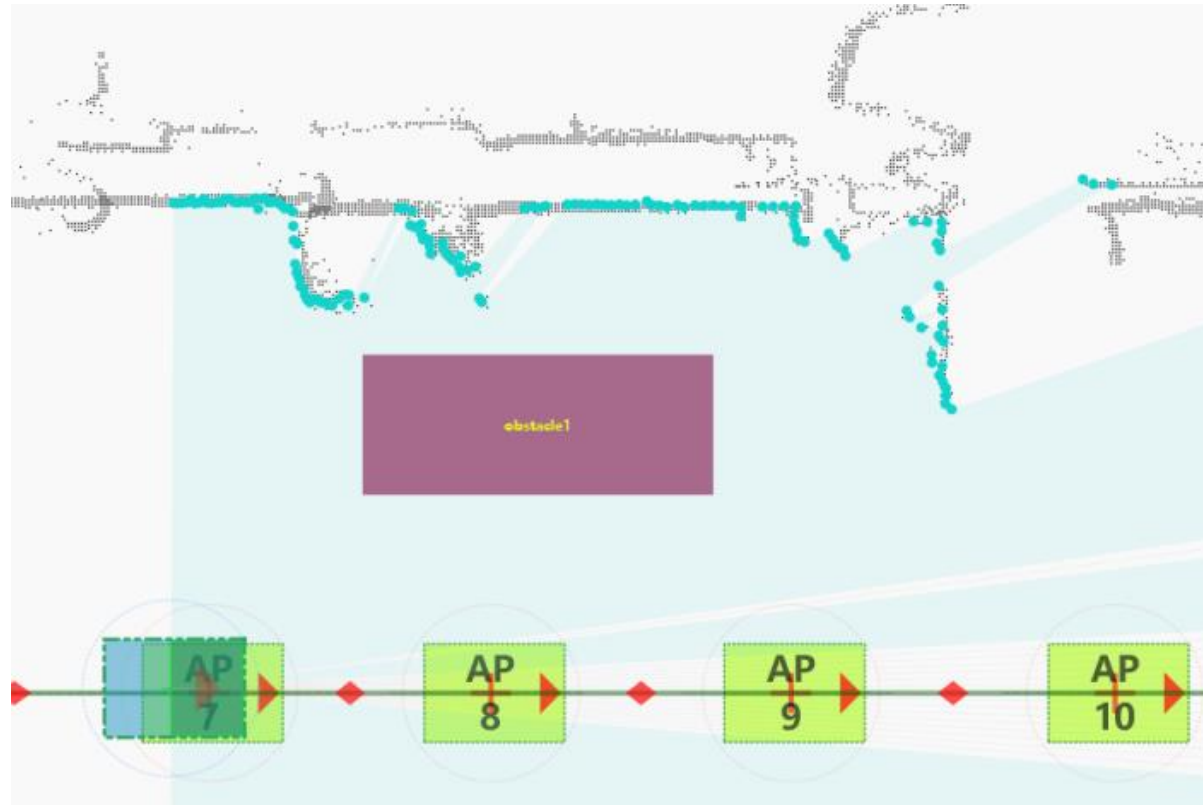
Request Example

```
{
  "name": "obstacle1",
  "x1": 12.0,
  "y1": -3.6,
  "x2": 14.0,
  "y2": -3.6,
  "x3": 14.0,
  "y3": -4.4,
  "x4": 12.0,
  "y4": -4.4
}
```

// Add dynamic obstacle1 in global coordinate system.

```
5A 01 00 01 00 00 00 64 10 FF 00 00 00 00 00 00
7B 22 6E 61 6D 65 22 3A 22 6F 62 73 74 61 63 6C
65 31 22 2C 22 78 31 22 3A 31 32 2E 30 2C 22 79
31 22 3A 2D 33 2E 36 2C 22 78 32 22 3A 31 34 2E
30 2C 22 79 32 22 3A 2D 33 2E 36 2C 22 78 33 22
3A 31 34 2E 30 2C 22 79 33 22 3A 2D 34 2E 34 2C
22 78 34 22 3A 31 32 2E 30 2C 22 79 34 22 3A 2D
34 2E 34 7D
```

When the server receives the detailed coordinates of the dynamic obstacle, it will add the obstacle to the corresponding coordinates in the global coordinate system.



As shown in the following figure:

Response

- API number: 14351 (0x380F)
- Name: robot_config_addgobstacle_res
- Description: Response of add dynamic obstacles (global coordinate system)
- JSON data area: See table below.

Field name	Type	Description	Whether default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 38 0F 10 FF 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Remove Dynamic Obstacles

Request

- API number: 4352 (0x1100)
- Name: robot_config_removeobstacle_req
- Description: Remove dynamic obstacles.
- JSON data area: See table below.

Field name	Type	Description
name	string	The name of the inserted obstacle, if default, remove all the inserted dynamic obstacles.

Request Example

```
{ "name": "1" }
// Remove the dynamic obstacle named "1".
5A 01 00 01 00 00 00 0C 11 00 00 00 00 00 00 00
7B 22 6E 61 6D 65 22 3A 22 31 22 7D
```

Response

- API number: 14352 (0x3810)
- Name: robot_config_removeobstacle_res
- Description: Response of removing dynamic obstacles.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES

Field Name	Type	Description	Whether Default
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 38 10 11 00 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Clear Shelf Description File

Request

- API number: 4356 (0x1104)
- Name: robot_config_clear_goodsshape_req
- Description: Clear shelf description file.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 11 04 00 00 00 00 00 00
```

Response

- API number: 14356 (0x3814)
- Name: robot_config_clear_goodsshape_res
- Description: Response of clear shelf description file
- JSON data area: See table below.

Field name	Type	Description	Whether default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	error message	YES

Response Example

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 38 14 11 04 00 00 00 00  
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Set Shelf Description File

Request

- API number: 4357 (0x1105)
- Name: robot_config_set_shelfshape_req
- Description: Set shelf description file
- JSON data area: See table below.

Field Name	Type	Description
object_path	string	Relative path of shelf description file (Contains only 0-9, a-z, A-Z, -, _ instead of illegal characters such as Chinese characters.)

Request Example

```
5A 01 00 00 00 00 00 23 11 05 00 00 00 00 00 00
{"object_path":"shelf/s0002.shelf"}
// The hexadecimal after serialization is:
7B 22 6F 62 6A 65 63 74 5F 70 61 74 68 22 3A 22
73 68 65 6C 66 2F 73 30 30 30 32 2E 73 68 65 6C
66 22 7D
```

Response

- API number: 14357 (0x3815)
- Name: robot_config_set_shelfshape_res
- Description: Response of setting shelf description file.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 38 15 11 05 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Set Third-Party Error

After setting, an error code of 52900 will be generated, the robot stops moving, and Roboshop can see the following alarm: "an error" refers to the content passed in msg.

Request

- API number: 4800 (0x12C0)
- Name: robot_config_seterror_req
- Description: Set third-party error.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
msg	string	Description of Error, English only	NO

Request Example

```
5A 01 00 01 00 00 00 2E 12 C0 00 00 00 00 00 00
{"msg": "an error from netprotocol tcp/ip api"}
// The hexadecimal after serialization is:
7B 22 6D 73 67 22 3A 22 61 6E 20 65 72 72 6F 72
20 66 72 6F 6D 20 6E 65 74 70 72 6F 74 6F 63 6F
6C 20 74 63 70 2F 69 70 20 61 70 69 22 7D
```

Response

- API number: 14800 (0x39D0)

- Name: robot_config_seterror_res
- Description: Response of setting third-party error.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 39 D0 12 C0 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Clear Third-Party Error

Clear the error code of 52900.

Request

- API number: 4801 (0x12C1)
- Name: robot_config_clearerror_req
- Description: Clear third-party error.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 12 C1 00 00 00 00 00 00
```

Response

- API number: 14801 (0x39D1)
- Name: robot_config_clearerror_res
- Description: Response of clear third party Error
- JSON data area: See table below.

Field name	Type	Description	Whether default
ret_code	Number	API error code	YES

Field name	Type	Description	Whether default
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 39 D1 12 C1 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Set Third-Party Warning

After setting, warning code of 54900 will be generated, and Roboshop can see the following alarm: "a warning" refers to the content passed in msg.

Request

- API number: 4802 (0x12C2)
- Name: robot_config_setwarning_req
- Description: Set third-party warning.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
msg	String	Description of Warning, English only	NO

Request Example

```
5A 01 00 01 00 00 00 2F 12 C2 00 00 00 00 00 00
{"msg": "a warning from netprotocol tcp/ip api"}
// The hexadecimal after serialization is:
7B 22 6D 73 67 22 3A 22 61 20 77 61 72 6E 69 6E
67 20 66 72 6F 6D 20 6E 65 74 70 72 6F 74 6F 63
6F 6C 20 74 63 70 2F 69 70 20 61 70 69 22 7D
```

Response

- API number: 14802 (0x39D2)
- Name: robot_config_setwarning_res
- Description: Response of setting third-party warning
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 39 D2 12 C2 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Clear Third-Party Warning

Clear the Warning code of 54900.

Request

- API number: 4803 (0x12C3)

- Name: robot_config_clearwarning_req
- Description: Clear third-party warning.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 12 C3 00 00 00 00 00 00
```

Response

- API number: 14803 (0x39D3)
- Name: robot_config_clearwarning_res
- Description: Response of clear third-party warning.
- JSON data area: See table below

Field name	Type	Description	Whether default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
```

```
"ret_code": 0
}
5A 01 00 01 00 00 00 0E 39 D3 12 C3 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Clear Robot's All Errors

Request

- API number: 4009 (0x0FA9)
- Name: robot_config_clearallerrors_req
- Description: Clear robot's all errors.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 0F A9 00 00 00 00 00 00
```

Response

- API number: 14009 (0x36B9)
- Name: robot_config_clearallerrors_res
- Description: Response of clear robot's all errors
- JSON data area: See table below.

Field name	Type	Description	Whether default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 36 B9 0F A9 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Configure Robot Push Port

Request

- API number: 4091 (0x0FFB)
- Name: robot_config_push_req
- Description: Configure robot push port 19301.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
interval	Interger	Message push interval (ms)	YES
included_fields	Array[string]	Set the fields included in the message.	YES
excluded_fields	Array[string]	Set excluded fields in messages.	YES

include_fields and exclude_fields cannot be set at the same time

Request Example

```
// Set interval and include_fields at the same time.
{"interval":2000,"included_fields":["vx","vy"]}
5A 01 00 01 00 00 00 2F 0F FB 00 00 00 00 00 00
7B 22 69 6E 74 65 72 76 61 6C 22 3A 32 30 30 30
2C 22 69 6E 63 6C 75 64 65 64 5F 66 69 65 6C 64
73 22 3A 5B 22 76 78 22 2C 22 76 79 22 5D 7D
```

Response

- API number: 14091 (0x370B)
- Name: robot_config_push_res
- Description: Response of configuring robot push port
- JSON data area: See table below

Field name	Type	Description	Whether default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 37 0B 0F FB 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Other APIs

Other APIs include various miscellaneous items, such as play audio, DO control, fork control, roller control, etc.

API List

Request

Serial Number	Name	Description
6000	robot_other_play_audio_req	Play audio
6001	robot_other_setdo_req	Set the DO
6002	robot_other_setdos_req	Batch Set DO
6003	robot_other_setrelay_req	Set relay
6004	robot_other_softemc_req	Soft emergency stop
6005	robot_other_setchargingrelay_req	Set charging relay
6010	robot_other_pause_audio_req	Pause the audio
6011	robot_other_resume_audio_req	Resume the audio

Serial Number	Name	Description
6012	robot_other_stop_audio_req	Stop audio
6020	robot_other_setvdi_req	Set virtual DI
6033	robot_other_audio_list_req	Get audio file list
6040	robot_other_set_fork_height_req	Set fork height
6041	robot_other_stop_fork_req	Stop fork
6042	robot_other_set_fork_forward_req	Set the forks forward or backward
6049	robot_other_write_peripheral_data_req	Write peripherals defined data
6051	robot_other_roller_front_roll_req	Roller front roll
6052	robot_other_roller_back_roll_req	Roller back roll
6053	robot_other_roller_left_roll_req	Roller left roll

Serial Number	Name	Description
6054	robot_other_roller_right_roll_req	Roller right roll
6055	robot_other_roller_front_load_req	Roller front load
6056	robot_other_roller_front_unload_req	Roller front unload
6057	robot_other_roller_front_pre_load_req	Roller front pre-load
6058	robot_other_roller_back_load_req	Roller back load
6059	robot_other_roller_back_unload_req	Roller back unload
6060	robot_other_roller_back_pre_load_req	Roller back pre-load
6061	robot_other_roller_left_load_req	Roller left load
6062	robot_other_roller_left_unload_req	Roller left unload
6063	robot_other_roller_right_load_req	Roller right load

Serial Number	Name	Description
6064	robot_other_roller_right_unload_req	Roller right unload
6065	robot_other_roller_left_pre_load_req	Roller left pre-load
6066	robot_other_roller_right_pre_load_req	Roller right pre-load
6067	robot_other_roller_stop_req	Stop roller
6068	robot_other_roller_left_right_inverse_req	Roller left and right reverse
6069	robot_other_roller_front_back_inverse_req	Roller front and back reverse
6070	robot_other_jack_load_req	Jacking load
6071	robot_other_jack_unload_req	Jacking unload
6072	robot_other_jack_stop_req	Jacking stop
6073	robot_other_jack_set_height_req	Jacking set height

Serial Number	Name	Description
6080	robot_other_reset_cargo_req	Reset cargo
6082	robot_other_hook_load_req	Hook load
6083	robot_other_hook_unload_req	Hook unload
6100	robot_other_slam_req	Start scanning the map
6101	robot_other_endslam_req	Stop scanning the map
6200	robot_other_set_motors_req	Motor debugging
6201	robot_other_set_motor_enable_req	Motor enabling and disabling
...	...	...

Response

Serial Number	Name	Description
16000	robot_other_play_audio_res	Response to 6000 request
16001	robot_other_setdo_res	Response to 6001 request
16002	robot_other_setdos_res	Response to 6002 request
16003	robot_other_setrelay_res	Response to 6003 request
16004	robot_other_softemc_res	Response to 6004 request
16005	robot_other_setchargingrelay_res	Response to 6005 request
16010	robot_other_pause_audio_res	Response to 6010 request
16011	robot_other_resume_audio_res	Response to 6011 request
16012	robot_other_stop_audio_res	Response to 6012 request

Serial Number	Name	Description
16020	robot_other_setvdi_res	Response to 6020 request
16033	robot_other_audio_list_res	Response to 6033 request
16040	robot_other_set_fork_height_res	Response to 6040 request
16041	robot_other_stop_fork_res	Response to 6041 request
16042	robot_other_set_fork_forward_res	Response to 6042 request
16049	robot_other_write_peripheral_data_res	Response to 6049 request
16051	robot_other_roller_front_roll_res	Response to 6051 request
16052	robot_other_roller_back_roll_res	Response to 6052 request
16053	robot_other_roller_left_roll_res	Response to 6053 request
16054	robot_other_roller_right_roll_res	Response to 6054 request

Serial Number	Name	Description
16055	robot_other_roller_front_load_res	Response to 6055 request
16056	robot_other_roller_front_unload_res	Response to 6056 request
16057	robot_other_roller_front_pre_load_res	Response to 6057 request
16058	robot_other_roller_back_load_res	Response to 6058 request
16059	robot_other_roller_back_unload_res	Response to 6059 request
16060	robot_other_roller_back_pre_load_res	Response to 6060 request
16061	robot_other_roller_left_load_res	Response to 6061 request
16062	robot_other_roller_left_unload_res	Response to 6062 request
16063	robot_other_roller_right_load_res	Response to 6063 request
16064	robot_other_roller_right_unload_res	Response to 6064 request

Serial Number	Name	Description
16065	robot_other_roller_left_pre_load_res	Response to 6065 request
16066	robot_other_roller_right_pre_load_res	Response to 6066 request
16067	robot_other_roller_stop_res	Response to 6067 request
16068	robot_other_roller_left_right_inverse_res	Response to 6068 request
16069	robot_other_roller_front_back_inverse_res	Response to 6069 request
16070	robot_other_jack_load_res	Response to 6070 request
16071	robot_other_jack_unload_res	Response to 6071 request
16072	robot_other_jack_stop_res	Response to 6072 request
16073	robot_other_jack_set_height_res	Response to 6073 request
16080	robot_other_reset_cargo_res	Response to 6080 request

Serial Number	Name	Description
16082	robot_other_hook_load_res	Response to 6082 request
16083	robot_other_hook_unload_res	Response to 6083 request
16100	robot_other_slam_res	Response to 6100 request
16101	robot_other_endslam_res	Response to 6101 request
16200	robot_other_set_motors_res	Response to 6200 request
16201	robot_other_set_motor_enable_res	Response to 6201 request
...	...	...

- [API List](#)
- [Request](#)

Play Audio

Request

- API number: 6000 (0x1770)
- Name: robot_other_speaker_req
- Description: Play audio and specified audio file.
- JSON data area: See table below.

Field Name	Type	Description
name	String	Audio file name to be played (without file extension)
loop	Boolean	Whether to play in a loop (true = loop, false = no loop, if default, not loop)

Request Example

```
5A 01 00 01 00 00 00 20 17 70 00 00 00 00 00 00
{"name":"collision","loop":true}
// loop play collision
7B 22 6E 61 6D 65 22 3A 22 63 6F 6C 6C 69 73 69
6F 6E 22 2C 22 6C 6F 6F 70 22 3A 74 72 75 65 7D
```

Response

- API number: 16000 (0x3E80)
- Name: robot_other_speaker_res
- Description: Response of playing audio
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E 80 17 70 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Set DO

This API is used to set the DO (Digital Output) status on the robot controller

Request

- API number: 6001 (0x1771)
- Name: robot_other_setdo_req
- Description: Set DO
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
id	Number	id of the DO	NO
status	Boolean	True is high level, and false is low level.	NO

Request Example

```
5A 01 00 01 00 00 00 17 17 71 00 00 00 00 00 00
{"id":3,"status":false}
// After serialization, the hex is:
7B 22 69 64 22 3A 33 2C 22 73 74 61 74 75 73 22
3A 66 61 6C 73 65 7D
```

Response

- API number: 16001 (0x3E81)

- Name: robot_other_setdo_res
- Description: Response of set DO
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

Omitted

Batch Set DO

This API is used to set the DO (Digital Output) status on the robot controller in batches.

Request

- API number: 6002 (0x1772)

- Name: robot_other_setdos_req
- Description: Batch set DO
- JSON data area: See below.

```
// The data area must be a JSON array, each object in the array is the same as in the 6001 instruction
// The following JSON sets DO 0,1,2,3 at the same time, respectively set to high, high, low, low
[{"id":0,"status":true}, {"id":1,"status":true}, {"id":2,"status":false}, {"id":3,"status":false}]
```

Request Example

```
[{"id":0,"status":true}, {"id":1,"status":true}, {"id":2,"status":false}, {"id":3,"status":false}, {"id":4,"status":true}, {"id":5,"status":true}, {"id":6,"status":false}]
```

Response

- API number: 16002 (0x3E82)
- Name: robot_other_setdos_res
- Description: Response of batch set DO
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES

Field Name	Type	Description	Whether Default
err_msg	String	error message	YES

Response Example

Omitted

Set Relay

This API is used to set the relay status on the robot controller.

Request

- API number: 6003 (0x1773)
- Name: robot_other_setrelay_req
- Description: Set relay.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
status	boolean	True is relay pull-in, and false is relay off.	NO

Request Example

```
5A 01 00 01 00 00 00 0F 17 73 00 00 00 00 00 00
{"status":true}
// After serialization, the hex is:
7B 22 73 74 61 74 75 73 22 3A 74 72 75 65 7D
```

Response

- API number: 16003 (0x3E83)
- Name: robot_other_setrelay_res
- Description: Response of setting relay
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E 83 17 73 00 00 00 00
```

7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Soft Estop

This API is used to control whether the robot controller outputs an emergency stop signal through software.

Request

- API number: 6004 (0x1774)
- Name: robot_other_softemc_req
- Description: Soft Estop
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
status	Boolean	True is output Estop, and false is not output Estop.	NO

Request Example

```
5A 01 00 01 00 00 00 0F 17 74 00 00 00 00 00 00
{"status":true}
// After serialization, the hex is:
7B 22 73 74 61 74 75 73 22 3A 74 72 75 65 7D
```

Response

- API number: 16004 (0x3E84)
- Name: robot_other_softemc_res
- Description: Response soft Estop
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E 84 17 74 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Set Charging Relay

This API is used to set the robot charging relay state.

Request

- API number: 6005 (0x1775)
- Name: robot_other_setchargingrelay_req
- Description: Set charging relay
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
status	boolean	true=the charging relay connected false=the charging relay disconnected	NO

Request Example

```
5A 01 00 01 00 00 00 0F 17 75 00 00 00 00 00 00
{"status":true}
// After serialization, the hex is:
7B 22 73 74 61 74 75 73 22 3A 74 72 75 65 7D
```

Response

- API number: 16005 (0x3E85)
- Name: robot_other_setchargingrelay_res
- Description: Response of setting charging relay

- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E 85 17 75 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Pause Audio

Request

- API number: 6010 (0x177A)
- Name: robot_other_pause_audio_req
- Description: Pause audio.

- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 7A 00 00 00 00 00 00
```

Response

- API number: 16010 (0x3E8A)
- Name: robot_other_pause_audio_res
- Description: Response of pause audio
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 3E 8A 17 7A 00 00 00 00
```

7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Resume Audio

Resume from where it was paused.

Request

- API number: 6011 (0x177B)
- Name: robot_other_resume_audio_req
- Description: Resume audio.
- JSON data area: None

Request Example

5A 01 00 01 00 00 00 00 17 7B 00 00 00 00 00 00

Response

- API number: 16011 (0x3E8B)
- Name: robot_other_resume_audio_res
- Description: Response of resuming audio
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	error message	YES

Response Example

```
{  
  "ret_code": 0  
}
```

Stop Audio

End audio directly.

Request

- API number: 6012 (0x177C)
- Name: robot_other_stop_audio_req
- Description: Stop audio
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 7C 00 00 00 00 00 00
```

Response

- API number: 16012 (0x3E8C)
- Name: robot_other_stop_audio_res
- Description: Response of stopping audio
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "ret_code": 0  
}
```

Set Virtual DI

This API is used to set the virtual DI (Virtual Digital Input) status on the robot controller.

Request

- API number: 6020 (0x1784)
- Name: robot_other_setvdi_req
- Description: Set virtual DI.
- JSON data area: See table below.

Field Name	Type	Description
id	Number	The id of the virtual DI (the index of the first virtual DI in the DI list is 0, the range of the id is [0, 7])
status	Boolean	True is high level, and false is low level.

Request Example

```
5A 01 00 01 00 00 00 17 17 84 00 00 00 00 00 00
{"id":1,"status":false}
// After serialization, the hex is:
7B 22 69 64 22 3A 31 2C 22 73 74 61 74 75 73 22
3A 66 61 6C 73 65 7D
```

Response

- API number: 16020 (0x3E94)
- Name: robot_other_setvdi_res
- Description: Response of set virtual DI
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E 94 17 84 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Get Audio File List

Request

- API number: 6033 (0x1791)
- Name: robot_other_audio_list_req
- Description: Get list of audio files.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 91 00 00 00 00 00 00
```

Response

- API number: 16033 (0x3EA1)
- Name: robot_other_audio_list_res
- Description: Response of getting list of audio files
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
audios	Array[string]	An array of all audio file names stored on the robot	NO

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "audios": [  
    "astern",  
    "block",  
    "collision",  
    "error",  
    "lowBattery",  
    "navigation",  
    "reached",  
    "rotate",  
    "taskfinished",  
    "turnleft",  
    "turnright"  
  ],  
  "ret_code": 0  
}
```

```

}
5A 01 00 01 00 00 00 91 3E A1 17 91 00 00 00 00
7B 22 61 75 64 69 6F 73 22 3A 5B 22 61 73 74 65
72 6E 22 2C 22 62 6C 6F 63 6B 22 2C 22 63 6F 6C
6C 69 73 69 6F 6E 22 2C 22 65 72 72 6F 72 22 2C
22 6C 6F 77 42 61 74 74 65 72 79 22 2C 22 6E 61
76 69 67 61 74 69 6F 6E 22 2C 22 72 65 61 63 68
65 64 22 2C 22 72 6F 74 61 74 65 22 2C 22 74 61
73 6B 66 69 6E 69 73 68 65 64 22 2C 22 74 75 72
6E 6C 65 66 74 22 2C 22 74 75 72 6E 72 69 67 68
74 22 5D 2C 22 72 65 74 5F 63 6F 64 65 22 3A 30
7D

```

Set Fork Height

This API is used to set the height of the fork of the forklift. This command will return immediately, so you need to query the `fork_height_in_place` in the 1028 or 1100 command to check whether the fork is in place.

Request

- API number: 6040 (0x1798)
- Name: `robot_other_set_fork_height_req`
- Description: Set fork height.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
height	Number	Fork height, unit: m For pallet track: 1 = Rise; 0 = fell	NO

Request Example

```
5A 01 00 01 00 00 00 0E 17 98 00 00 00 00 00 00
{"height":1.2}
// After serialization, the hex is:
7B 22 68 65 69 67 68 74 22 3A 31 2E 32 7D
```

Response

- API number: 16040 (0x3EA8)
- Name: robot_other_set_fork_height_res
- Description: Response of settig fork height
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES

Field Name	Type	Description	Whether Default
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E A8 17 98 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Stop Fork Motion

This API is used to stop fork motion

Request

- API number: 6041 (0x1799)

- Name: robot_other_stop_fork_req
- Description: Stop fork motion
- JSON data area: None

Response Example

```
5A 01 00 01 00 00 00 00 17 99 00 00 00 00 00 00
```

Response

- API number: 16041 (0x3EA9)
- Name: robot_other_stop_fork_res
- Description: Response of stoppinf fork motion
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
```

```
"ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E A9 17 99 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Set Fork Forward or Backward

This API is used to set the fork of the forklift to move forward and backward, and the command will return immediately, so you need to query the forward_in_place in the 1028 or 1100 command to check whether the fork is in place.

Request

- API number: 6042 (0x179A)
- Name: robot_other_set_fork_forward_req
- Description: Set fork forward or backward
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
forward	Number	① For the case without cable encoder, 2 = move forward, 1 = move backward, 0 = stop ② For the case with cable encoder, a positive number indicates the forward distance, while a negative number indicates the backward distance, unit m	NO

Request Example

```
5A 01 00 00 00 00 00 0F 17 9A 00 00 00 00 00 00
{"forward":1.0}
// After serialization, the hex is:
7B 22 66 6F 72 77 61 72 64 22 3A 31 2E 30 7D
```

Response

- API number: 16042 (0x3EAA)
- Name: robot_other_set_fork_forward_res
- Description: Response of setting fork forward or backward
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E AA 17 9A 00 00 00 00
```

7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Reset Cargo

Clear the load status.

Request

- API number: 6080 (0x17C0)
- Name: robot_other_reset_cargo_req
- Description: Reset cargo.
- JSON data area: None

Request Example

5A 01 00 01 00 00 00 00 17 C0 00 00 00 00 00 00

Response

- API number: 16080 (0x3ED0)
- Name: robot_other_reset_cargo_res
- Description: Response of resetting cargo
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E D0 17 C0 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Write Peripheral User-Defined Data

Request

- API number: 6049 (0x17A1)
- Name: robot_other_write_peripheral_data_req
- Description: Write peripheral user-defined data.

- JSON data area: See table below.

Field Name	Type	Description	Whether Default
id	uint32_t	id, 0~15	NO
data	uint16_t	Register data	NO

Response Example

```
5A 01 00 01 00 00 00 13 17 A1 00 00 00 00 00 00
{"id":0,"data":233}
// After serialization, the hex is:
7B 22 69 64 22 3A 30 2C 22 64 61 74 61 22 3A 32
33 33 7D
```

Response

- API number: 16049 (0x3EB1)
- Name: robot_other_write_peripheral_data_res
- Description: Response of writing peripheral user-defined data
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E B1 17 A1 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Roller Front Roll

When rolling, both sides of the baffle are down.



The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. Rolling will not fail due to timeout, and the roller will keep rolling until receiving a new command.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6051 (0x17A3)
- Name: robot_other_roller_front_roll_req
- Description: Roller rolls forward.
- JSON data area: None

Request Eexample

```
5A 01 00 01 00 00 00 00 17 A3 00 00 00 00 00 00
```

Response

- API number: 16051 (0x3EB3)
- Name: robot_other_roller_front_roll_res
- Description: Response of roller front roll
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E B3 17 A3 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Roller Back Roll

When rolling, both sides of the baffle are down.



The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. Rolling will not fail due to timeout, and the roller will keep rolling until receiving a new command.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6052 (0x17A4)
- Name: robot_other_roller_back_roll_req
- Description: Roller rolls backward.
- JSON data area: None

Request example

```
5A 01 00 01 00 00 00 00 17 A4 00 00 00 00 00 00
```

Response

- API number: 16052 (0x3EB4)
- Name: robot_other_roller_back_roll_res
- Description: Response of roller back roll
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E B4 17 A4 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Roller Left Roll

When rolling, both sides of the baffle are down.



The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. Rolling will not fail due to timeout, and the roller will keep rolling until receiving a new command.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6053 (0x17A5)
- Name: robot_other_roller_left_roll_req
- Description: Roller rolls to the left.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 A5 00 00 00 00 00 00
```

Response

- API number: 16053 (0x3EB5)
- Name: robot_other_roller_left_roll_res
- Description: Response of roller rolls to the left
- JSON data area: See table below

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E B5 17 A5 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Roller Right Roll

When rolling, both sides of the baffle are down.



The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. Rolling will not fail due to timeout, and the roller will keep rolling until receiving a new command.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6054 (0x17A6)
- Name: robot_other_roller_right_roll_req
- Description: Roller rolls to the right.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 A6 00 00 00 00 00 00
```

Response

- API number: 16054 (0x3EB6)
- Name: robot_other_roller_right_roll_res
- Description: Response of roller rolling to the right
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E B6 17 A6 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Roller Front Load

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The loading may fail due to timeout.



Roller motion commands are not accepted during robot movement.

Request

- API number: 6055 (0x17A7)
- Name: robot_other_roller_front_load_req
- Description: Roller front load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 A7 00 00 00 00 00 00
```

Response

- API number: 16055 (0x3EB7)
- Name: robot_other_roller_front_load_res
- Description: Response of roller front load
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 3E B7 17 A7 00 00 00 00  
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller Front Unload

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The unloading may fail due to timeout.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6056 (0x17A8)
- Name: robot_other_roller_front_unload_req
- Description: Roller front unload
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 A8 00 00 00 00 00 00
```

Response

- API number: 16056 (0x3EB8)
- Name: robot_other_roller_front_unload_res
- Description: Response of roller front unload

- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E B8 17 A8 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller Front Pre-Load

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The pre-loading will not fail due to timeout, and the roller will keep rolling until receiving a new command.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6057 (0x17A9)
- Name: robot_other_roller_front_pre_load_req
- Description: Roller front pre-load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 A9 00 00 00 00 00 00
```

Response

- API number: 16057 (0x3EB9)
- Name: robot_other_roller_front_pre_load_res
- Description: Response of roller front pre-load
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E B9 17 A9 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Roller Back Load

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The loading may fail due to timeout.



Roller motion commands are not accepted during robot movement.

Request

- API number: 6058 (0x17AA)
- Name: robot_other_roller_back_load_req
- Description: Roller back load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 AA 00 00 00 00 00 00
```

Response

- API number: 16058 (0x3EBA)
- Name: robot_other_roller_back_load_res
- Description: Response of roller back load
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 3E BA 17 AA 00 00 00 00  
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller Back Unload

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The back unloading may fail due to timeout.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6059 (0x17AB)
- Name: robot_other_roller_back_unload_req
- Description: Roller back unload
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 AB 00 00 00 00 00 00
```

Response

- API number: 16059 (0x3EBB)
- Name: robot_other_roller_back_unload_res
- Description: Response of roller back unload

- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E BB 17 AB 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller Back Pre-Load

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The pre-loading will not fail due to timeout, and the roller will keep rolling until receiving a new command.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6060 (0x17AC)
- Name: robot_other_roller_back_pre_load_req
- Description: Roller back pre-load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 AC 00 00 00 00 00 00
```

Response

- API number: 16060 (0x3EBC)
- Name: robot_other_roller_back_pre_load_res
- Description: Response of roller back pre-load
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES
err_msg	string	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E BC 17 AC 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller left load

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The loading may fail due to timeout.



Roller motion commands are not accepted during robot movement.

Request

- API number: 6061 (0x17AD)
- Name: robot_other_roller_left_load_req
- Description: Roller left load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 AD 00 00 00 00 00 00
```

Response

- API number: 16061 (0x3EBD)
- Name: robot_other_roller_left_load_res
- Description: Response of roller left load
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E BD 17 AD 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller Left Unload

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The unloading may fail due to timeout.

Roller motion commands are not accepted during robot movement

Request

- API number: 6062 (0x17AE)
- Name: robot_other_roller_left_unload_req
- Description: Roller left unload
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 AE 00 00 00 00 00 00
```

Response

- API number: 16062 (0x3EBE)
- Name: robot_other_roller_left_unload_res
- Description: Response of roller left unload

- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E BE 17 AE 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller right load

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The loading may fail due to timeout.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6063 (0x17AF)
- Name: robot_other_roller_right_load_req
- Description: Roller right load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 AF 00 00 00 00 00 00
```

Response

- API number: 16063 (0x3EBF)
- Name: robot_other_roller_right_load_res
- Description: Response of roller right load
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E BF 17 AF 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Roller Right Unload

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The unloading may fail due to timeout.



Roller motion commands are not accepted during robot movement.

Request

- API number: 6064 (0x17B0)
- Name: robot_other_roller_right_unload_req
- Description: Roller right unload
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 B0 00 00 00 00 00 00
```

Response

- API number: 16064 (0x3EC0)
- Name: robot_other_roller_right_unload_res
- Description: Response of roller right unload
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E C0 17 B0 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller Left Pre-Load

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The pre-loading will not fail due to timeout, and the roller will keep rolling until receiving a new command.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6065 (0x17B1)
- Name: robot_other_roller_left_pre_load_req
- Description: Roller left pre-load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 B1 00 00 00 00 00 00
```

Response

- API number: 16065 (0x3EC1)
- Name: robot_other_roller_left_pre_load_res
- Description: Response of roller left pre-load

- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E C1 17 B1 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller Right Pre-Load

The command will return immediately, so you need to check whether the loading is completed by the roller_state in the 1029 or 1100 command. The pre-loading will not fail due to timeout, and the roller will keep rolling until receiving a new command.

Roller motion commands are not accepted during robot movement.

Request

- API number: 6066 (0x17B2)
- Name: robot_other_roller_right_pre_load_req
- Description: Roller right pre-load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 B2 00 00 00 00 00 00
```

Response

- API number: 16066 (0x3EC2)
- Name: robot_other_roller_right_pre_load_res
- Description: Response of roller right pre-load
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E C2 17 B2 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Roller Stop

Request

- API number: 6067 (0x17B3)
- Name: robot_other_roller_stop_req
- Description: Roller stop

- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 B3 00 00 00 00 00 00
```

Response

- API number: 16067 (0x3EC3)
- Name: robot_other_roller_stop_res
- Description: Response of roller stop
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	error message	YES

Response Example

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 3E C3 17 B3 00 00 00 00
```


7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Roller Inverse Left and Right

After using this command to set the inverse , the left and right logic of the roller will be inversed, that is, the left load becomes the right load, and the right load becomes the left load.

Request

- API number: 6068 (0x17B4)
- Name: robot_other_roller_left_right_inverse_req
- Description: Roller inverse left and right
- JSON data area: See table below.

Field name	Type	Description
inverse	Boolean	Whether to inverse , true = inverse , false = not inverse (ie return to normal)

Request Example

```
5A 01 00 01 00 00 00 10 17 B4 00 00 00 00 00 00
{"inverse":true}
// inverse left and right
7B 22 69 6E 76 65 72 73 65 22 3A 74 72 75 65 7D
```

Response

- API number: 16068 (0x3EC4)
- Name: robot_other_roller_left_right_inverse_res
- Description: Response of roller inverse left and right
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 3E C4 17 B4 00 00 00 00  
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Roller Inverse Front and Back

After using this command to set the inverse, the front and back logic of the roller will be inverted, that is, the front load becomes the back load, and the back load becomes the front load.

Request

- API number: 6069 (0x17B5)
- Name: robot_other_roller_front_back_inverse_req
- Description: Roller inverse front and back
- JSON data area: See table below.

Field Name	Type	Description
inverse	boolean	Whether to inverse , true = inverse , false = not inverse (ie return to normal)

Request Example

```
5A 01 00 01 00 00 00 10 17 B5 00 00 00 00 00 00
{"inverse":true}
// inverse front and back
7B 22 69 6E 76 65 72 73 65 22 3A 74 72 75 65 7D
```

Response

- API number: 16069 (0x3EC5)

- Name: robot_other_roller_front_back_inverse_res
- Description: Response of roller inverse front and back
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E C5 17 B5 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Jacking Load

This command will return immediately, so you need to query jack_state in the 1027 or 1100 command.

Request

- API number: 6070 (0x17B6)
- Name: robot_other_jack_load_req
- Description: Jacking load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 B6 00 00 00 00 00 00
```

Response

- API number: 16070 (0x3EC6)
- Name: robot_other_jack_load_res
- Description: Response of jacking load
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES

Field Name	Type	Description	Whether Default
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E C6 17 B6 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Jacking Unload

This commnd will return immediately, so you need to query jack_state in the 1027 or 1100 command.

Request

- API number: 6071 (0x17B7)
- Name: robot_other_jack_unload_req
- Description: Jack unload
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 B7 00 00 00 00 00 00
```

Response

- API number: 16071 (0x3EC7)
- Name: robot_other_jack_unload_res
- Description: Response of jacking unload
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	number	API error code	YES
create_on	string	API upload timestamp	YES

Field Name	Type	Description	Whether Default
err_msg	string	Error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E C7 17 B7 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Jacking Stop

This commnd will return immediately, so you need to query jack_state in the 1027 or 1100 command.

Request

- API number: 6072 (0x17B8)
- Name: robot_other_jack_stop_req
- Description: Jacking stop
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 B8 00 00 00 00 00 00
```


Response

- API number: 16072 (0x3EC8)
- Name: robot_other_jack_stop_res
- Description: Response of jacking stop
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E C8 17 B8 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Jacking Height

This command will return immediately, so you need to query jack_state in the 1027 or 1100 command.

Request

- API number: 6073 (0x17B9)
- Name: robot_other_jack_set_height_req
- Description: Jacking moves to a fixed height.
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
height	Number	Jack height, unit: m	NO

Request Example

```
5A 01 00 01 00 00 00 0E 17 B9 00 00 00 00 00 00
{"height":0.6}
// The hexadecimal after serialization is:
7B 22 68 65 69 67 68 74 22 3A 30 2E 36 7D
```

Response

- API number: 16073 (0x3EC9)
- Name: robot_other_jack_set_height_res

- Description: Response of jack height
- JSON data area: See table below

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "ret_code": 0  
}  
5A 01 00 01 00 00 00 0E 3E C9 17 B9 00 00 00 00  
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Hook Load

Rear hook mechanism grips the carrier.

Request

- API number: 6082 (0x17C2)
- Name: robot_other_hook_load_req
- Description: Hook load
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 C2 00 00 00 00 00 00
```

Response

- API number: 16082 (0x3ED2)
- Name: robot_other_hook_load_res
- Description: Response of hook load
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	error message	YES

Response Example

```
{
  "ret_code": 0
}
5A 01 00 01 00 00 00 0E 3E D2 17 C2 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Hook Unload

Rear hook mechanism releases the carrier.

Request

- API number: 6083 (0x17C3)
- Name: robot_other_hook_unload_req
- Description: Hook unload
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 C3 00 00 00 00 00 00
```

Response

- API number: 16083 (0x3ED3)
- Name: robot_other_hook_unload_res
- Description: Response of hook unload
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E D3 17 C3 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

Begin SLAM

If you want to accept real-time scan data, please listen to UDP port 19301, the data format is binary [rbk.protocol.Message_MapLog](#)

Request

- API number: 6100 (0x17D4)

- Name: robot_other_slam_req
- Description: Begin SLAM
- JSON data area: See table below.

Field Name	Type	Description
real_time	bool	Whether to enable real-time scan data transmission, the default is false
screen_width	number	Screen width (pixels)
screen_height	number	Screen height (pixels)

Request Example

```
5A 01 00 01 00 00 00 12 17 D4 00 00 00 00 00 00
{"real_time":true}
// After serialization, the hex is:
7B 22 72 65 61 6C 5F 74 69 6D 65 22 3A 74 72 75
65 7D
```

Response

- API number: 16100 (0x3EE4)
- Name: robot_other_slam_res
- Description: Response of beginning SLAM
- JSON data area: See table below.

Field Name	Type	Description	Whether default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

5A 01 00 01 00 00 00 0E 3E E4 17 D4 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D

End SLAM

Request

- API number: 6101 (0x17D5)
- Name: robot_other_endslam_req
- Description: End SLAM.
- JSON data area: None

Request Example

```
5A 01 00 01 00 00 00 00 17 D5 00 00 00 00 00 00
```

Response

- API number: 16101 (0x3EE5)
- Name: robot_other_endslam_res
- Description: Response of ending SLAM
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
real_time	Bool	Whether to enable the real-time data transfer	YES
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{  
  "real_time": true,  
  "ret_code": 0
```

```
}  
5A 01 00 01 00 00 00 1F 3E E5 17 D5 00 00 00 00  
7B 22 72 65 61 6C 5F 74 69 6D 65 22 3A 74 72 75  
65 2C 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Motor Enabling and Disabling

Request

- API number: 6201 (0x1839)
- Name: robot_other_set_motor_enable_req
- Description: Motor enabling and disabling
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
motor_name	String	Motor name	NO
enable	Boolean	true=enable; false=disable	NO

Request Example

```
5A 01 00 01 00 00 00 26 18 39 00 00 00 00 00 00
// disable motor1:
{"motor_name":"motor1","enable":false}
// After serialization, the hex is:
7B 22 6D 6F 74 6F 72 5F 6E 61 6D 65 22 3A 22 6D
6F 74 6F 72 31 22 2C 22 65 6E 61 62 6C 65 22 3A
66 61 6C 73 65 7D
```

Response

- API number: 16201 (0x3F49)
- Name: robot_other_set_motor_enable_res
- Description: Response of motor enabling and disabling
- JSON data area: See table below.

Field Name	Type	Description	Whether Default
ret_code	Number	API error code	YES
create_on	String	API upload timestamp	YES
err_msg	String	Error message	YES

Response Example

```
{
  "ret_code": 0
}
```

```
5A 01 00 01 00 00 00 0E 3F 49 18 39 00 00 00 00
7B 22 72 65 74 5F 63 6F 64 65 22 3A 30 7D
```

Robot Push API

The robot push API is used to actively push data to connected clients.

API List

Request

API Number	Name	Description
19301	robot_push	Robot data push
...	...	...

Robot Data Push

This API is used by the robot to actively push data to the connected client.

- API number: 19301 (0x4B65)
- Name: robot_push
- Description: Robot data push
- JSON data area: See table below.

Field Name	Type	Description
controller_temp	number	Controller temperature, unit °C
x	number	The x coordinate of the robot, unit m
y	number	The y coordinate of the robot, unit m
angle	number	Robot's angle, unit rad
current_station	string	The id of the station where the robot is currently located (the robot must be very close to a station (<50cm), otherwise it is a blank character, that is, it is not at any station.)
vx	number	The speed of the robot in the x direction of the robot coordinate system, unit m/s
vy	number	The speed of the robot in the y direction of the robot coordinate system, unit m/s
w	number	The angular velocity of the robot in the robot coordinate system (i.e. clockwise is negative, counterclockwise is positive), unit rad/s
steer	number	The current steering wheel angle of the single steering wheel robot, unit rad
blocked	boolean	Whether the robot is blocked.
battery_level	number	Robot battery level, range [0, 1]
charging	boolean	the battery charging status
emergency	boolean	True means the emergency stop button is active (pressed), and false means the emergency stop button is inactive (pull up).

Field Name	Type	Description
DI	array[Object]	id: id number source: source normal = normal DI virtual = virtual DI modbus = modbus DI status: high or low levels true = high level false = low level valid: Whether the corresponding DI is enabled. true = enable false = disable
DO	array[Object]	id: id number source: source normal = normal DO modbus = modbus DO status: high or low levels true = high level false = low level
fatals	array[Object]	Error Code Fatal array, all Fatal alarms that appear will appear in the data.
errors	array[Object]	Error Code Error array, all Error alarms that appear will appear in the data.
warnings	array[Object]	Alarm Code Warning array, all warnings that appear will appear in the data.

Field Name	Type	Description
notices	array[Object]	Notice array, all notice alarms that appear will appear in the data.
current_map	string	current map name
vehicle_id	string	robot name
create_on	string	Timestamp (2020-10-28T14:29:32.153Z)
requestVoltage	double	Request charging current voltage. These two variables are named after the hump, so don't change them for compatibility with older versions of the program.
requestCurrent	double	Request charging current
brake	boolean	Whether to brake
soft_emc	boolean	Whether it is in soft emergency stop state.
task_status_package	object	Return to the list of the robot's last completed task and all uncompleted tasks (including failed tasks). For details, see API 1110 in code19204.
map	object	Internal (SEED) use, see map userData.
charge_spot	object	Forklift automatic charging pile status data
reloc_status	number	0 = FAILED, 1 = SUCCESS, 2 = RELOCING, 3=COMPLETED

Field Name	Type	Description
loadmap_status	number	0 = FAILED (Load map fails.), 1 = SUCCESS (Map loading succeeds.), 2 = LOADING (Map is loading.)
slam_status	number	0 = No sweep, 1 = Sweeping (offline), 2 = Real-time sweep (SLAM)
robot_note	string	Robot remarks
current_map_md5	string	MD5 value of the current map
current_lock	object	Queries the current control owner, see 1060 .
confidence	number	Location confidence of robot, range [0, 1]
correction_errs	array[number]	
reliabilities	array [number]	
in_forbidden_area	boolean	
loc_state	number	0 = Normal 1 = Skidding 2 = LowConfidence
similarity	number	

Field Name	Type	Description
update_reason	number	0 = None 1 = OdoUpdate 2 = LaserCorrec 3 = LaserThenOdo
battery_temp	number	Temperature of robot battery, unit °C
battery_user_data	string	Battery User-Defined data
battery_cycle	number	Number of battery cycles (It is obtained from battery BMS, so it may not correct.)
voltage	number	Voltage, unit V
current	number	Current, unit A
r_vx	number	The velocity received by the robot in the x direction of the robot coordinate system, unit m/s
r_vy	number	The velocity received by the robot in the y direction of the robot coordinate system, unit m/s
r_w	number	The angular velocity received by the robot in the robot coordinate system (that is, clockwise means negative, while counterclockwise positive), unit rad/s
r_steer	number	Steering wheel angle received by the steering robot, unit rad
r_spin	number	Rotational speed of pallet received by the pallet robot, units rad/s
r_steer_angles	array[number]	The steering angle received by multi-wheel robot, double wheel =[front wheel angle, rear wheel Angle], unit rad

Field Name	Type	Description
motor_steer_angles	array[object]	
di_max_node	number	
do_max_node	number	
tasklist_status	object	Task chain status JSON object, see 3101 .
odo	number	Accumulated mileage, unit m
today_odo	number	Today's total mileage, unit m
time	number	The running time (from the startup time to the current time), unit ms
total_time	number	Cumulative running time, unit ms
spin	number	Robot pallet (if it exists) angle, unit rad
motor_info	array[Object]	Motor status information. For details, see 1040 .
is_stop	boolean	Whether the robot chassis is stationary (walking motor).
detect_skid	boolean	
steer_angles	array[number]	Current steering angle of multi-wheel robot, double wheel =[front wheel angle, rear wheel angle], unit rad
target_x	number	
target_y	number	

Field Name	Type	Description
tracking_status	object	
user_objects	object	Robot obstacle
fork	object	For the status of cargo forks, see 1028 .
move_status_info	string	Additional information about navigation tasks
area_ids	array[string]	An array of id for the regions where the robot is located (Since regions on the map can overlap, the robot may be in more than one region at a time). The array may null.
block_x	number	The x-coordinate of the nearest obstacle, unit m
block_y	number	The y-coordinate of the nearest obstacle, unit m
block_reason	number	Reasons for blocking, 0 = Ultrasonic, 1 = Laser, 2 = Fallingdown, 3 = Collision, 4 = Infrared, 5 = Lock, 6 = Dynamic obstacle, 7 = Virtual laser point, 8 = 3D camera, 9 = Distance sensor, 10 = DI ultrasonic

Field Name	Type	Description
block_id	number	The ID where block occurs. This parameter is valid only when block_Reason = 0, 2, 3, 4, 9, or 10. Add in 3.2.8.0.
slowed	boolean	Whether the robot slows down.
slow_x	number	The x-coordinate of the nearest obstacle position causing deceleration, unit m
slow_y	number	The y-coordinate of the nearest obstacle position causing deceleration, unit m
slow_reason	number	Reasons for slowing, 0 = Ultrasonic, 1 = Laser, 2 = Fallingdown, 3 = Collision, 4 = Infrared, 5 = Lock, 6 = Dynamic obstacle, 7 = Virtual laser point, 8 = 3D camera, 9 = Distance sensor, 10 = DI ultrasonic
slow_id	number	The id where the slow occurs. Only available when the slow_reason = 0, 2, 3, 4, 9, or 10. Add in 3.2.8.0.
task_status	number	0 = NONE, 1 = WAITING (currently impossible),

Field Name	Type	Description
		2 = RUNNING, 3 = SUSPENDED, 4 = COMPLETED, 5 = FAILED, 6 = CANCELED
running_status	object	
task_type	number	Navigation type, 0 = no navigation, 1 = free navigation to any point, 2 = free navigation to site, 3 = path navigation to site, 7 = translational rotation, and 100 = Other.
target_dist	number	
target_id	string	This field is valid only if the task_type value is 2 or 3. When task_status is RUNNING, the site is being visited. When task_status is COMPLETED, the site has been arrived. If task_status is "FAILED", the navigation to the site fails. If task_status is "SUSPENDED", the navigation to the site is SUSPENDED.
target_label	string	
target_point	array[number]	If task_type is 1, task_status is RUNNING; if task_status is RUNNING, task_status is RUNNING; if task_status is RUNNING, task_status is RUNNING. If task_status is COMPLETED, the point has been reached, if task_status is FAILED, the point FAILED, and if task_status is SUSPENDED, the navigation to the point is SUSPENDED
finished_path	array[string]	An array of sites that have been passed on the current navigation path. This field is valid only when task_type is 3 or 4. All the intermediate points that haven't been passed are listed here.

Field Name	Type	Description
unfinished_path	array[string]	An array of sites that have not been passed on the current navigation path. This field is valid only when task_type is 3 or 4. All the intermediate points that haven't been passed are listed here.
roller	object	For the status of rollers, see 1029 .
jack	object	For the status of jacking, see 1027 .
yaw	number	Yaw angle, unit rad
roll	number	Roll angle, unit rad
pitch	number	Pitch angle, unit rad
rfids	array[number]	Ids of all scanned RFID tags. If no RFID tag is detected, the array is empty.
hook	object	Hook related state
pgvs	array[Object]	QR code recognition data. See 1017 for data examples.
peripheral_data	array[number]	User-Defined data in a peripheral
calib_status	object	Robot calibration
driver_emc	boolean	True indicates that the motor driver has an emergency stop. False indicates that the motor driver has not stopped.
src_release	boolean	SRC monitor mode
controller_humi	number	Controller humidity (unit: %)

Field Name	Type	Description
controller_voltage	number	Controller voltage, unit V
manual_charge	boolean	It only indicates whether the robot is connected to the manual charger, but cannot ensure whether the power supply is switched on. When the robot is connected to the manual charger, it cannot move (only supported by SRC-2000).
auto_charge	boolean	It only indicates whether the robot is connected to the automatic charging pile, but cannot ensure whether the power supply is switched on (only supported by SRC-2000).
electric	boolean	True indicates that the relay is on. False indicates that the relay is off.
ssid	string	SSID currently connected to WIFI (The robot needs to support WIFI connection and should have already connected, otherwise it is null.)
rssi	number	Signal strength from 0 to 100 percent currently connected to WIFI (The robot needs to support WIFI connection and should have already connected, otherwise it is 0.)
ap_addr	string	MAC address of the AP currently connected

- [Robot Data Push](#)

API Error Code

The API error code is the error code (`ret_code`) contained in the JSON object in the response data area, used to indicate errors such as unsuccessful execution of the request or request error. If `ret_code` is 0 or default, it means there is no error. If it is not 0, it means that an error has occurred. The error codes are shown in the following table:

The error code will only appear in the `JSON` data in the response data area.

Error code	Name	Description
40000	req_unavailable	Request is not available.
40001	param_missing	Required request parameters are missing.
40002	param_type_error	Request parameter type error
40003	param_illegal	Request parameter is invalid.
40004	mode_error	Running mode error
40005	illegal_map_name	Map name is illegal.
40006	programming_dsp	Firmware is programming.
40007	program_dsp_error	Firmware programming error
40008	illegal_filename	File name is illegal.
40010	shutdown_error	Shutdown command error

Error code	Name	Description
40011	reboot_error	Restart command error
40012	dispatching	It is controlled by dispatching system.
40013	robod_error	Robod is wrong.
40014	robod_warning	Robod warning error
40015	manual_charging	It is charging manually and can't move.
40016	emc_status	It is in the state of emergency stop.
40020	locked	The control right is preempted.
40050	map_parse_error_	Map parsing error
40051	map_not_exists	Map does not exist.
40052	load_map_error	Load map error

Error code	Name	Description
40053	load_mapobj_error	Reload map error
40054	empty_map	Empty map
40055	file_not_exists	File does not exist.
40056	map_convert_error	Map conversion failed.
40057	rawmap_not_exists	No rawmap file currently available
40058	calib_file_not_exists	No calib file currently available
40060	audio_not_exists	Audio file does not exist.
40061	audio_play_error	Play audio error
40062	upload_audio_error	Failed to upload audio file.
40063	audio_is_playing	Audio is playing.

Error code	Name	Description
40069	model_save_error	Error saving model file
40070	model_parse_error	Model file parsing error
40071	calibration_parse_error	Calibration data parsing error
40072	calibration_save_error	Error saving calibration file
40073	calibration_clear_error	Error clearing calibration data
40100	req_timeout	Request execution timeout
40101	req_forbidden	Request is forbidden
40102	robot_busy	Robot busy
40199	robot_internal_error	internal error
40200	tasklist_parse_error	Parse task list error

Error code	Name	Description
40201	illegal_tasklist_name	Illegal task list name
40202	tasklist_not_exists	Task list does not exist.
40203	tasklist_executing	Task list is being executed.
40300	set_param_type_error	Setting parameter type error
40301	set_param_not_exists	The set parameter does not exist.
40302	set_param_error	Error setting parameters
40310	save_param_type_error	Set and save parameter type error
40311	save_param_not_exists	Set and save the parameter does not exist.
40312	save_param_error	Error setting and saving parameters
40320	reload_param_type_error	Reload parameter type error

Error code	Name	Description
40321	reload_param_not_exists	Reload parameter does not exist.
40322	reload_param_error	Error reloading parameters
40400	src_require_error	Require control error
40401	src_release_error	Release control error
41000	init_status_error	Initialization status error
41001	loadmap_status_error	Map loading status error
41002	reloc_status_error	Relocation status error
41003	reloc_no_robot_home	No robotHome to No relocate
41004	confidence_too_low	Low confidence
41100	no_start_pos	No starting station

Error code	Name	Description
41101	no_ready_pos	No preparation station
41102	no_end_pod	No end station
41103	no_charge_pos	No charge station
41200	speed_illegal	Illegal speed value
42000	roller_connect_error	Incorrect roller connection
42001	roller_type_unknown	Unknown roller type
42002	roller_cmd_unsupported	Roller does not support this instruction.
42003	jack_connect_error	Incorrect jack connection
42004	jack_type_unknown	Unknown jack type
42005	jack_cmd_unsupported	Jack does not support this instruction.

Error code	Name	Description
43000	lift_failed	Error in lifting operation
44000	redis_conn_error	redis connection error
44001	subchannel_error	redis subscription error
60000	robot_error_wtype_res	Wrong message type. If the user sends a certain type of message to the wrong port, this response will be received.
60001	robot_error_utype_res	Unknown message type. Message type number is not defined above.
60002	robot_error_data_res	Wrong data area. You will get this response when the data area cannot be deserialized into a JSON object.
60003	robot_error_version_res	Response when the protocol version is wrong.
60004	robot_error_hugedata_res	The data area is too large, and the server will automatically disconnect due to limit 200 MB.
...	...	...

Map Format

Check this link for the map format of .smap <https://shimo.im/docs/0m8AZVlgBJOF5Aba/>

Robot Detection

We use Bonjour for robot detection.

For more details of Bonjour, please refer to the Apple Developer website <https://developer.apple.com/bonjour/>.

After installing Bonjour, you can find the domain of the robot in the LAN by searching for the service type `_robokitV2._tcp`, and then the IP address of the robot can be resolved by domain.

Example

Get all services of type `_robokitV2._tcp`:

```
dns-sd -B _robokitV2._tcp
```

The list of services is as follows:

```
Browsing for _robokitV2._tcp
Timestamp      A/R  Flags  if  Domain      Service Type      Instance Name
16:45:27.577   Add    3  10  local.      _robokitV2._tcp.   SeerRobot
```

Query the domain of a service:

```
dns-sd -L SeerRobot _robokitV2._tcp
```

The result of domain:

```
Lookup SeerRobot._robokitV2._tcp.local
```

16:46:45.624 SeerRobot._robokitV2._tcp.local. can be reached at seer-yyss-win.local.:19204 (interface 10) Flags: 1

Get IP address based on domain:

```
dns-sd -G v4 seer-yyss-win.local
```

The result of IP address:

Timestamp	A/R	Flags	if	Hostname	Address	TTL
16:47:49.544	Add	2	8	seer-yyss-win.local.	192.168.4.129	120

The result is 192.168.4.129.